33 Cosmology I: Basics

33.1 Preliminary Remarks

We now turn away from considering isolated systems (stars) to some (admittedly very idealised) description of the universe as a whole. This subject is known as Cosmology. It is certainly one of the most fascinating subjects of theoretical physics, dealing with such issues as the origin and ultimate fate and the large-scale structure of the universe.

Due to the difficulty of performing cosmological experiments and making precise measurements at large distances, many of the most basic questions about the universe are still unanswered today:

1. PAST:

   What actually happened at (or even before) what is usually called the Big Bang?

2. PRESENT:

   is our universe spatially finite or infinite?

3. FUTURE:

   Will our universe keep expanding forever or will it recollapse?

4. MATTER:

   What is Dark Matter?

5. ENERGY

   - Is Dark Energy, responsible for what appears to be a current phase of accelerated expansion of the universe, a cosmological constant?
   - If Dark Energy is indeed a cosmological constant, why is the cosmological constant so small and what determines its value?

While recent precision data, e.g. from supernovae surveys and detailed analysis of the cosmic microwave background radiation, suggest answers to at least some of these questions, these answers leave less wiggle-room for philosophical prejudices or esthetic preferences and actually just make the universe more mysterious than ever.

Of course, we cannot study any of these questions in detail, in particular because an important role in studying these questions is played by the interaction of cosmology with astronomy, astrophysics and elementary particle physics, each of these subjects deserving at least a course of its own.

Fortunately, however, many of the important features any realistic cosmological model should display are already present in some very simple models, the so-called *Friedmann-Lemaître-Robertson-Walker Models* (FLRW models) already studied in the 20's and

30's of the last century. They are based on the simplest possible ansatz for the metric compatible with the assumption that on large scales the universe is roughly homogeneous and isotropic (cf. the next section for a more detailed discussion of this *Cosmological Principle*) and have become the 'standard model' of cosmology.

We will see that they already display all the essential features such as

1. a Big Bang
2. an expanding universes (Hubble expansion) and a cosmological redshift.
3. different long-term behaviour (eternal expansion versus recollapse)

Our first aim will be to make maximal use of the symmetries that simple cosmological models should have to find a simple ansatz for the metric. Our guiding principle will be the *Cosmological Principle*.

33.2 Fundamental Assumption: The Cosmological Principle

At first, it may sound impossibly difficult to find solutions of the Einstein equations describing the universe as a whole. However: if one looks at the universe at large (very large) scales, in that process averaging over galaxies and even clusters of galaxies, then the situation simplifies a lot in several respects:

1. First of all, at those scales non-gravitational interactions can be completely ignored because they are either short-range (the nuclear forces) or compensate each other at large distances (electro-magnetism).
2. Furthermore we assume that the earth, and our solar system, or even our galaxy, have no privileged position in the universe (this is occasionally referred to as the *Copernican Principle*). This means that at large scales the universe should look the same from any point in the universe. Mathematically this means that there should be translational symmetries from any point of space to any other, in other words, space should be homogeneous.
3. Also, we assume that, at large scales, the universe looks the same in all directions. Thus there should be rotational symmetries and hence space should be isotropic.

Together, the second and third assumptions form the *Cosmological Principle*, which is the starting point for our discussion of cosmology and on which much of the work in cosmology is based. It is plausible (and true) that the assumption of isotropy (around

us) can be tested experimentally / observationally, while testing the assumption of homogeneity is evidently going to be more tricky.[138]

Making the above assumptions, it follows from our discussion in section 14, that the n-dimensional space (of course $n = 3$ for us) has n translational and $n(n-1)/2$ rotational Killing vectors, i.e. that the spatial metric is *maximally symmetric*. For $n = 3$, we will thus have six Killing vectors, two more than for the Schwarzschild metric, and the ansatz for the metric will simplify accordingly.

Note that, since we know from observation that the universe expands, we do not require a priori a maximally symmetric space-*time* as this would imply that there is also a timelike Killing vector.

What simplifies life considerably is the fact that (cf. the discussion in section 14) there are only three species of maximally symmetric spaces (for any n), namely

- flat space $\mathbb{R}^n$ (with its standard Euclidean metric),
- the sphere S^n (with tis standard "round" metric),
- and its negatively curved counterpart, the n-dimensional pseudosphere or hyperboloid we will call H^n.

Thus, for a space-time metric with maximally symmetric spacelike 'slices', the only unknown is the time-dependence of the overall size of the metric. More concretely, the metric can (now fixing the number of spatial dimensions to be $n = 3$) be chosen to be

$$ds^2 = -dt^2 + a^2(t)\left(\frac{dr^2}{1-kr^2} + r^2 d\Omega^2\right) \ , \tag{33.1}$$

where $k = 0, \pm 1$ corresponds to the three possibilities mentioned above. Thus the metric contains only one unknown function, the 'radius' or cosmic scale factor $a(t)$. This function will be determined by the Einstein equations via the matter content of the universe (we will of course be dealing with a non-vanishing energy-momentum tensor), modelled by a perfect fluid.

33.3 Fundamental Observations I: Olbers' Paradox

One paradox, popularised by Olbers (1826) but noticed before by others is the following. He asked the seemingly innocuous question "Why is the sky dark at night?". According to his calculation, reproduced below, the sky should instead be infinitely bright.

[138] The extent to which these assumptions, in particular homogeneity, can be observationally tested is discussed e.g. in C. Clarkson, R. Maartens, *Inhomogeneity and the foundations of concordance cosmology*, `arXiv:1005.2165 [astro-ph.CO]`, G. Ellis, *Inhomogeneity effects in Cosmology*, `arXiv:1103.2335 [astro-ph.CO]`, R. Maartens, *Is the Universe homogeneous?*, `arXiv:1104.1300 [astro-ph.CO]`.

The simplest assumption one could make in cosmology (prior to the discovery of the Hubble expansion) is that the universe is static, infinite and homogeneously filled with stars. In fact, this is probably the naive picture one has in mind when looking at the stars at night, and certainly for a long time astronomers had no reason to believe otherwise.

However, these simple assumptions immediately lead to a paradox, namely the conclusion that the night-sky should be infinitely bright (or at least very bright) whereas, as we know, the sky is actually quite dark at night. This is a nice example of how very simple observations can actually tell us something deep about nature (in this case, the nature of the universe). The argument runs as follows.

1. Assume that there is a star of brightness (luminosity) L at distance r. Then, since the star sends out light into all directions, the apparent luminosity A (neglecting absorption) will be
$$A(r) = L/4\pi r^2 \quad . \tag{33.2}$$

2. If the number density ν of stars is constant, then the number of stars at distances between r and $r + dr$ is
$$dN(r) = 4\pi\nu r^2 dr \quad . \tag{33.3}$$
Hence the total energy density due to the radiation of all the stars is
$$E = \int_0^\infty A(r)dN(r) = L\nu \int_0^\infty dr = \infty \quad . \tag{33.4}$$

3. Therefore the sky should be infinitely bright.

Now what is one to make of this? Clearly some of the assumptions in the above are much too naive. The way out suggested by Olbers is to take into account absorption effects and to postulate some absorbing interstellar medium, but this is also too naive because in an eternal universe we should now be in a stage of thermal equilibrium. Hence the postulated interstellar medium should emit as much energy as it absorbs, so this will not reduce the radiant energy density either.

Of course, the stars themselves are not transparent, so they could block out light completely from distant sources, but if this is to rescue the situation, one would need to postulate so many stars that every line of sight ends on a star, but then the night sky would be bright (though not infinitely bright) and not dark.

Modern cosmological models can resolve this problem in a variety of ways. For instance, the universe could be static but finite (there are such solutions, but this is nevertheless an unlikely scenario) or the universe is not eternal since there was a 'Big Bang' (and this is a more likely scenario).

33.4 Fundamental Observations II: The Hubble(-Lemaître) Expansion

We have already discussed one of the fundamental inputs of simple cosmological models, namely the cosmological principle. This led us to consider space-times with maximally-symmetric spacelike slices. One of the few other things that is definitely known about the universe, and that tells us something about the time-dependence of the universe, is that it expands or, at least that it appears to be expanding.

In fact, in the 1920's and 1930's, the astronomer Edwin Hubble made a remarkable discovery regarding the motion of galaxies. He found that light from distant galaxies is systematically redshifted (increased in wave-length λ), the increase being proportional to the distance d of the galaxy,

$$z := \frac{\Delta\lambda}{\lambda} \propto d \ . \tag{33.5}$$

Hubble interpreted this redshift as due to a Doppler effect and therefore ascribed a *recessional velocity* $v = cz$ to the galaxy. While, as we will see, this pure Doppler shift explanation is not tenable or at least not always the most useful way of phrasing things, the terminology has stuck, and Hubble's law can be written in the form

$$v = Hd \ , \tag{33.6}$$

where H is *Hubble's constant*. To set the historical record straight: credit for this fundamental discovery should perhaps (also) go to G. Lemaître.[139]

We will see later that in most cosmological models H is actually a function of time, so the H in the above equation should then be interpreted as the value H_0 of H today. It is one of the main goals of observational cosmology to determine H_0 and H as precisely as possible, and the main problem here is naturally a precise determination of the distances of distant galaxies. This is a complex and fascinating issue in its own right, but one that we will not go into here (safe for a brief mention of the *luminosity distance* at the end of section 34.9).[140] I will just conclude this section with one comment on the units usually employed to express galactic distances and the Hubble constant H_0.

Galactic distances are frequently measured in *mega-parsecs* (Mpc). A parsec is the distance from which a star subtends an angle of 2 arc-seconds at the two diametrically opposite ends of the earth's orbit. This unit arose because of the old trigonometric

[139] See e.g. M. Way, H. Nussbaumer, *The linear redshift-distance relationship: Lemaître beats Hubble by two years*, `arXiv:1104.3031v1 [physics.hist-ph]`; J.-P. Luminet, *Editorial note to "The beginning of the world from the point of view of quantum theory"*, `arXiv:1105.6271v1 [physics.hist-ph]`, and in particular also M. Livio, *The Expanding Universe: Lost (in Translation) and Found*, `http://hubblesite.org/pubinfo/pdf/2011/36/pdf.pdf` for an important addition to this debate.

[140] For a beautiful introduction to this subject of the *Cosmic Distance Ladder*, see the pdf-slides of a public talk by Fields Medalist (the mathematics counterpart of a Nobel Laureate!) Terry Tao at `http://terrytao.files.wordpress.com/2010/10/cosmic-distance-ladder.pdf` (you can also find a video of the talk on youtube).

method of measuring stellar distances (a triangle is determined by the length of one side and the two adjacent angles). 1 parsec is approximately 3×10^{18} cm, a little over 3 light-years. The Hubble constant is therefore often expressed in units of km s^{-1} $(\mathrm{Mpc})^{-1}$. The best currently available estimates point to a value of H_0 in the range (using a standard parametrisation)

$$\begin{aligned} H_0 &= 100h\mathrm{km/s/Mpc} \ , \\ h &= 0.71 \pm 0.06 \ . \end{aligned} \tag{33.7}$$

We will usually prefer to express it just in terms of inverse units of time. The above result leads to an order of magnitude range of

$$H_0^{-1} \approx 10^{10} \text{ years} \tag{33.8}$$

(whereas Hubble's original estimate was more in the 10^9 year range).

34 Cosmology II: Geometry and Physics of Robertson-Walker Metrics

34.1 Mathematical Model: the Robertson-Walker Metric

Having determined that the metric of a maximally symmetric space is of the simple form (14.27), we can now deduce that a space-time metric satisfying the Cosmological Principle can be chosen to be of the form (33.1),

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2 d\Omega^2\right] \quad . \tag{34.1}$$

Here we have used the fact that (as in the ansatz for a spherically symmetric metric) non-trival g_{tt} and g_{tr} can be removed by a coordinate transformation. This metric is known as the *Friedmann-Robertson-Walker metric* or just the Robertson-Walker metric, and spatial coordinates in which the metric takes this form are called *comoving coordinates*, for reasons that will become apparent below. The function $a(t)$, the "radius of the universe", is known as the *cosmic (or cosmological) scale factor*.

It is perhaps useful at this point to recall the symmetries of this metric (the isometries). See sections 14.4 and 39.1 for a more detailed and general discussion:

1. For $k = 0$, the spatial part of the metric is just (up to the overall factor $a(t)^2$), the Euclidean metric on $\mathbb{R}^3$. The isometry group of the Euclidean metric is the 6-parameter Euclidean group $\mathsf{E}(3)$, consisting of spatial translations and spatial rotations,

   $$k = 0: \quad \text{Isometry Group} = \mathsf{E}(3) = \mathsf{SO}(3) \ltimes \mathbb{R}^3 \quad . \tag{34.2}$$

   These are also symmetries of the space-time metric (34.1) for $k = 0$. For generic choices of $a(t)$, there are no further isometries of the metric.

2. For $k = +1$, the spatial part of the metric is just (up to the overall factor $a(t)^2$), the standard metric on the 3-sphere S^3, the one induced on S^3 from the Euclidean metric

   $$ds^2 = dx^2 + dy^2 + dz^2 + dw^2 \tag{34.3}$$

   on $\mathbb{R}^4$ via the embedding

   $$S^3 = \{(x, y, z, w) \in \mathbb{R}^4 : \; x^2 + y^2 + z^2 + w^2 = 1\} \quad . \tag{34.4}$$

   The isometry group of this metric on S^3 is therefore the group $\mathsf{SO}(4)$ of 4-dimensional rotations leaving the 4-dimensional Euclidean metric (and norm) invariant,

   $$k = +1: \quad \text{Isometry Group} = \mathsf{SO}(4) \quad . \tag{34.5}$$

This group is also 6-dimensional. These are also symmetries of the space-time metric (34.1) for $k = +1$. For generic choices of $a(t)$, there are no further isometries of the metric.

3. For $k = -1$, the spatial part of the metric is just (up to the overall factor $a(t)^2$), the standard metric on the 3-hyperboloid H^3, the one induced on H^3 from the Lorentzian metric

$$ds^2 = dx^2 + dy^2 + dz^2 - dw^2 \tag{34.6}$$

on $\mathbb{R}^4$ via the embedding

$$H^3 = \{(x, y, z, w) \in \mathbb{R}^4 : \; x^2 + y^2 + z^2 - w^2 = -1\} \; . \tag{34.7}$$

The isometry group of this metric on H^3 is therefore the group $\mathsf{SO}(3, 1)$ of 4-dimensional rotations leaving the 4-dimensional Lorentzian metric (and norm) invariant,

$$k = -1 : \quad \text{Isometry Group} = \mathsf{SO}(3, 1) \; . \tag{34.8}$$

This group is also 6-dimensional. These are also symmetries of the space-time metric (34.1) for $k = -1$. For generic choices of $a(t)$, there are no further isometries of the metric.

REMARKS:

1. The isometry group for $k = -1$ happens to be isomorphic to the Lorentz group of 4-dimensional Minkowski space. However, here it is realised purely on the spatial part of the metric, while the time coordinate t is not transformed at all. We will later on write the $k = -1$ metric in such a way, via a suitable coordinate transformation, that the $\mathsf{SO}(3, 1)$ symmetries are realised precisely by standard Lorentz transformations on the coordinates (cf. section 34.6).

2. As mentioned in the discussion around (14.29), a rescaling of k by a positive constant, $k \to k/L^2$, is equivalent to an overall constant rescaling of the spatial metric,

$$ds^2 = \frac{dr^2}{1 - kr^2/L^2} + r^2 d\Omega^2 = L^2 \left(\frac{d\tilde{r}^2}{1 - k\tilde{r}^2} + \tilde{r}^2 d\Omega^2 \right) \; , \tag{34.9}$$

where $\tilde{r} = r/L$. In the context of the Robertson-Walker metric (34.1), a scaling of k is thus equivalent to a scaling of the cosmic scale factor $a(t)$. This scaling freedom can be used

   - either to fix $k = 0, \pm 1$ once and for all,
   - or to normalise $a(t)$ in a convenient way, e.g. by the condition $a(t_0) = 1$ for t_0 the time "today".

In the latter case, the 3 different possibilities for the spatial geometry, are distinguished by $k < 0, k = 0, k > 0$. For the most part we work with the first option, but for certain questions (like "why is space so close to being flat today?") it is convenient to rephrase and express this in terms of k ("why is k so close to zero?"), which is evidently only meaningful if one does not restrict k to the 3 discrete values $k = 0, \pm 1$.

3. Another convenient way of representing this metric, that we will occasionally make use of below, is as

$$ds^2 = -dt^2 + a(t)^2(d\psi^2 + g_k(\psi)^2 d\Omega^2) \quad . \tag{34.10}$$

where (cf. (14.34) and (14.35))

$$g_k(\psi) = \begin{cases} \psi & k = 0 \\ \sin\psi & k = +1 \\ \sinh\psi & k = -1 \end{cases} \tag{34.11}$$

4. The metric of the three-space at constant t is

$$g_{ij} = a^2(t)\tilde{g}_{ij} \quad , \tag{34.12}$$

where $\tilde{g}_{ij}$ is the maximally symmetric spatial metric. Thus for $k = +1$, $a(t)$ directly gives the size (radius) of the universe. For $k = -1$, space is infinite, so no such interpretation is possible, but nevertheless $a(t)$ still sets the scale for the geometry of the universe, e.g. in the sense that the curvature scalar $R^{(3)}$ of the metric g_{ij} is related to the curvature scalar $\tilde{R}^{(3)}$ of $\tilde{g}_{ij}$ by

$$R^{(3)}(t) = \frac{1}{a^2(t)}\tilde{R}^{(3)} \quad . \tag{34.13}$$

Finally, for $k = 0$, three-space is flat and also infinite, but one could replace $\mathbb{R}^3$ by a three-torus T^3 (still flat but now compact) and then $a(t)$ would once again be related directly to the size of the universe at constant t.

5. Through the dependence of $a(t)$ on t, proper length scales and distances in the constant time surfaces depend on time. Thus $a(t)$ changes or sets the scale, i.e. $a(t)$ plays the role of a cosmological scale factor.

6. Note that the case $k = +1$ opened up for the very first time the possibility of considering, even conceiving, an unbounded but finite universe! These and other generalisations made possible by a general relativistic approach to cosmology are important as more naive (Newtonian) models of the universe immediately lead to paradoxes or contradictions (as we have seen e.g. in the discussion of Olbers' paradox in section 33.3).

34.2 Timelike Geodesics and Comoving Observers

We now look at timelike geodesics in the Robertson-Walker geometry. We will analyse the general case momentarily, but we can already identify one privileged class of geodesic observers by inspection. Indeed, as already noted (way back) in the discussion around (3.11), in a space-time of the Robertson-Walker form there is a particularly simple class of timelike geodesics, given by fixing once and for all the spatial coordinates of the observer.

Let us quickly rederive this result here. Note that, since $g_{tt} = -1$ is a constant and the off-diagonal time - space components of the metric are zero, $g_{tk} = 0$, one has

$$\Gamma_{\mu tt} = 0 \ . \tag{34.14}$$

Therefore the vector field ∂_t is geodesic, which can be expressed as the statement that

$$\nabla_t \partial_t := \Gamma^{\mu}_{tt} \partial_\mu = 0 \ . \tag{34.15}$$

In simpler terms this means that the curves $\vec{x} = const.$ ($\vec{x}$ referring to the spatial coordinates),

$$\tau \to (t(\tau), \vec{x}(\tau)) = (\tau, \vec{x}_0) \tag{34.16}$$

are geodesics.

Hence, in this coordinate system, observers remaining at fixed values of the spatial coordinates are in free fall. In other words, the coordinate system is falling with them or *comoving*, and the proper time τ along such geodesics coincides with the coordinate time or cosmic time t, $d\tau = dt$. It is these observers of constant $\vec{x}$ or constant (r, θ, ϕ) who all see the same isotropic universe at a given value of t.

Remarks:

1. This may sound a bit strange but a good way to visualise such a coordinate system is, as in Figure 44, as a mesh of coordinate lines drawn on a balloon that is being inflated or deflated (according to the behaviour of $a(t)$). Draw some dots on that balloon (that will eventually represent galaxies or clusters of galaxies). As the balloon is being inflated or deflated, the dots will move but the coordinate lines will move with them and the dots remain at fixed spatial coordinate values. Thus, as we now know, regardless of the behaviour of $a(t)$, these dots follow a geodesic, and we will thus think of galaxies in this description as being in free fall.

2. Recall that we had already encountered analogous comoving coordinates in our discussion of Lemaître coordinates for the Schwarzschild metric in section 27.3 and subsequently in equation (29.26) of section 29.3, when we had introduced proper time of the freely falling particles on the surface of the star to describe the metric induced on the surface of the star.

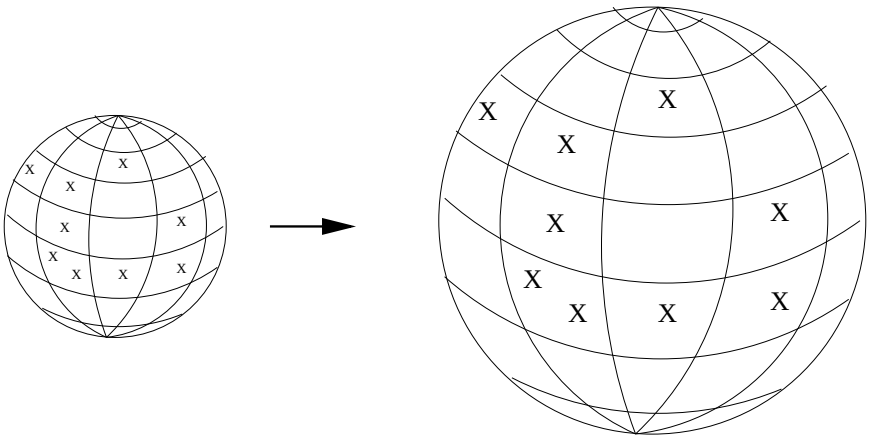


Figure 44: Illustration of a comoving coordinate system: Even though the sphere (universe) expands, the X's (galaxies) remain at the same spatial coordinates. These trajectories are geodesics and hence the X's (galaxies) can be considered to be in free fall. The figure also shows (cf. the discussion in section 6.8) that it is the number density per unit coordinate volume that is conserved, *not* the density per unit proper volume.

3. Another advantage of the comoving coordinate system is that the six-parameter family of isometries just acts on the spatial part of the metric. Indeed, let $K^i\partial_i$ be a Killing vector of the maximally symmetric spatial metric. Then $K^i\partial_i$ is also a Killing vector of the Robertson-Walker metric. This would not be the case if one had e.g. made an x-dependent coordinate transformation of t or a t-dependent coordinate transformation of the x^i. In those cases there would of course still be six Killing vectors, but they would have a more complicated form.

The worldlines of comoving observers discussed above are special timelike geodesics. To discuss the general case, it will be convenient to write the metric in the form (34.10)

$$ds^2 = -dt^2 + a(t)^2(d\psi^2 + g_k(\psi)^2 d\Omega^2) \ . \tag{34.17}$$

By spatial maximal symmetry and the associated conserved (angular) momenta, we can without loss of generality consider motion in the (t,ψ)-direction, so that $\dot{t}$ and $\dot{\psi}$ are related by

$$\dot{t}^2 - a(t)^2\dot{\psi}^2 = 1 \ , \tag{34.18}$$

where, as usual, an overdot refers to differentiation with respect to proper time.

Even though we do not have a timelike Killing vector (and its associated conserved energy) to further simplify this, in the case at hand we have plenty of spacelike Killing vectors $V^\alpha\partial_\alpha$ with $V^t = 0$. Among them there will be ψ-translational Killing vectors which have the form

$$V = f(\theta,\phi)\partial_\psi + \dots \tag{34.19}$$

(the 3-dimensional counterparts of the Killing vectors $V_{(1)}, V_{(2)}$ (9.55) of the 2-sphere, say). Associated to any such Killing vector and the timelike geodesic there is the conserved momentum

$$P = \dot{x}^\alpha V_\alpha = a(t)^2 \dot{\psi} f(\theta, \phi) \ . \tag{34.20}$$

Since θ and ϕ are constant along the lightray, we can absorb $f(\theta, \phi)$ into the definition of P, and thus we have

$$\dot{\psi} = P/a^2 \ . \tag{34.21}$$

Therefore (34.18) can be written as

$$\dot{t} = \sqrt{1 + P^2/a^2} \ . \tag{34.22}$$

In particular, we see that comoving observers are characterised by $P = 0$,

$$P = 0 \quad \Rightarrow \quad \dot{\psi} = 0 \quad \text{(comoving)} \ , \tag{34.23}$$

and that precisely for these observers the cosmic time t coincides with their proper time,

$$P = 0 \quad \Leftrightarrow \quad \dot{t} = 1 \quad \Leftrightarrow \quad dt = d\tau \ . \tag{34.24}$$

Nevertheless, even in the general case it is useful to combine the two previous equations to obtain an equation for ψ as a function of t, namely (assuming $P \neq 0$)

$$\frac{d\psi(t)}{dt} = \frac{\dot{\psi}}{\dot{t}} = \frac{P}{\sqrt{a(t)^4 + P^2 a(t)^2}} = \frac{1}{a(t)\sqrt{1 + a(t)^2/P^2}} \ . \tag{34.25}$$

In general, even for simple power-law behaviours for $a(t)$ (which we will typically find as solutions to the Einstein equations in the spatially flat case $k = 0$), this equation cannot be solved in closed form (but can be approximated by a tractable, even elementary, integral when $a(t) \gg |P|$ or when $a(t) \ll |P|$).

34.3 Velocity - Distance Relation, Recessional Velocities and the Hubble Sphere

The Robertson-Walker metric immediately, and in complete generality, implies a crude distance - velocity relation reminiscent of Hubble's law. Namely, let us ego- or geocentrically place ourselves at the origin $r = 0$ (remember that because of maximal symmetry this point is as good as any other and in no way privileged). Consider another galaxy following the comoving geodesic at the fixed value $r = r_1$. Its "instantaneous" proper distance $R_p(t)$ at time t can be calculated from

$$dR_p = a(t) \frac{dr}{(1 - kr^2)^{1/2}} \quad \Rightarrow \quad R_p(t) = a(t) f_k(r_1) \ , \tag{34.26}$$

where

$$f_k(r) = \begin{cases} r & k = 0 \\ \arcsin r & k = +1 \\ (\sinh)^{-1} r & k = -1 \end{cases} \tag{34.27}$$

Note that the $f_k(r)$ are the inverses of the functions $g_k(\psi)$ defined in (34.11) (the precise form of $f_k(r)$ will however be irrelevant for this argument). If we use the coordinates (t, ψ) instead of (t, r), the instantaneous proper distance to a point with coordinate ψ_1 simply has the form

$$R_p(t) = a(t)\psi_1 \quad . \tag{34.28}$$

It follows that its proper velocity is

$$V_p(t) \equiv \frac{d}{dt} R_p(t) = \dot{a}(t)\psi_1 = H(t) R_p(t) \quad . \tag{34.29}$$

Here we have introduced the *Hubble parameter*

$$H(t) = \frac{\dot{a}(t)}{a(t)} \quad , \tag{34.30}$$

which plays a pivotal role in cosmology.

The relation (34.29) clearly expresses something like Hubble's law $v = Hd$ (33.6): all objects run away from each other with velocities proportional to their distance. And the above derivation shows that such an (at first sight perhaps surprising) relationship between (some kind of) velocity and (some kind of) distance can arise as a general purely geometric statement about the geometry of space-time. We will have much more to say about $H(t)$, and about the relation between distance and redshift z (which is what is actually observed), below.

Calculating the second time-derivative of $R_p(t)$, one finds

$$\frac{d^2}{dt^2} R_p(t) = (\frac{d}{dt} H(t)) R_p(t) + H(t) \frac{d}{dt} R_p(t) = \frac{\ddot{a}(t)}{a(t)} R_p(t) \tag{34.31}$$

since

$$\frac{d}{dt} H(t) = \frac{\ddot{a}(t)}{a(t)} - \frac{\dot{a}(t)^2}{a(t)^2} = \frac{\ddot{a}(t)}{a(t)} - H(t)^2 \quad . \tag{34.32}$$

Thus the cosmological expansion or contraction can be visualised as acting like a linear harmonic oscillator force on the separation of comoving objects, with (in general time-dependent) real or imaginary frequency $\omega(t)$,

$$\frac{d^2}{dt^2} R_p(t) + \omega(t)^2 R_p(t) = 0 \quad , \quad \omega(t)^2 = -\frac{\ddot{a}(t)}{a(t)} \quad . \tag{34.33}$$

Universes with an accelerating expansion thus lead to imaginary frequencies, and hence to an exponential-like rather than harmonic motion (over periods of time during which the time-dependence of the frequency can be neglected).

If the object is not comoving, i.e. is not sitting at a fixed value of ψ, say, one has

$$R_p(t) = a(t)\psi(t) \quad , \tag{34.34}$$

and then its proper velocity is

$$V_p(t) = \frac{d}{dt}R_p(t) = H(t)R_p(t) + a(t)\frac{d\psi(t)}{dt} \quad . \tag{34.35}$$

Here "proper" refers to the fact that it is the t-derivative of the instantaneous proper distance, but it should be kept in mind that t is not the proper time for non-comoving observers. This proper velocity can be decomposed into what are known as its *recessional velocity* (equal to the proper velocity for a comoving object) and its *peculiar velocity* (relative to the recessional velocity or *Hubble flow*),

$$V_p(t) = H(t)R_p(t) + a(t)\frac{d\psi(t)}{dt} \equiv V_{rec}(t) + V_{pec}(t) \quad . \tag{34.36}$$

As a consequence of (34.18), one has

$$V_{pec}(t)^2 = a(t)^2 \left(\frac{d\psi(t)}{dt}\right)^2 < 1 \quad . \tag{34.37}$$

Alternatively this follows from (34.25), which allows us to express the peculiar velocity as

$$V_{pec}(t) = a(t)\frac{d\psi(t)}{dt} = \frac{1}{\sqrt{1 + a(t)^2/P^2}} \quad . \tag{34.38}$$

Either way we see that the peculiar velocity is always "subluminal".

However, the recessional velocity is not restricted in this way and in most cosmological models superluminal recessional velocities will occur for objects which are suffficiently far away (in the sense of having a sufficiently large proper distance $R_p(t)$).

REMARKS:

1. There is absolutely nothing illegal or pathological about this because V_{rec} measures the rate of an increase in distance between two objects, not any locally measurable velocity. For example, even in Special Relativity, if in your rest-frame you send off two objects in opposite directions at speeds $> c/2$ each, then the distance between them grows at a rate (measured with respect to your proper time) larger than c, but clearly you have not violated or disproven Special Relativity by doing this.

2. Moreover, in a curved space-time a comparison of vectors (say velocity vectors) at different points requires some care. Indeed, as explained in section 5.8, this requires some notion of parallel transport in order to have two vectors at the same point that one can meaningfully compare.[141]

[141] See e.g. A. Kaya, *Hubble's law and faster than light expansion speeds*, `arXiv:1107.5168 [physics.gen-ph]` for an analysis of this issue from the parallel transport point of view.

3. For a further illustration of the harmlessness of superluminal recessional velocities in the present context, it suffices to consider the special case $a(t) = t$, in which case one has

$$a(t) = t \quad \Rightarrow \quad H(t) = 1/t \quad \Rightarrow \quad V_{rec}(t) = R_p(t)/t \ . \tag{34.39}$$

At any given time t, objects further away than t appear to have superluminal recession velocities. On the other hand, as shown in section 37.1, the space-time with $a(t) = t$ and $k = -1$ is just a part of Minkowski space. Thus even Minkowski space can be foliated in such a way (by hyperboloids in the future lightcone) that events appear to have superluminal recession velocities, but evidently there is nothing here that violates any of the postulates of Special Relativity.

4. Cosmologists frequently refer to the sphere beyond which the recessional velocity exceeds the speed of light as the *Hubble sphere*. Its radius $R_H(t)$ at time t, the *Hubble radius*, is (restoring for once and temporarily the speed of light c)

$$V_{rec}(t) = c \quad \Leftrightarrow \quad R_H(t) = c/H(t) \ . \tag{34.40}$$

Misleadingly, this surface is also often referred to as the *Hubble horizon*, the reason for this apparently being the idea or belief that we can never observe objects outside the Hubble sphere, but this is in general not correct. In particular, it is not correct to say (perhaps based on a mistaken analogy with special relativistic reasoning) that objects with recessional velocities $V_{rec} > c$ are infinitely redshifted and therefore invisible to us. There is indeed a cosmological redshift, worked out in section 34.8, and there is also an ensuing Hubble-like redshift - distance relation in Robertson-Walker geometries, derived in section 34.9. However, this cannot be written as a standard special relativistic recessional velocity - redshift relation involving the recessional velocity.[142]

5. It is true that there are limits to how much of the universe one can observe at any given time, and it is also true that in certain situations the Hubble radius $R_H(t)$ provides one with an order of magnitude estimate of the size of the visible universe.

However, it is certainly misleading (even though some people appear to be obsessed with this) to think of the visible universe (or the inside of the Hubble sphere) as somehow being like the inside of a Schwarzschild black hole or some such nonsense. In fact, with any standard definition of a black hole going beyond pop-sci culture wisdom this statement is so obviously wrong or misleading in so many respects that I don't even know where to start (so don't get me started). Nevertheless I will briefly revisit this claim in section 36.9. For the time you may enjoy poking holes into this statement yourself ...

[142] See also T. Davis, C. Lineweaver, *Superluminal Recession Velocities*, `arXiv:astro-ph/0011070` for a nice discussion of this.

We will return to some of these issues in a (slightly) more quantitative way later on, in sections 36.7 – 36.9, with the Friedmann equations (i.e. the Einstein equations for the standard model of cosmology we are in the process of developing) at our disposal.

34.4 Painlevé-Gullstrand-like Coordinates for Comoving Observers

In section 27.2 we introduce coordinates for the Schwarzschild metric that are adapted to radial geodesic observers, known as Painlevé-Gullstrand coordinates. We can do something analogous for the comoving observers of the Robertson-Walker metrics.

For $k = 0$ the Robertson-Walker metric is

$$ds^2 = -dt^2 + a(t)^2(dr^2 + r^2 d\Omega^2) \ . \tag{34.41}$$

To put this into PG-like form, we keep t (which is, after all, already the proper time of comoving observers) but introduce, instead of the comoving coordinate r the area radius

$$\tilde{r}(t, r) = a(t)r \ . \tag{34.42}$$

In terms of this the Robertson-Walker metric takes the PG-like form

$$\begin{aligned} ds^2 &= -(1 - \tilde{r}^2 H(t)^2)dt^2 - 2\tilde{r}H(t)dtd\tilde{r} + (d\tilde{r}^2 + \tilde{r}^2 d\Omega^2) \\ &= -dt^2 + (d\tilde{r} - \tilde{r}H(t)dt)^2 + \tilde{r}^2 d\Omega^2 \ , \end{aligned} \tag{34.43}$$

where, as above, $H(t) = \dot{a}(t)/a(t)$ is the Hubble parameter.

Remarks:

1. This is clearly analogous to the Schwarzschild metric in PG coordinates (27.12)

$$\begin{aligned} ds^2 &= -(1 - 2m/r)dT^2 + 2\sqrt{2m/r}dTdr + (dr^2 + r^2 d\Omega^2) \\ &= -dT^2 + \left(dr + \sqrt{2m/r}dT\right)^2 + r^2 d\Omega^2 \ . \end{aligned} \tag{34.44}$$

2. Just as the Schwarzschild metric in PG coordinates is adapted to observers with $dr = -\sqrt{2m/r}dT$ (for which T is proper time, $\dot{r} = -\sqrt{2m/r}$ describing geodesic radial free fall with $E = 1$), the metric (34.43) is adapted to observers with

$$\frac{d}{dt}\tilde{r} = H(t)\tilde{r} \tag{34.45}$$

which are precisely the comoving observers $\tilde{r}(t) = a(t)r$ with r fixed obeying the Hubble law (34.29).

3. Performing the same coordinate transformation to the area radius $\tilde{r} = a(t)r$ for $k \neq 0$, one finds the metric

$$ds^2 = -dt^2 + \frac{1}{1 - k\tilde{r}^2/a^2}\left(d\tilde{r} - \tilde{r}H(t)dt\right)^2 + \tilde{r}^2 d\Omega^2 \ . \tag{34.46}$$

4. While these PG-like coordinates are not widely used in the cosmological context, we make use of them in section 29.5, in the description of the interior geometry of a collapsing star (because this PG-like form of the metric makes it particularly easy to match the interior metric to the exterior Schwarzschild metric).

34.5 Conformal Time η

Writing the Robertson-Walker metric as

$$ds^2 = -dt^2 + a(t)^2 d\tilde{s}^2 = a(t)^2[-dt^2/a(t)^2 + d\tilde{s}^2] \ , \tag{34.47}$$

where a tilde refers to the maximally symmetric spatial metric, we see that it is natural to introduce a new time-coordinate η through

$$d\eta = dt/a(t) \ , \tag{34.48}$$

in terms of which the Robertson-Walker metric takes the simple form

$$d\eta = dt/a(t) \quad \Rightarrow \quad ds^2 = a^2(\eta)(-d\eta^2 + d\tilde{s}^2) \tag{34.49}$$

($a(\eta)$ is short (and sloppy) for $a(t(\eta))$). In terms of polar coordinates (34.10), this becomes

$$ds^2 = a^2(\eta)(-d\eta^2 + d\psi^2 + g_k(\psi)^2 d\Omega^2) \ . \tag{34.50}$$

Remarks:

1. In particular, "radial" null lines are determined by $d\eta = \pm d\psi$, as in flat space, and η is also known as *conformal time*. This coordinate is very convenient for discussing the causal structure of the Friedmann-Lemaître-Robertson-Walker universes.

2. Since the η-dependence resides exclusively in the overall conformal factor of the metric, the vector ∂_η is a *conformal Killing vector* of the Robertson-Walker metric in the sense of (10.5),

$$C = \partial_\eta : \quad \nabla_\alpha C_\beta + \nabla_\beta C_\alpha = 2\frac{a'(\eta)}{a(\eta)}\, g_{\alpha\beta} = 2\dot{a}(t)\, g_{\alpha\beta} \ . \tag{34.51}$$

3. We will use conformal time immediately below to establish the conformal flatness of the Robertson-Walker metrics, in section 37.3 to solve the cosmological Einstein equations in a particular case, and we will use the above conformal Killing vector and the associated conserved charge for null geodesics (10.9) in the discussion of the cosmological redshift in section 34.8.

In the spatially flat case $k = 0$, (34.50) shows that Robertson-Walker metrics are conformally flat, here written in radial polar coordinates with $\psi = r$,

$$k = 0 \quad \Rightarrow \quad ds^2 = a^2(\eta)(-d\eta^2 + dr^2 + r^2 d\Omega^2) = a(\eta)^2 \eta_{\alpha\beta} dx^\alpha dx^\beta \ . \tag{34.52}$$

This is actually true for all Robertson-Walker metrics, i.e. also for $k \neq 0$, but the coordinate transformation required to exhibit this as explicitly as in the $k = 0$ case is somewhat more involved (and, as we will see below, the conformal factor does not depend just on the Minkowski time coordinate).

As a first step it will be convenient to introduce null coordinates

$$u = \eta - \psi \quad , \quad v = \eta + \psi \ , \tag{34.53}$$

in terms of which the metric (34.50) takes the form

$$ds^2 = a^2(\eta) \left[-du\, dv + g_k((v-u)/2)^2 \, d\Omega^2 \right] \ . \tag{34.54}$$

This metric is conformally flat if the line element in brackets, which does not depend on the cosmic scale factor $a(t)$ or $a(\eta)$, is conformally flat, so let us focus on

$$d\bar{s}^2 = ds^2/a^2(\eta) = -du\, dv + g_k((v-u)/2)^2 \, d\Omega^2 \ , \tag{34.55}$$

where, recall, $g_k(\psi) = \sin(\mathrm{h})\psi$ for $k = \pm 1$ respectively.

Given a metric of this form, it is natural to consider transformations of the form

$$U = U(u) \quad , \quad V = V(v) \tag{34.56}$$

because under such transformations $dudv$ and $dUdV$ are just related by an overall conformal factor. It is now straightforward to check, using some basic trigonometric identities (or their hyperbolic counterparts) that the specific coordinate transformation

$$U = 2\tan(\mathrm{h})u/2 \quad , \quad V = 2\tan(\mathrm{h})v/2 \tag{34.57}$$

(we will consider an alternative transformation for $k = -1$ below) is such that it transforms $d\bar{s}^2$ into

$$d\bar{s}^2 = \left(\cos(\mathrm{h})^2 u/2\right) \left(\cos(\mathrm{h})^2 v/2\right) \left[-dU\, dV + \tfrac{1}{4}(V-U)^2 d\Omega^2 \right] \ . \tag{34.58}$$

Now the metric in brackets is just the Minkowski metric, written in radial null coordinates, as can be seen by undoing the transformation to null coordinates through

$$U = T - R \quad , \quad V = T + R \ , \tag{34.59}$$

which results in

$$-dU\, dV + \tfrac{1}{4}(V-U)^2 d\Omega^2 = -dT^2 + dR^2 + R^2 d\Omega^2 \ , \tag{34.60}$$

as claimed. This establishes explicitly the conformal flatness of the Robertson-Walker metrics even for $k \neq 0$. In particular, therefore, the Weyl tensor of a Robertson-Walker metric is zero (section 11.4), and the Riemann tensor can be expressed in terms of its traces, the Ricci tensor and the Ricci scalar.

Note that for $k \neq 0$ the conformal factor relating the Robertson-Walker metric to the Minkowski metric is not just a function of the Minkowski time T alone, but a particular function of T and R,

$$k \neq 0: \qquad ds^2_{FRW} = \Omega^2(T,R)\left(-dT^2 + dR^2 + R^2 d\Omega^2\right) \quad . \tag{34.61}$$

This could hardly be otherwise because (recall the discussion in section 34.1) for $k \neq 0$ the isometry group is not the Euclidean group, the symmetry group of the Euclidean metric $dR^2 + R^2 d\Omega^2$, but rather it is $\mathsf{SO}(4)$ for $k = +1$ and $\mathsf{SO}(3,1)$ for $k = -1$.

In the latter case, we can make this symmetry manifest by slightly modifying the above procedure. To that end consider, instead of (34.57), the transformation

$$k = -1: \quad U = T - R = \mathrm{e}^{\,u} \quad , \quad V = T + R = \mathrm{e}^{\,v} \quad , \tag{34.62}$$

leading to

$$T = \mathrm{e}^{\,\eta}\cosh\psi \quad , \quad R = \mathrm{e}^{\,\eta}\sinh\psi \quad . \tag{34.63}$$

Then a straightforward calculation, using

$$R^2 = \mathrm{e}^{\,2\eta}\sinh^2\psi = UV\sinh^2\psi \quad , \tag{34.64}$$

shows that

$$\begin{aligned} ds^2 &= -dt^2 + a(t)^2(d\psi^2 + \sinh^2\psi\, d\Omega^2) \\ &= a(\eta)^2\left[-du\, dv + \sinh^2\psi\, d\Omega^2\right] \\ &= \frac{a(\eta)^2}{UV}\left[-dU\, dV + UV\sinh^2\psi\, d\Omega^2\right] \\ &= \mathrm{e}^{\,-2\eta}a(\eta)^2\left[-dT^2 + dR^2 + R^2 d\Omega^2\right] \quad . \end{aligned} \tag{34.65}$$

Here the term in brackets is just the 4-dimensional Minkowski metric,

$$-dT^2 + dR^2 + R^2 d\Omega^2 = -dT^2 + d\vec{X}^2 \quad , \tag{34.66}$$

and the conformal prefactor is only a function of η (or t) which, via

$$T^2 - R^2 = \mathrm{e}^{\,2\eta} \quad , \tag{34.67}$$

is itself only a function of the Lorentz invariant quantity

$$T^2 - R^2 = T^2 - \vec{X}^2 \quad . \tag{34.68}$$

Written in this way, the complete $\mathsf{SO}(3,1)$ symmetry of the $k = -1$ metrics is manifest. In the original comoving coordinates, it was realised as the spatial isometry group of the hyperboloid H^3, with t not transformed. In the new form of the metric, it is explicitly realised as the Lorentz transformations of (an auxiliary) 4-dimensional Minkowski space-time with coordinates $(T, \vec{X})$, but with the original time coordinate t or η transforming as a Lorentz scalar under these transformations, as it should.

34.7 Area Measurements and Number Counts

The aim of this and the subsequent sections is to learn as much as possible about the general properties of Robertson-Walker geometries (without using the Einstein equations) with the aim of looking for observational means of distinguishing e.g. among the models with $k = 0, \pm 1$.

To get a feeling for the geometry of the Schwarzschild metric, we studied the properties of areas and lengths in the Schwarzschild geometry. Spatial length measurements are rather obvious in the Robertson-Walker geometry, so here we focus on the properties of areas.

We write the spatial part of the Robertson-Walker metric in polar coordinates as (34.10)

$$ds^2 = a(t)^2[d\psi^2 + g_k^2(\psi)d\Omega^2] \ , \tag{34.69}$$

where $g_k(\psi) = \psi, \sin\psi, \sinh\psi$ for $k = 0, +1, -1$ (see (14.35)). Now the radius of a surface $\psi = \psi_0$ around the point $\psi = 0$ (or any other point, our space is isotropic and homogeneous) is given by

$$\rho = a\int_0^{\psi_0} d\psi = a\psi_0 \ . \tag{34.70}$$

On the other hand, the area of this surface is determined by the induced metric $a^2 g_k^2(\psi_0)d\Omega^2$ and is

$$A(\rho) = a^2 g_k^2(\psi_0)\int_0^{2\pi} d\phi \int_0^{\pi} d\theta \sin\theta = 4\pi a^2 g_k^2(\rho/a) \ . \tag{34.71}$$

For $k = 0$, this is just the standard behaviour

$$A(\rho) = 4\pi\rho^2 \ , \tag{34.72}$$

but for $k = \pm 1$ the geometry looks quite different. For $k = +1$, we have

$$A(\rho) = 4\pi a^2 \sin^2(\rho/a) \ . \tag{34.73}$$

Thus the area reaches a maximum for $\rho = \pi a/2$ (or $\psi = \pi/2$), then *decreases* again for larger values of ρ and goes to zero as $\rho \to \pi a$. Already the maximal area, $A_{max} = 4\pi a^2$ is much smaller than the area of a sphere of the same radius in Euclidean space, which would be $4\pi\rho^2 = \pi^3 a^2$.

This behaviour is best visualised by replacing the three-sphere by the two-sphere and looking at the circumference of circles as a function of their distance from the origin (see Figure 45).

For $k = -1$, we have

$$A(\rho) = 4\pi a^2 \sinh^2(\rho/a) \ , \tag{34.74}$$

so in this case the area grows much more rapidly with the radius than in flat space.

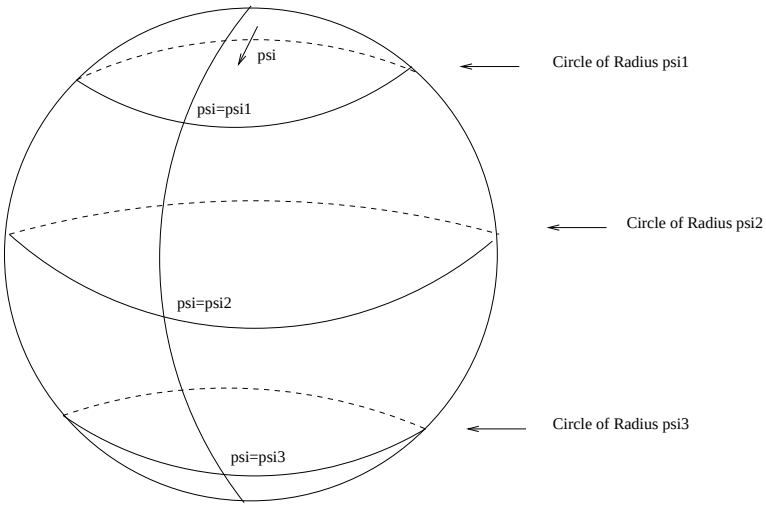


Figure 45: Visualisation of the $k = +1$ Robertson-Walker geometry via a two-sphere of unit radius: Circles of radius ψ, measured along the two-sphere, have an area which grows at first, reaches a maximum at $\psi = \pi/2$ and goes to zero when $\psi \to \pi$. E.g. the maximum value of the circumference, at $\psi = \pi/2$, namely 2π, is much smaller than the circumference of a circle with the same radius $\pi/2$ in a flat geometry, namely $2\pi \times \pi/2 = \pi^2$. Only for ψ very small does one approximately see a standard Euclidean geometry.

In principle, this distinct behaviour of areas in the models with $k = 0, \pm 1$ might allow for an empirical determination of k. For instance, one might make the assumption that there is a homogeneous distribution of the number and brightness of galaxies, and one could try to determine observationally the number of galaxies as a function of their apparent luminosity. As in the discussion of Olbers' paradox, the radiation flux would be proportional to $F \propto 1/\rho^2$. In Euclidean space ($k = 0$), one would expect the number $N(F)$ of galaxies with flux greater than F, i.e. distances less than ρ to behave like ρ^3, so that the expected Euclidean behaviour would be

$$N(F) \propto F^{-3/2} \quad . \tag{34.75}$$

Any empirical departure from this behaviour could thus be an indication of a universe with $k \neq 0$, but clearly, to decide this, many other factors (redshift, evolution of stars, etc.) would have to be taken into account. This illustrates as a matter of principle how the geometry of the spatial slices influences, and can be encoded in, observable quantities. In practice, however,

> [...] the statistical uncertainties, together with source evolution (which affects the detection probabilities), prevent this from being a useful test of k.[143]

[143] G. Ellis, R. Maartens, M. MacCallum, *Relativistic Cosmology*, section 7.6.

34.8 Cosmological Redshift

The most important information about the cosmic scale factor $a(t)$ comes from the observation of shifts in the frequency of light emitted by distant sources.

To calculate the expected shift in a Robertson-Walker geometry, let us again place ourselves at the origin $r = 0$. We consider a radially travelling electro-magnetic wave (a lightray) and consider the equation $d\tau^2 = 0$ or

$$dt^2 = a^2(t)\frac{dr^2}{1-kr^2} \quad . \tag{34.76}$$

Since the cosmological scale factor $a(t)$ sets the length scale, one may expect that wave lengths at different times are related by

$$\frac{\lambda(t_1)}{\lambda(t_0)} = \frac{a(t_1)}{a(t_0)} \quad , \tag{34.77}$$

leading to the relation

$$\frac{\omega(t_0)}{\omega(t_1)} = \frac{a(t_1)}{a(t_0)} \quad , \tag{34.78}$$

among the frequencies. This is indeed the correct result.

As in our discussion of the gravitational redshift in section 3.5, I will analyse this situation in two ways, in a geometric optics approach, where we trace the lightrays in the above geometry, and in a slightly more covariant language using the geodesic equation and the symmetries and associated conserved charges. I will also give a third, essentially one-line (but perhaps at first somewhat obscure looking), derivation based on the conformal Killing vector (34.51) and its associated conserved charge.

1. Let us assume that the wave leaves a galaxy located at $r = r_1$ at the time t_1. Then it will reach us at $r = 0$ at a time t_0 given by

$$f_k(r_1) = \int_{r_1}^{0} \frac{dr}{\sqrt{1-kr^2}} = \int_{t_1}^{t_0} \frac{dt}{a(t)} \quad . \tag{34.79}$$

   Note that there will only be a solution to this equation if the light from the galaxy at $r = r_1$ actually reaches us at a time t_0. In this sense galaxies whose light has not yet reached us (or may perhaps never reach us) are implicitly (and now, having said this, explicitly) excluded from the analysis - after all, such galaxies are not particularly useful for analysing redshifts.

   As typical galaxies will be comoving, i.e. have have constant spatial coordinates, $f_k(r_1)$ (34.27) is time-independent. If the next wave crest leaves the galaxy at r_1 at time $t_1 + \delta t_1$, it will arrive at a time $t_0 + \delta t_0$ determined by

$$f_k(r_1) = \int_{t_1+\delta t_1}^{t_0+\delta t_0} \frac{dt}{a(t)} \quad . \tag{34.80}$$

Subtracting these two equations and making the (eminently reasonable) assumption that the cosmic scale factor $a(t)$ does not vary significantly over the period δt given by the frequency of light, we obtain

$$\frac{\delta t_0}{a(t_0)} = \frac{\delta t_1}{a(t_1)} \quad . \tag{34.81}$$

Indeed, say that $b(t)$ is the integral of $1/a(t)$. Then we have

$$b(t_0 + \delta t_0) - b(t_1 + \delta t_1) = b(t_0) - b(t_1) \quad , \tag{34.82}$$

and Taylor expanding to first order, we obtain

$$b'(t_0)\delta t_0 = b'(t_1)\delta t_1 \quad , \tag{34.83}$$

which is the same as (34.81). Therefore the observed frequency ω_0 is related to the emitted frequency ω_1 by

$$\frac{\omega_0}{\omega_1} = \frac{a(t_1)}{a(t_0)} \quad , \tag{34.84}$$

precisely as anticipated in (34.78). I will comment further on this result below.

2. As in derivation 2 of section 3.5, we describe a lightray by the null wave vector $k^\mu = (\omega, \vec{k})$. The frequency measured by an observer with velocity u^μ is then $\omega = -u^\mu k_\mu$ (3.117).

   Adapting the discussion of timelike geodesics in the Robertson-Walker geometry in section 34.2 to null rays, we can choose the wave vector to be of the form

   $$k^\mu = (\dot{t}, \dot{\psi}, 0, 0) \tag{34.85}$$

   with

   $$-\dot{t}^2 + a(t)^2\dot{\psi}^2 = 0 \quad . \tag{34.86}$$

   In section 3.5 we used the timelike Killing vector of the static spherically symmetric metric, and its associated conserved energy, to relate the measured frequencies for static observers at different radial positions and to determine the gravitational redshift. Here we use one of the spatial Killing vectors to deduce, as in section 34.2, that

   $$\dot{\psi} = P/a^2 \tag{34.87}$$

   for some constant P. Thus

   $$\dot{t} = |P|/a \quad \Rightarrow \quad k^\alpha = (|P|/a)\,(1, \pm 1/a, 0, 0) \quad . \tag{34.88}$$

   The observers we are interested in are the comoving observers at fixed values of the spatial coordinates, i.e. with $u^\alpha = (1, 0, 0, 0)$. Thus these measure the frequency

   $$\omega(t) = -k^\alpha u_\alpha = |P|/a(t) \quad . \tag{34.89}$$

In particular, for the ratio of frequencies at times t_0 and t_1 one has

$$\frac{\omega(t_0)}{\omega(t_1)} = \frac{a(t_1)}{a(t_0)} \quad , \tag{34.90}$$

in complete agreement with the result (34.78) found previously in (34.84).

3. Alternatively, and even more quickly, one can use the *conformal Killing vector* (34.51)

$$C = \partial_\eta = a(t)\partial_t \quad \text{or} \quad C^\alpha = a(t)u^\alpha \tag{34.91}$$

of the Robertson-Walker metric, and the associated conserved quantity $C_\alpha \dot{x}^\alpha$ (10.9) for null geodesics, to deduce

$$C_\alpha k^\alpha = -a(t)\omega(t) = \text{const.} \quad , \tag{34.92}$$

leading imediately to the same conclusion (34.90).

Astronomers like to express this result in terms of the redshift parameter (see the discussion of Hubble's law above)

$$z = \frac{\lambda_0 - \lambda_1}{\lambda_1} \quad , \tag{34.93}$$

which in view of the above result we can write as

$$z = \frac{a(t_0)}{a(t_1)} - 1 \quad . \tag{34.94}$$

Thus if the universe expands one has $z > 0$ and there is a redshift while in a contracting universe with $a(t_0) < a(t_1)$ the light of distant glaxies would be blueshifted.

REMARKS:

1. This cosmological redshift has nothing to do with the star's own gravitational field - that contribution to the redshift is completely negligible compared to the effect of the cosmological redshift.

2. Unlike the gravitational redshift we discussed before, this cosmological redshift is symmetric between receiver and emitter, i.e. light sent from the earth to the distant galaxy would likewise be redshifted if we observe a redshift of the distant galaxy.

3. However, like the gravitational redshift, the final result depends only on the position (time) of emission and arrival of the lightray, not on the intermediate gravitational field (cosmic scale factor). This illustrates that fundamentally the redshift is due to the different reference frames used by emitter and observer, not due to the fact that something happens to the lightray along the way. We will briefly return to this matter from a slightly different perspective in section 34.11.

4. While the previous remark suggests a purely Doppler-like explanation of the cosmological redshift it is best to think of the redshift as a combined effect of gravitational and Doppler redshifts. Without additional choices (like preferred families of intermediate observers) it is not very meaningful to separate this into the two and/or to interpret this only in terms of one of them.[144]

5. Nowadays, astronomers tend to express the distance of a galaxy not in terms of light-years or megaparsecs, but directly in terms of the observed redshift factor z, the conversion to distance then following from some version of Hubble's law. It is good to keep in mind that when cosmologists talk about small distances, i.e. small redshifts $z \approx 0.1$, this corresponds to a distance of approximately 1 billion light-years! The largest observed redshift of a galaxy is currently $z \approx 10$, corresponding to a distance of the order of 13 billion light-years.

6. The cosmic microwave background radiation (CMBR), which originated just a couple of 100.000 years after the Big Bang ($\approx$ 370.000 years), has $z \gtrsim 1000$. This was the time when atoms were formed, and the CMBR photons were decoupled and emitted. This happened at a temperature of $T_{dec} \approx 3000K$. Comparing with the fact the temperature of the CMBR today is $T_{cmbr} \approx 3K$ and using $T \propto a^{-1}$ (this is essentially a reformulation of our above result for the redshift, since - up to conversion factors $\hbar$ and k - frequency = energy = temperature), one then finds the above-quoted estimate for z.[145]

34.9 Redshift - Distance Relation (Hubble's Law)

We have seen that there is a cosmological redshift in Robertson-Walker geometries. Our aim will now be to see if and how these geometries are capable of explaining Hubble's law that the redshift is approximately proportional to the distance and how the Hubble constant is related to the cosmic scale factor $a(t)$.

For a long time, reliable data for cosmological redshifts as well as for distance measurements were only available for small values of z, and thus it was common to consider the case where $t_0 - t_1$ and r_1 are small, i.e. small on cosmological scales. This allows one to find a redshift-distance relation which can e.g. be written as a power-series in z. Such a formula is not quite good enough for modern purposes, however, and I will come back to this below.

[144] For a lucid discussion of this issue, actually in the end arguing in favour of the pure Doppler interpretation, see E. Bunn, D. Hogg, *The kinematic origin of the cosmological redshift*, `arXiv:0808.1081`.

[145] For an introduction to the rich physics of the CMBR see e.g. the review articles D. Samtleben, S. Staggs, B. Winstein, *The Cosmic Microwave Background for Pedestrians: A Review for Particle and Nuclear Physicists*, `arXiv:0803.0834 [astro-ph]`, or A. Jones, A. Lasenby, *The Cosmic Microwave Background*, Living Rev. Relativity 1, (1998), 11; `http://www.livingreviews.org/lrr-1998-11`.

Assuming the validity of such an expansion, this allows us in particular to expand $a(t)$ in a Taylor series,

$$a(t) = a(t_0) + (t-t_0)\dot{a}(t_0) + \tfrac{1}{2}(t-t_0)^2\ddot{a}(t_0) + \dots \tag{34.95}$$

Let us introduce the *Hubble parameter* $H(t)$ (which already made a brief appearance in (34.30) of section 34.3) and the *deceleration parameter* $q(t)$ by

$$\begin{aligned} H(t) &= \frac{\dot{a}(t)}{a(t)} \\ q(t) &= -\frac{a(t)\ddot{a}(t)}{\dot{a}(t)^2} \quad , \end{aligned} \tag{34.96}$$

and denote their present day values by a subscript zero, i.e. $H_0 = H(t_0)$ and $q_0 = q(t_0)$. $H(t)$ measures the expansion velocity as a function of time while $q(t)$ measures whether the expansion velocity is increasing or decreasing. We will also denote $a_0 = a(t_0)$ and $a(t_1) = a_1$. In terms of these parameters, the Taylor expansion can be written as

$$a(t) = a_0(1 + H_0(t-t_0) - \tfrac{1}{2}q_0H_0^2(t-t_0)^2 + \dots) \quad . \tag{34.97}$$

Higher order terms in this expansion are known as *jerk* (3rd derivative) and *snap* (4th derivative).[146]

We can use the expansion (34.97) to express z as a function of $(t_0 - t_1)$, and we can in principle use (34.79) to express r_1 as a function of $(t_0 - t_1)$. Combining the two results and eliminating $(t_0 - t_1)$, one therefore obtains the sought-for relation between the redshift z and the (coordinate-) distance r_1. The result is given in (34.104). The derivation is primarily an exercise in inverting series expansions and not *per se* particularly enlightning.

From (34.97) one finds that the redshift parameter z, as a power series in time, is

$$\frac{1}{1+z} = \frac{a_1}{a_0} = 1 + (t_1 - t_0)H_0 - \tfrac{1}{2}q_0H_0^2(t_1-t_0)^2 + \dots \tag{34.98}$$

or

$$z = (t_0 - t_1)H_0 + (1 + \tfrac{1}{2}q_0)H_0^2(t_0-t_1)^2 + \dots \tag{34.99}$$

For small $H_0(t_0 - t_1)$ this can be inverted,

$$t_0 - t_1 = \frac{1}{H_0}[z - (1 + \tfrac{1}{2}q_0)z^2 + \dots] \quad . \tag{34.100}$$

We can also use (34.79) to express $(t_0 - t_1)$ in terms of r_1. On the one hand we have

$$\int_{t_1}^{t_0} \frac{dt}{a(t)} = r_1 + \mathcal{O}(r_1^3) \quad , \tag{34.101}$$

[146] If you want to know why the 5th and 6th order derivative terms are known as *crackle* and *pop* respectively, look at M. Dunajski, G. Gibbons, *Cosmic Jerk, Snap and Beyond*, `arXiv:0807.0207 [gr-qc]`.

while expanding $a(t)$ in the denominator we get

$$\begin{aligned}\int_{t_1}^{t_0}\frac{dt}{a(t)} &= \frac{1}{a_0}\int_{t_1}^{t_0}\frac{dt}{(1+(t-t_0)H_0+\ldots)}\\ &= \frac{1}{a_0}\int_{t_1}^{t_0}dt\,[1+(t_0-t)H_0+\ldots]\\ &= \frac{1}{a_0}[(t_0-t_1)+t_0(t_0-t_1)H_0-\tfrac{1}{2}(t_0^2-t_1^2)H_0+\ldots]\\ &= \frac{1}{a_0}[(t_0-t_1)+\tfrac{1}{2}(t_0-t_1)^2H_0+\ldots]\ .\end{aligned}\tag{34.102}$$

Therefore we get

$$r_1=\frac{1}{a_0}[(t_0-t_1)+\tfrac{1}{2}(t_0-t_1)^2H_0+\ldots]\ .\tag{34.103}$$

Using (34.100), we finally obtain

$$a_0r_1=\frac{1}{H_0}[z-\tfrac{1}{2}(1+q_0)z^2+\ldots]\ .\tag{34.104}$$

This clearly indicates to first order a linear dependence of the redshift on the distance of the galaxy and identifies H_0, the present day value of the Hubble parameter, as playing the role of the Hubble constant introduced in (33.6).

REMARKS:

1. Note that the linear relation (34.29) between recessional velocity and distance of comoving objects is exact while (34.104) shows that the relation between redshift and distance is only approximately linear for small z.

2. Nowadays, cosmologists routinely deal with objects with redshifts $z>1$. For such objects, the relation (34.104), a power-series exansion in z, is evidently not appropriate. In section 38.1 we will derive a "non-perturbative" formula for $H=H(z)$ (the value of the Hubble parameter at the time an object emitted the light that we now observe with redshift z), namely (38.8)

$$H(z)=H_0(1+z)\left[1+(\Omega_M)_0z+(\Omega_\Lambda)_0\left(\frac{1}{(1+z)^2}-1\right)\right]^{1/2}\ .\tag{34.105}$$

   Here Ω_M and Ω_Λ are the so-called *density parameters* associated to matter and a cosmological constant, the subscript 0 denoting their value today (so that $H(z)$ is expressed in terms of quantities that are in principle directly or indirectly observable).

3. Returning to the case of small z, even in that case (34.104) is not yet a very useful way of expressing Hubble's law even in that case. First of all, the distance a_0r_1 that appears in this expression is not the proper distance (unless $k=0$), but is at least equal to it in our approximation. Note that a_0r_1 is the present distance to the galaxy, not the distance at the time the light was emitted.

4. Even proper distance is not directly measurable or observable and thus, to compare this formula with experiment, one needs to relate r_1 to the measures of distance used by astronomers.

One practical way of doing this is based on the so-called *luminosity distance* d_L. If for some reasons one knows the absolute luminosity of a distant star (for instance because it shows a certain characteristic behaviour known from other stars nearby whose distances can be measured by direct means - such objects are known as *standard candles*), then one can compare this absolute luminosity L with the apparent luminosity A. Then one can define the luminosity distance d_L by (cf. (33.2))

$$d_L^2 = \frac{L}{4\pi A} \quad . \tag{34.106}$$

We thus need to relate d_L to the coordinate distance r_1. The key relation is

$$\frac{A}{L} = \frac{1}{4\pi a_0^2 r_1^2} \frac{1}{1+z} \frac{a_1}{a_0} = \frac{1}{4\pi a_0^2 r_1^2 (1+z)^2} \quad . \tag{34.107}$$

Here the first factor arises from dividing by the area of the sphere at distance $a_0 r_1$ and would be the only term in a flat geometry (see the discssion of Olbers' paradox). In a Robertson-Walker geometry, however, the photon flux will be diluted. The second factor is due to the fact that each individual photon is being redshifted. And the third factor (identical to the second) is due to the fact that as a consequence of the expansion of the universe, photons emitted a time δt apart will be measured a time $(1+z)\delta t$ apart. Hence the relation between r_1 and d_L is

$$d_L = (L/4\pi A)^{1/2} = r_1 a(t_0)(1+z) \quad . \tag{34.108}$$

Intuitively, the fact that for z positive d_L is larger than the actual (proper) distance of the galaxy can be understood by noting that the redshift makes an object look darker (further away) than it actually is.

This can be inserted into (34.104) to give an expression for the redshift in terms of d_L, *Hubble's law*

$$d_L = H_0^{-1}[z + \tfrac{1}{2}(1-q_0)z^2 + \ldots] \quad . \tag{34.109}$$

The program would then be to collect as much astronomical information as possible on the relation between d_L and z in order to determine the parameters q_0 and H_0.

34.10 Klein-Gordon Scalar Field in a Cosmological Background

The study of fields propagating in a cosmological background space-time plays an important role in cosmology, in particular when these fields arise as perturbations of the metric itself (cosmological perturbations). Here we will be content with simply deriving

the equations of motion of a free massive Klein-Gordon scalar field Ψ in a spatially flat ($k = 0$) Robertson-Walker background

$$ds^2 = -dt^2 + a(t)^2 d\vec{x}^2 \quad , \tag{34.110}$$

described by the standard action (6.11)

$$S[\Psi] = \int \sqrt{g} d^4x \left[-\tfrac{1}{2} g^{\alpha\beta} \partial_\alpha \Psi \partial_\beta \Psi - \tfrac{1}{2} m^2 \Psi^2 \right] \quad , \tag{34.111}$$

and very briefly qualitatively discussing some of the properties of these equations. This a continuation and variation of the theme begun in section 6.3 (general formalism for scalar fields in a gravitational field), section 7.8 (scalar field in Rindler coordinates), and section 26.8 (scalar field in the Schwarzschild space-time).

In the original coordinates $(t, \vec{x})$, the action explicitly takes the form

$$S[\Psi] = \tfrac{1}{2} \int dt \; d^3x \; a(t)^3 \left(\dot{\Psi}^2 - (\vec{\nabla}\Psi)^2 / a(t)^2 - m^2 \Psi^2 \right) \quad , \tag{34.112}$$

leading to the equation of motion

$$\ddot{\Psi} + 3(\dot{a}/a)\Psi - (\Delta\Psi)/a^2 + m^2 \Psi = 0 \quad . \tag{34.113}$$

This exhibits a characteristic drag/friction term proportional to the Hubble parameter $H(t)$.

However, in order to make the setting as close as possible to that of a scalar field in Minkowski space (which is useful e.g. if one is intent on quantising the scalar field afterwards), it is useful to employ the conformal time coordinate η, already introduced in (34.48) and defined by

$$\begin{aligned} ds^2 &= -dt^2 + a(t)^2 d\vec{x}^2 = a(t)^2 (-dt^2/a(t)^2 + d\vec{x}^2) \\ &\equiv A(\eta)^2 (-d\eta^2 + d\vec{x}^2) \quad , \end{aligned} \tag{34.114}$$

with $A(\eta(t)) = a(t)$ and $d/d\eta = a(d/dt)$. Denoting an η-derivative by a prime, $d\Psi/d\eta = \Psi'$, the action can then more explicitly be written as

$$S[\Psi] = \tfrac{1}{2} \int d^3x \; d\eta \; A(\eta)^2 \left[(\Psi')^2 - (\vec{\nabla}\Psi)^2 - m^2 A(\eta)^2 \Psi^2 \right] \quad . \tag{34.115}$$

We see that the field Ψ has a non-canonical kinetic term $\sim A(\eta)^2 (\Psi')^2$, leading to Euler-Lagrange equations of motion containing "friction" terms,

$$\tfrac{d}{d\eta}(A^2 \Psi') = A^2 (\Psi'' + 2(A'/A)\Psi') \quad , \tag{34.116}$$

which are awkward (in the classical theory, but even more so for quantisation). Happily, these non-canonical terms can be eliminated by the field redefinition

$$\phi(\eta, \vec{x}) = A(\eta) \Psi(\eta, \vec{x}) \quad . \tag{34.117}$$

Indeed, up to a total derivative term one then finds

$$S[\Psi] \to S[\phi] = \tfrac{1}{2}\int d^3x\, d\eta\, \left[(\phi')^2 - (\vec{\nabla}\phi)^2 - (m^2A^2 - A''/A)\phi^2\right] \ . \tag{34.118}$$

Remarks:

1. Observe that $S[\phi]$ is the standard action for a Klein-Gordon field in Minkowski space, its only mildly exotic feature being the time-dependent mass term with an effective mass

$$m^2_{\text{eff}}(\eta) = m^2A(\eta)^2 - A''(\eta)/A(\eta) \ . \tag{34.119}$$

Thus the interaction of the scalar field ϕ with the gravitational background is entirely encoded in this time-dependent mass term. Note that this effective mass term is even present when the original field is massless, $m^2 = 0$.

2. This purely geometric contribution to the mass term can be interpreted as an induced non-minimal coupling to the scalar curvature R of the space-time. Indeed, we have

$$A''/A = (1/a)(a\frac{d}{dt})(a\frac{d}{dt})a = \frac{d}{dt}(a\dot{a}) = a\ddot{a} + \dot{a}^2 \ . \tag{34.120}$$

Comparison with the result (35.7) for the Ricci scalar of the Robertson-Walker metric for $k = 0$ shows that

$$A''/A = Ra^2/6 = RA^2/6 \ , \tag{34.121}$$

so that the effective mass can also be written as

$$m^2_{\text{eff}}(\eta) = A(\eta)^2(m^2 - R(\eta)/6) \ . \tag{34.122}$$

3. This non-minimal coupling to the scalar curvature, and the factor 1/6, are (or should be) reminiscent of the *conformal coupling* $\xi R\phi^2$ of a scalar field discussed in section 22.3. If instead of with the action (34.111) we start off with the non-minimally coupled action (22.102)

$$S_\xi[\Psi] = -\tfrac{1}{2}\int \sqrt{g}d^4x\, \left(g^{\alpha\beta}\partial_\alpha\Psi\partial_\beta\Psi + \xi R\Psi^2\right) \ , \tag{34.123}$$

then instead of (34.118) we will find the action

$$S_\xi[\phi] = \tfrac{1}{2}\int d^3x\, d\eta\, \left[(\phi')^2 - (\vec{\nabla}\phi)^2 + A(\eta)^2(\xi - 1/6)R(\eta)\phi^2\right] \ . \tag{34.124}$$

In particular, when ξ takes the value (22.103)

$$\xi = \frac{D-2}{4(D-1)} = +1/6 \ , \tag{34.125}$$

for a conformal coupling, the action

$$S_{\xi=1/6}[\phi] = \tfrac{1}{2}\int d^3x\, d\eta\, \left[(\phi')^2 - (\vec{\nabla}\phi)^2\right] \tag{34.126}$$

is simply the action of a free massless scalar field in Minkowski space. This is as it should be: the action with $\xi = 1/6$ (and $m^2 = 0$) is conformally invariant, and the Robertson-Walker metric is conformally flat (this is manifest in (34.114)). Thus the action with this value of ξ must reduce to the free action in Minkowski space, the rescaling (34.117) of the scalar field reflecting the non-trivial conformal weight of a scalar field in $D = 4$.

When one introduces the non-mimimal ξ-coupling in the action together with a non-zero explicit mass term, then everything goes through as above, the only difference being that the effective mass is now ξ-dependent,

$$m^2_{\text{eff}}(\eta) = A(\eta)^2\left(m^2 + (\xi - 1/6)R(\eta)\right) \quad . \tag{34.127}$$

4. The equations of motion are

$$\phi''(\eta,\vec{x}) - \Delta\phi(\eta,\vec{x}) + m^2_{\text{eff}}(\eta)\phi(\eta,\vec{x}) = 0 \quad . \tag{34.128}$$

Spatial flatness $k = 0$ brings with it the simplifying feature that we can expand the spatial dependence of the fields in standard Fourier modes. Upon spatial Fourier expansion,

$$\phi(\eta,\vec{x}) \sim \int d^3k\, \phi_{\vec{k}}(\eta)\, \mathrm{e}^{i\vec{k}.\vec{x}} \quad , \tag{34.129}$$

one sees that each mode $\phi_{\vec{k}}$ satisfies the equation

$$\phi''_{\vec{k}}(\eta) + \omega^2_k(\eta)\phi_{\vec{k}}(\eta) = 0 \tag{34.130}$$

of a time-dependent harmonic oscillator, with time-dependent frequency

$$\omega^2_k(\eta) = m^2_{\text{eff}}(\eta) + k^2 \quad . \tag{34.131}$$

5. The crucial feature of this action and the mode equations are their explicit time-dependence which means that the energy of ϕ is not conserved. This in turn will lead to the important phenomena of particle or mode production in a cosmological background.

6. For example, one can consider the (evidently highly idealised) situation where the cosmic scale factor is asymptotically constant in the remote past and in the remote future. During these early and late periods the metric is essentially the Minkowski metric (possibly up to a rescaling of the coordinates), and one thus has a preferred notion of particles during those eras, uniquely determined by the asymptotic Poincaré symmetry. However, these definitions of particles need

not (and will amost invariably not) agree when there is an intermediate time-dependent phase. For instance, the early time vacuum Heisenberg state would not be interpreted or seen as a vacuum by the late time observer, and this disagreement about the particle content is then interpreted as a particle production due to the time-dependent gravitational field. See the references in footnote 87 (section 27.7) for a detailed discussion of these and other related fascinating issues.

34.11 Comments on Cosmic Expansion as "Expansion of Space"

As an aside, and as a conclusion to this section, let me make some comments on (and issue a *caveat* regarding) the seductive picture of an inflating balloon as the model for an expanding universe, as depicted e.g. in Figure 44 of section 34.2.

1. For many purposes, this picture certainly provides an instructive and illuminating analogy:

   - it illustrates how an expanding universe can look the same to all comoving observers;
   - in particular, it illustrates how it is possible for everything to move away from any given (comoving) observer without that observer actually being singled out as special;
   - it shows that a spatially homogeneous expansion naturally gives rise to a universal velocity-distance relation;
   - it illustrates (or is at least meant to illustrate) that expansion of the universe is something intrinsic and does not mean expansion in and into some space into which the universe has somehow been embedded.

2. The above picture of the inflating balloon is often used to describe the expansion of the universe as "an expansion of space itself", and it is then, in view of the success of this picture, tempting to *ascribe* the behaviour of light and test particles in the universe (which are mathematically described by the geodesic equations) to such an expansion of space.

   This manner of speaking and this imagery may provide some further useful intuitive understanding for some effects in the general relativistic description of cosmology. However, unless one defines precisely what one means by this, the notion of expanding space is not without its pitfalls and as with all analogies here one runs the risk of pushing this analogy too far. In particular, the danger hides in the above word "ascribe", i.e. in the risk of confusing cause and effect, or cause and effective description. Fundamentally, there is no (new?) force that (somehow) acts on space to (somehow) make it expand, and that can therefore be invoked

to "explain" (in a Newtonian way) the behaviour of particles and light in such a space-time.[147]

3. The danger of thinking of the cosmological expansion in terms of the expansion of space and a corresponding agent responsible for this is well-illustrated by the cosmological redshift discussed in section 34.8, and derived there on the basis of the null geodesic equations.

   If (somehow) the expansion of space were fundamentally responsible for this effect, one would expect this effect to be cumulative and to depend on the gravitational field (or cosmic scale factor) $a(t)$ during the entire period of propagation. However, as we have seen, and as stressed at the end of section 34.8, the final result (34.94) for the redshift, $z = a(t_0)/a(t_1) - 1$, depends only on the values of the cosmological scale factor at the times t_0 and t_1, indicating that the redshift is *not* a cumulative effect due to the expansion of space while it was traversed by the lightray, but that it can (equally intuitively and perhaps more correctly) be ascribed to the fact that emitter and observer do not share the same inertial frame.

4. The issue also, and in particular, arises when it comes to frequently asked questions such as "which objects participate in the cosmic expansion?" (do you expand with the universe? does a hydrogen atom? does our solar system?) or, to use cosmologists' jargon, "which objects join the Hubble flow?", which have generated a lot of confusion over the decades. Here again the "expanding space" image may lead to a misleading intuition, in particular when space is then viewed as some kind of viscous fluid which will invariably drag other objects along with it when it expands.[148]

5. These kinds of questions have a long history in general relativity, dating back at least to an article by Einstein and Straus in 1945 entitled *The Influence of the Expansion of Space on the Gravitational Fields Surrounding the Individual Stars*. The Einstein-Straus solution, known as the *Einstein-Straus vacuole* is a space-time that is obtained by a cut-and-paste procedure from a suitable cosmological solution, removing a ball of mass M and replacing it by a Schwarzschild solution of the same mass. This is essentially an inside-out version of the Oppenheimer-Snyder collapse solution (removing a ball from Schwarzschild and replacing it

[147] I strongly recommend the lucid and engaging discussion of these issues in M. Francis, L. Barnes, J. Berian James, G. Lewis, *Expanding Space: the Root of all Evil?*, `arXiv:0707.0380 [astro-ph]`. See also A. Whiting, *The Expansion of Space: Free Particle Motion and the Cosmological Redshift*, `arXiv:astro-ph/0404095`, and J. Peacock, *A diatribe on expanding space*, `arXiv:0809.4573 [astro-ph]` for related discussions.

[148] For a clear discussion of these issues for geodesic but non-comoving particles, highlighting the necessity to define precisely and carefully what one means by "joining the Hubble flow", see L. Barnes, M. Francis, J. Berian James, G. Lewis, *Joining the Hubble Flow: Implications for Expanding Space*, `arXiv:astro-ph/0609271`.

by a contracting cosmological solution modelling the collapsing star) discussed in section 29. The Einstein-Straus procedure can also be applied multiple times around various "centers" and can then be used to model inhomogeneities in an otherwise homogeneous universe (and in this context the model is then known to cosmologists as the *Swiss cheese model*).

6. Since the work of Einstein and Straus, a lot of work has gone into finding exact solutions of the Einstein equations that describe gravitational objects like stars or black holes somehow "embedded" into cosmological backgrounds. On the basis of such exact solutions one can then (try to) answer the question if a given bound object takes part in the cosmic expansion or not, and try to to develop some intuition for this issue that complements intuition coming from more Newtonian considerations.

   There is a common folklore statement or rule of thumb to the effect that "gravitationally (or otherwise) bound systems do not expand with the universe", and while this statement undoubtedly has a certain validity it requires a more precise formulation to decide if or when such a statement is not only true but also has some non-trivial content.

7. The most prominent class of solutions among these hybrid star-cosmology solutions (apart from black holes in (anti-)de Sitter space, the Schwarzschild (anti-)de Sitter metrics (30.7)) is the so-called McVittie solution, found already in 1933. It consists of a crude superposition of a (in the simplest case $k = 0$) Robertson-Walker metric,

$$ds^2 = -dt^2 + a(t)^2 d\vec{x}^2 \quad , \tag{34.132}$$

   with the Schwarzschild metric in isotropic coordinates (24.46)

$$ds^2 = -\frac{(1 - \frac{m}{2|\vec{x}|})^2}{(1 + \frac{m}{2|\vec{x}|})^2} dt^2 + (1 + \frac{m}{2|\vec{x}|})^4 \, d\vec{x}^2 \quad , \tag{34.133}$$

   and thus has the form

$$ds^2 = -\frac{(1 - \frac{m}{2a(t)|\vec{x}|})^2}{(1 + \frac{m}{2a(t)|\vec{x}|})^2} dt^2 + (1 + \frac{m}{2a(t)|\vec{x}|})^4 \, a(t)^2 \, d\vec{x}^2 \quad . \tag{34.134}$$

   While this metric is easy to write down, it leads to a somewhat peculiar energy-momentum tensor, and therefore its physical interpretation and significance are somewhat obscure. These issues, as well as aspects of the global structure of the McVittie space-time, continue to be debated in the literature to this day.[149]

[149] For a pedagogical review of these topics and a discussion of the literature see e.g. M. Carrera, D. Giulini, *On the influence of the global cosmological expansion on the local dynamics in the Solar System*, `arXiv:gr-qc/0602098`, *Influence of global cosmological expansion on local dynamics and kinematics*, `arXiv:0810.2712 [gr-qc]`. For more on the on-going McVittie debate, see also N. Kaloper, M. Kleban, D. Martin, *McVittie's Legacy: Black Holes in an Expanding Universe* and references therein and thereto.

35 Cosmology III: Friedmann-Lemaître-Robertson-Walker Cosmology

So far, we have only used the kinematical framework provided by the Robertson-Walker metrics and we never used the Einstein equations. The benefit of this is that it allows one to deduce relations betweens observed quantities and assumptions about the universe which are valid even if the Einstein equations are not entirely correct, perhaps because of higher derivative or other quantum corrections in the early universe.

Now, on the other hand we will have to be more specific, specify the matter content and solve the Einstein equations for $a(t)$. We will see that a lot about the solutions of the Einstein equations can already be deduced from a purely qualitative analysis of these equations, without having to resort to explicit solutions (section 36). Exact solutions will then be the subject of section 37.

35.1 Curvature and Einstein Tensor of the Robertson-Walker Metric

Of course, the first thing we need to discuss solutions of the Einstein equations is the Ricci tensor of the Robertson-Walker metric. Since we already know the curvature tensor of the maximally symmetric spatial metric entering the Robertson-Walker metric (and its contractions), this is not difficult.

1. First of all, we write the Robertson-Walker metric as

$$ds^2 = -dt^2 + a(t)^2\, d\tilde{s}^2 = -dt^2 + a(t)^2\, \tilde{g}_{ij} dx^i dx^j \quad . \tag{35.1}$$

   In this section all objects with a tilde, $\tilde{}$, will refer to 3-dimensional quantities calculated with respect to the maximally symmetric metric $\tilde{g}_{ij}$.

2. One can then calculate the Christoffel symbols in terms of $a(t)$ and $\tilde{\Gamma}^i{}_{jk}$. The non-vanishing components are (we had already established that $\Gamma^\mu_{00} = 0$)

$$\Gamma^i{}_{jk} = \tilde{\Gamma}^i{}_{jk} \quad , \quad \Gamma^i{}_{j0} = \frac{\dot{a}}{a}\delta^i_j \quad , \quad \Gamma^0{}_{ij} = \dot{a} a \tilde{g}_{ij} \quad . \tag{35.2}$$

3. The non-zero components of the Riemann tensor are

$$\begin{aligned} R^i{}_{0j0} &= -\frac{\ddot{a}}{a}\delta^i_j \quad \Leftrightarrow \quad R^0{}_{i0j} = a\ddot{a}\tilde{g}_{ij} \\ R^k{}_{ilj} &= \tilde{R}^k{}_{ilj} + \dot{a}^2(\delta^k_l \tilde{g}_{ij} - \delta^k_j \tilde{g}_{il}) \quad . \end{aligned} \tag{35.3}$$

   where, by maximal symmetry, $\tilde{R}_{kilj}$ has the form (14.10)

$$\tilde{R}^k{}_{ilj} = k(\delta^k_l \tilde{g}_{ij} - \delta^k_j \tilde{g}_{il}) \quad . \tag{35.4}$$

4. The partial contraction of the purely spatial components of the Riemann tensor over the spatial indices is thus

$$\begin{aligned} R^k_{ikj} &= \tilde{R}_{ij} + 2\dot{a}^2\tilde{g}_{ij} \\ &= 2k\tilde{g}_{ij} + 2\dot{a}^2\tilde{g}_{ij} \ . \end{aligned} \tag{35.5}$$

5. Therefore the non-zero components of the space-time Ricci tensor are

$$\begin{aligned} R_{00} &= -3\frac{\ddot{a}}{a} \\ R_{ij} &= (a\ddot{a} + 2\dot{a}^2 + 2k)\tilde{g}_{ij} \\ &= (\frac{\ddot{a}}{a} + 2\frac{\dot{a}^2}{a^2} + \frac{2k}{a^2})g_{ij} \ . \end{aligned} \tag{35.6}$$

6. Thus the Ricci scalar is

$$R = \frac{6}{a^2}(a\ddot{a} + \dot{a}^2 + k) \ , \tag{35.7}$$

7. Finally, therefore, putting everything together, we find that the Einstein tensor has the components

$$\begin{aligned} G_{00} &= 3(\frac{\dot{a}^2}{a^2} + \frac{k}{a^2}) \\ G_{0i} &= 0 \\ G_{ij} &= -(\frac{2\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{k}{a^2})g_{ij} \ . \end{aligned} \tag{35.8}$$

REMARKS:

1. It follows on symmetry grounds (spatial maximal symmetry) alone

   - that the only potentially non-vanishing components of the Einstein tensor are G_{00} and G_{ij};
   - that $G_{ij} \sim g_{ij}$
   - and that the coefficients are only functions of t, not functions of the spatial coordinates.

   A formal proof of this is given in section 35.3. It is phrased there as a statement about the energy-momentum tensor in a Robertson-Walker metric, but the result is a general statement about the structure of spatially maximally symmetric space-time tensors. Thus in a sense the only non-trivial content of the above calculation is in the precise form of the t-dependent coefficients of G_{00} and G_{ij}.

2. We already know that in a maximally symmetric space not only can we express the Ricci tensor in terms of the Riemann tensor (namely as a contraction thereof) but we can also write the Riemann tensor algebraically in terms of the Ricci tensor (and even just in terms of the Ricci scalar), as is obvious from (35.4).

Even though the Robertson-Walker metrics are not space-time maximally symmetric, it is nevertheless true that even in this case the Riemann tensor can be expressed algebraically in terms of the Ricci tensor. Indeed, it is easy to see that the components of the Riemann tensor given in (35.3) can be written in terms of the components of the Ricci tensor in (35.6) simply as

$$\begin{aligned} R^i{}_{0j0} &= \tfrac{1}{3} R_{00} \delta^i{}_j \\ R^k{}_{ilj} &= \tfrac{1}{2}(\delta^k{}_l R_{ij} - \delta^k{}_j R_{il}) \quad . \end{aligned} \tag{35.9}$$

On general grounds this follows from the fact, established in section 34.6, that the Robertson-Walker metrics are conformally flat so that the Weyl tensor vanishes.

3. The significance of this statement lies in the fact that it shows that a vacuum solution of the Einstein equations with spatial maximal symmetry is necessarily flat Minkowski space. This is perhaps as it should be, and at least vaguely Machian, but it is still good to have established this here once and for all since by just solving the vacuum equations one may (and will) find a solution that at first sight appears to be non-trivial, namely the *Milne universe* to be discussed in section 37.1, but which can then be shown to be just Minkowski space written in some non-inertial coordinates. The above result (35.9) shows that this had to be true.

4. Occasionally, in particular for the canonical analysis (i.e. developing the Hamiltonian formalism), it is useful to know the Ricci scalar (i.e. the Einstein-Hilbert Lagrangian) for the slightly more general metric

$$ds^2 = -N^2(t')(dt')^2 + a^2(t') d\tilde{s}^2 \quad , \tag{35.10}$$

where the function $N(t')$ is known as the *lapse function* (cf. (21.18)). Instead of redoing the calculation of the scalar curvature in this case, one can simply use the change of variable

$$dt = N(t')dt' \quad \Rightarrow \quad \frac{d}{dt} = \frac{1}{N}\frac{d}{dt'} \tag{35.11}$$

to rewrite the final result (35.7) as (a prime on a or N denoting a derivative with respect to t')

$$R = \frac{6}{a^2}(a\ddot{a} + \dot{a}^2 + k) = \frac{6}{a^2 N^3}(N(aa'' + (a')^2) - aa'N' + kN^3) \quad . \tag{35.12}$$

We will come back to and make use of this result in section 35.8.

5. The results for the Christoffel symbols (35.2) and the Riemann tensor (35.3) are true in any dimension, i.e. for a general n-dimensional maximally symmetric space, and the first time that a dimension-dependence enters is in the factors of 2 and 3 in equations (35.5) and (35.6), which arise from taking traces. If one replaces the

spatial dimension $3 \to n$, equations (35.6) - (35.8) take the form

$$\begin{aligned} R_{00} &= -n\frac{\ddot{a}}{a} \\ R_{ij} &= \left(\frac{\ddot{a}}{a} + (n-1)\frac{k+\dot{a}^2}{a^2}\right) g_{ij} \\ R &= 2n\frac{\ddot{a}}{a} + n(n-1)\frac{k+\dot{a}^2}{a^2} \\ G_{00} &= \frac{n(n-1)}{2}\frac{k+\dot{a}^2}{a^2} \\ G_{ij} &= -(n-1)\left(\frac{\ddot{a}}{a} + \frac{(n-2)}{2}\frac{k+\dot{a}^2}{a^2}\right) g_{ij} \quad . \end{aligned} \tag{35.13}$$

35.2 Matter Content: A Perfect Fluid

Next we need to specify the matter content. On physical grounds one might perhaps like to argue that in the approximation underlying the cosmological principle galaxies (or clusters) should be treated as non-interacting particles or a perfect fluid (first discussed in section 7.2). As it turns out, we do not need to do this as either the symmetries of the metric or comparison with the Einstein tensor determined above fix the energy-momentum tensor to be that of a perfect fluid anyway.

In section 35.3 I will give a formal argument for this using Killing vectors. Informally we can already deduce this from the structure of the Einstein tensor obtained above. Comparing (35.8) with the Einstein equation $G_{\alpha\beta} = 8\pi G_N T_{\alpha\beta}$, we deduce that the Einstein equations can only have a solution with a Robertson-Walker metric if the energy-momentum tensor is of the form

$$\begin{aligned} T_{00} &= \rho(t) \\ T_{0i} &= 0 \\ T_{ij} &= p(t) g_{ij} \quad , \end{aligned} \tag{35.14}$$

where $p(t)$ and $\rho(t)$ are some functions of time. A covariant way of writing this tensor is as

$$T_{\alpha\beta} = (p+\rho) u_\alpha u_\beta + p g_{\alpha\beta} \quad , \tag{35.15}$$

where $u^\alpha = (1,0,0,0)$ in a comoving coordinate system. This is precisely the energy-momentum tensor of a *perfect fluid* (cf. sections 7.2 and 7.5). In this context u^α is known as the *velocity field* of the fluid, and the comoving coordinates are those with respect to which the fluid is at rest. ρ is the energy-density of the perfect fluid and p is the pressure.

In general, this matter content has to be supplemented by an equation of state. This is usually assumed to be that of a *barytropic fluid*, i.e. one whose pressure depends only

on its density, $p = p(\rho)$. The most useful toy-models of cosmological fluids arise from considering a linear relationship between p and ρ, of the type

$$p = w\rho \ , \tag{35.16}$$

where w is known as the equation of state parameter. Occasionally also more exotic equations of state are considered, but the above covers a wide variety of commonly considered fluids and gases and other simple thermodynamic systems.

Consider e.g. a system whose entropy S is some function of the (internal) energy E and the (spatial) volume V,

$$S = S(E, V) \ . \tag{35.17}$$

Then the 1st law of thermodynamics

$$TdS = dE + pdV \tag{35.18}$$

implies

$$T = \left(\frac{\partial S}{\partial E}\right)^{-1}_{V} \quad , \quad p = T\left(\frac{\partial S}{\partial V}\right)_{E} \quad , \tag{35.19}$$

and thus

$$p = w\rho = wE/V \quad \Leftrightarrow \quad V\partial_V S = wE\partial_E S \ . \tag{35.20}$$

Thus the condition $p = w\rho$ is simply the statement that S is a function of $V^w E$,

$$p = w\rho \quad \Rightarrow \quad S = S(V^w E) \ . \tag{35.21}$$

Here are the most common and useful special cases of the equation of state $p = w\rho$.

1. Dust

   For non-interacting particles, there is no pressure, $p = 0$, i.e. $w = 0$, the energy-momentum tensor has the simple form

   $$T_{\alpha\beta} = \rho u_\alpha u_\beta \tag{35.22}$$

   and such matter is usually referred to as *dust*,

   $$\text{dust:} \qquad p = 0 \quad \Rightarrow \quad w = 0 \ . \tag{35.23}$$

   This is generally considered to be a good description of baryonic matter (and cold dark matter) today.

2. Radiation

   This corresponds to $w = 1/3$ (in 1+3 dimensions). One way to see this is to note that the trace of a perfect fluid energy-momentum tensor is

   $$T^\alpha_{\ \alpha} = -\rho + 3p \ . \tag{35.24}$$

For electro-magnetic radiation, for example, the energy-momentum tensor is that of Maxwell theory and hence traceless (7.121). Therefore electromagnetic *radiation* in an FLRW universe (in particular compatibility with the symmetries implies neglecting all anisotropies) has the equation of state

$$\text{radiation:} \qquad p = \rho/3 \quad \Rightarrow \quad w = 1/3 \ . \tag{35.25}$$

Alternatively, this can be deduced from familiar statements about the thermodynamics of electromagnetic radiation (i.e. the photon gas). E.g. $S \sim E/T$ and $\rho \sim T^4$ imply

$$S \sim E/T \sim E(E/V)^{-1/4} = E^{3/4}V^{1/4} = (V^{1/3}E)^{3/4} \ , \tag{35.26}$$

which (by (35.21)) also implies $w = 1/3$.

As an aside, note that one generalisation of this equation of state for radiation in $d = 3$ spatial dimensions to general spatial dimension d is a perfect fluid with a traceless energy-momentum tensor (describing what one might call a "Weyl invariant" or "conformal" fluid - cf. the discussion in section 7.7). Thus the energy-momentum tensor has to satisfy

$$g^{\alpha\beta}T_{\alpha\beta} = T^\alpha_\alpha = -(\rho + p) + (d+1)p = 0 \ , \tag{35.27}$$

leading to the equation of state

$$\text{conformal fluid:} \qquad p = \rho/d \quad \Rightarrow \quad w = 1/d \ . \tag{35.28}$$

and

$$T_{\alpha\beta} = \frac{\rho}{d}(g_{\alpha\beta} + (d+1)u_\alpha u_\beta) \ . \tag{35.29}$$

3. Cosmological Constant

A cosmological constant Λ, on the other hand, corresponds, as we will see, to a matter contribution with $p = -\rho$, i.e. $w = -1$,

$$\text{cosmological constant:} \qquad p = -\rho \quad \Rightarrow \quad w = -1 \ . \tag{35.30}$$

Thus either ρ is negative or p is negative.

To see the relation between a $w = -1$ perfect fluid and a cosmological constant, note that the Einstein equations with a cosmological constant (19.46) give a contribution to the energy-momentum tensor proportional to $g_{\mu\nu}$. Comparing this with the covariant form (35.15) of the energy-momentum tensor, one deduces that a cosmological constant Λ is tantamount to adding matter with $p = -\rho$. Specifically, one has

$$\begin{aligned} G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu} \quad &\Leftrightarrow \quad G_{\mu\nu} = 8\pi G_N \left(T_{\mu\nu} - (\Lambda/8\pi G_N) g_{\mu\nu}\right) \\ &\Rightarrow \quad T^\Lambda_{\mu\nu} = -\frac{\Lambda}{8\pi G_N} g_{\mu\nu} \\ &\Rightarrow \quad T^\Lambda_{00} \equiv \rho_\Lambda = \frac{\Lambda}{8\pi G_N} = -p_\Lambda \ . \end{aligned} \tag{35.31}$$

This is the identification anticipated in (19.50).

REMARKS:

1. In section 22.1 we had introduced various *energy conditions*, the *null energy condition* (NEC), the *weak energy condition* (WEC), the *dominant energy condition* (DEC), and the *strong energy condition* (SEC), and had also analysed their implications for a perfect fluid energy-momentum tensor. In particular, the conditions that we found were

$$\begin{aligned} \text{NEC:} \quad & \rho + p \geq 0 \\ \text{WEC:} \quad & \rho \geq 0 \quad , \quad \rho + p \geq 0 \\ \text{DEC:} \quad & \rho \geq |p| \\ \text{SEC:} \quad & \rho + p \geq 0 \quad , \quad \rho + 3p \geq 0 \ . \end{aligned} \tag{35.32}$$

With the equation of state $p = w\rho$ these energy conditions can now be written as conditions on w. For physical (gravitating instead of anti-gravitating) matter one usually requires at least the condition $\rho > 0$ (positive energy density). With this condition, the NEC and the WEC are equivalent and require

$$\rho > 0 \quad \text{and NEC/WEC} \quad \Rightarrow \quad w \geq -1 \ , \tag{35.33}$$

while the SEC requires (the 2nd condition is then stronger than the 1st)

$$\rho > 0 \quad \text{and SEC} \quad \Rightarrow \quad w \geq -1/3 \ . \tag{35.34}$$

Finally, the DEC implies $\rho > 0$ as well as

$$\text{DEC} \quad \Rightarrow \quad |w| \leq 1 \ . \tag{35.35}$$

Some of the conclusions about the qualitative behaviour of the solutions to the Einstein equations in section 36 rely on the strict validity of the SEC, i.e. on the assumption that (at least in the era of interest) the matter content of the universe is dominated by stuff with $w > -1/3$.

On the other hand, as we had seen, a cosmological constant has $w = -1$, and thus either ρ is negative or p is negative. Therefore this violates either $\rho > 0$ (the WEC and the DEC) or $\rho + 3p > 0$ (the SEC).

2. The equation of state parameter need not necessarily be a constant. Consider for instance a scalar field ϕ. Such a field will respect the symmetries of a Robertson-Walker metric (and hence can potentially give rise to a solution of the Einstein equations of the Robertson-Walker form) if it depends only on t and not on the spatial coordinates, $\phi = \phi(t)$. For such a scalar field, the energy-momentum tensor (7.75),

$$T_{\alpha\beta} = \partial_\alpha\phi\partial_\beta\phi - \tfrac{1}{2} g_{\alpha\beta}\, g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - g_{\alpha\beta}V(\phi) \ , \tag{35.36}$$

reduces to

$$T_{00} = \tfrac{1}{2}\dot{\phi}^2 + V(\phi) \quad , \quad T_{ij} = \left(\tfrac{1}{2}\dot{\phi}^2 - V(\phi)\right) \, g_{ij} \quad , \tag{35.37}$$

which is thus indeed of the general perfect fluid form (35.14). The energy and pressure density are

$$\rho(t) = \tfrac{1}{2}\dot{\phi}^2 + V(\phi) \quad , \quad p(t) = \tfrac{1}{2}\dot{\phi}^2 - V(\phi) \quad , \tag{35.38}$$

leading to the time-dependent equation of state parameter

$$w(t) = \frac{p(t)}{\rho(t)} = \frac{\frac{1}{2}\dot{\phi}^2 - V(\phi)}{\frac{1}{2}\dot{\phi}^2 + V(\phi)} \quad . \tag{35.39}$$

Note that here we are treating the scalar field as a source term for the Einstein equations. This should be contrasted with (and should not be confused with) what what we did in section 6.3, where we developed the general formalism for scalar fields in a fixed gravitational field, neglecting the backreaction of the matter fields on the gravitational field / metric.

Note also that (35.39) mimics a cosmological constant, i.e. $w = -1$, during periods (of the scalar field slowly rolling down a very flat potential) where the kinetic energy term is negligible compared with the potential energy term. This can also be seen directly from the action (6.14), a constant potential leading to a constant contribution to the matter or gravity Lagrangian, a.k.a. as a cosmological constant, and plays an important role in models of *inflation*.

3. Occasionally, more exotic equations of state are also considered in cosmology. For example, extended objects like (cosmic) strings, or membranes (domain walls) or their higher-dimensional generalisations typically have a positive tension, and therefore a negative (contribution to the) pressure.[150] In particular, the equation of state of a gas of non-relativistic strings in d spatial dimensions is

$$\text{non-relativistic string gas:} \qquad p = -\rho/d \quad \Rightarrow \quad w = -1/d \quad . \tag{35.40}$$

4. Another exotic object that appears occasionally in the context of cosmology is the so-called *Chaplygin gas*, with equation of state

$$\text{Chaplygin gas:} \qquad p = -A/\rho \qquad (A > 0) \quad . \tag{35.41}$$

We will briefly return to the properties of the Chaplygin gas at the end of section 35.6 - cf. (35.99) and (35.100), but for the most part we will concentrate on the linear equations of state $p = w\rho$ in the following.

[150] The equation of state for such extended objects, affectionately and collectively known as "p-branes" or just "branes", is derived e.g. in T. Boehm, R. Brandenberger, *On T-Duality in Brane Gas Cosmology*, `arXiv:hep-th/0208188`.

5. A null variant of the dust energy-momentum tensor (35.22), with u^α replaced by a null vector k^α (*null dust*) will appear as the source term for a toy model of a radiating star, the *Vaidya metric*,

$$T_{\alpha\beta} = \rho k_\alpha k_\beta \tag{35.42}$$

to be discussed in sections 40 - 42 (cf. in particular section 40.4). Since this energy-momentum tensor is manifestly traceless,

$$T^\alpha_{\ \alpha} = \rho k^\alpha k_\alpha = 0 \ , \tag{35.43}$$

it can alternatively be regarded as a variant of the energy-momentum tensor for radiation (also with $\rho = 3p$).

35.3 Appendix: Space-Time Tensors with Maximal Spatial Symmetry

Here is the formal argument that the energy-momentum tensor necessarily has the form given in (35.14).[151]

It is of course a consequence of the Einstein equations that any symmetries of the Ricci (or Einstein) tensor also have to be symmetries of the energy-momentum tensor. Now we know that the metric $\tilde{g}_{ij}$ has six Killing vectors $K^{(a)}$ and that (in the comoving coordinate system) these are also Killing vectors of the Robertson-Walker metric,

$$L_{K^{(a)}} \tilde{g}_{ij} = 0 \quad \Rightarrow \quad L_{K^{(a)}} g_{\mu\nu} = 0 \ . \tag{35.44}$$

Therefore also the Ricci and Einstein tensors have these symmetries,

$$L_{K^{(a)}} g_{\mu\nu} = 0 \quad \Rightarrow \quad L_{K^{(a)}} R_{\mu\nu} = 0 \ , \ \ L_{K^{(a)}} G_{\mu\nu} = 0 \ . \tag{35.45}$$

To prove this one can either (non-covariantly) choose, for each Killing vector, an adapted coordinate system, or one generalises the argument given in section 13.4 for the Ricci scalar, $L_K R = 0$, to the Ricci tensor.

The Einstein equations then imply that $T_{\mu\nu}$ should have these symmetries,

$$L_{K^{(a)}} G_{\mu\nu} = 0 \quad \Rightarrow \quad L_{K^{(a)}} T_{\mu\nu} = 0 \ . \tag{35.46}$$

Moreover, since the $L_{K^{(a)}}$ act like three-dimensional coordinate transformations, in order to see what these conditions mean we can make a $(3+1)$-decomposition of the energy-momentum tensor. From the three-dimensional point of view, T_{00} transforms like a scalar under coordinate transformations (and Lie derivatives), T_{0i} like a vector, and T_{ij} like a symmetric tensor. Thus we need to determine what are the three-dimensional

[151] This elementary argument, which requires no "higher knowledge" about Lie groups and invariants, is adapted from the discussion in section 13.4 of S. Weinberg, *Gravitation and Cosmology*.

scalars, vectors and symmetric tensors that are invariant under the full six-parameter group of the three-dimensional isometries.

For scalars ϕ we thus require (calling K now any one of the Killing vectors of $\tilde{g}_{ij}$),

$$L_K\phi = K^i\partial_i\phi = 0 \ . \tag{35.47}$$

Since $K^i(x)$ can take any value in a maximally symmetric space (homogeneity), this implies that ϕ has to be constant (as a function on the three-dimensional space) and therefore T_{00} can only be a function of time,

$$T_{00} = \rho(t) \ . \tag{35.48}$$

For vectors, it is almost obvious that no invariant vectors can exist because any vector would single out a particular direction and therefore spoil isotropy. The formal argument (as a warm up for the argument for tensors) is the following. We have

$$L_K V^i = K^j\tilde{\nabla}_j V^i + V^j\tilde{\nabla}_j K^i \ . \tag{35.49}$$

We now choose the Killing vectors such that $K^i(x) = 0$ but $\tilde{\nabla}_i K_j \equiv K_{ij}$ is an arbitrary anti-symmetric matrix. Then the first term disappears and we have

$$L_K V^i = 0 \ \ \Rightarrow \ \ K_{ij}V^j = 0 \ . \tag{35.50}$$

To make the anti-symmetry manifest, we rewrite this as

$$K_{ij}V^j = K_{kj}\delta^k_i V^j = \tfrac{1}{2}K_{kj}(\delta^k_i V^j - \delta^j_i V^k) = 0 \ . \tag{35.51}$$

If this is to hold for all anti-symmetric matrices, we must have

$$\delta^k_i V^j = \delta^j_i V^k \ \ , \tag{35.52}$$

and by contraction one obtains $nV^j = V^j$, and hence $V_j = 0$. Therefore, as expected, there is no invariant vector field and

$$T_{0i} = 0 \ . \tag{35.53}$$

We now come to symmetric tensors. Once again we choose our Killing vectors to vanish at a given point x and such that K_{ij} is an arbitrary anti-symmetric matrix. Then the condition

$$L_K T_{ij} = K^k\tilde{\nabla}_k T_{ij} + \tilde{\nabla}_i K^k T_{kj} + \tilde{\nabla}_j K^k T_{ik} = 0 \tag{35.54}$$

reduces to

$$K_{mn}(\tilde{g}^{mk}\delta^n_i T_{kj} + \tilde{g}^{mk}\delta^n_j T_{ik}) = 0 \ . \tag{35.55}$$

If this is to hold for all anti-symmetric matrices K_{mn}, the anti-symmetric part of the term in brackets must be zero or, in other words, it must be symmetric in the indices m and n, i.e.

$$\tilde{g}^{mk}\delta^n_i T_{kj} + \tilde{g}^{mk}\delta^n_j T_{ik} = \tilde{g}^{nk}\delta^m_i T_{kj} + \tilde{g}^{nk}\delta^m_j T_{ik} \ . \tag{35.56}$$

Contracting over the indices n and i, one obtains

$$n\tilde{g}^{mk}T_{kj} + \tilde{g}^{mk}T_{jk} = \tilde{g}^{mk}T_{kj} + \delta^m_j T^k_k \quad . \tag{35.57}$$

Therefore

$$T_{ij} = \frac{\tilde{g}_{ij}}{n} T^k_k \quad . \tag{35.58}$$

Now we already know that the scalar T^k_k has to be a constant. Thus we conclude that the only invariant tensor is the metric itself, and therefore the T_{ij}-components of the energy-momentum tensor can only be a function of t times $\tilde{g}_{ij}$. Writing this function as $p(t)a^2(t)$, we arrive at

$$T_{ij} = p(t)g_{ij} \quad . \tag{35.59}$$

We thus see that the energy-momentum tensor is determined by two functions, $\rho(t)$ and $p(t)$, precisely as in (35.14).

35.4 Conservation Laws for Perfect Fluids

The same arguments as above show that a current J^μ in a Robertson-Walker metric has to be of the form $J^\mu = (n(t), 0, 0, 0)$ in comoving coordinates, or

$$J^\mu = n(t)u^\mu \tag{35.60}$$

in covariant form. Here $n(t)$ could be a number density like a galaxy number density. It gives the number density per unit proper volume. The conservation law $\nabla_\mu J^\mu = 0$ is equivalent to

$$\nabla_\mu J^\mu = 0 \quad \Leftrightarrow \quad \partial_t(\sqrt{g}n(t)) = 0 \quad . \tag{35.61}$$

Thus we see that $n(t)$ is not constant, but the number density per unit coordinate volume is (as we had already anticipated in the picture of the balloon, Figure 44). For a Robertson-Walker metric, the time-dependent part of $\sqrt{g}$ is $a(t)^3$, and thus the conservation law says

$$n(t)a(t)^3 = const. \tag{35.62}$$

Let us now turn to the conservation laws associated with the energy-momentum tensor,

$$\nabla_\mu T^{\mu\nu} = 0 \quad . \tag{35.63}$$

The spatial components of this conservation law,

$$\nabla_\mu T^{\mu i} = 0 \quad , \tag{35.64}$$

turn out to be identically satisfied, by virtue of the fact that the u^μ are geodesic and that the functions ρ and p are only functions of time.

This could hardly be otherwise because $\nabla_\mu T^{\mu i}$ transforms as a vector under spatial rotations, and rotational invariance (isotropy) now implies that this vector has to be zero identically. Nevertheless it is of course instructive to check this explicitly.

The only interesting conservation law is thus the zero-component

$$\nabla_\mu T^{\mu 0} = \partial_\mu T^{\mu 0} + \Gamma^\mu_{\mu\nu} T^{\nu 0} + \Gamma^0_{\mu\nu} T^{\mu\nu} = 0 \quad , \tag{35.65}$$

which for a perfect fluid with $T_{00} = \rho(t)$ and $T_{ij} = p(t) g_{ij}$ becomes

$$\partial_t \rho(t) + \Gamma^\mu_{\mu 0} \rho(t) + \Gamma^0_{00} \rho(t) + \Gamma^0_{ij} T^{ij} = 0 \quad . \tag{35.66}$$

Inserting the explicit expressions (35.2) for the Christoffel symbols, one finds

$$\dot\rho = -3(\rho + p)\frac{\dot a}{a} \quad . \tag{35.67}$$

Introducing some fixed comoving volume $\sim v$ and its associated proper volume

$$V(t) = a(t)^3 v \quad , \tag{35.68}$$

and noting that

$$\frac{dV(t)}{dt} = 3a(t)^2 \dot a(t) v = 3H(t) V \quad , \tag{35.69}$$

the conservation equation (35.67) can be written in a perhaps more suggestive and familiar (mechanical or thermodynamical) form as

$$\frac{dE}{dt} = -p\frac{dV}{dt} \tag{35.70}$$

where

$$E(t) = \rho(t) V(t) \tag{35.71}$$

is the total energy in the volume V. Comparing with the 1st law of thermodynamics (35.18), this equation thus encodes the statement that the time evolution of the perfect fluid is *adiabatic* in the sense that its entropy remains constant,

$$\frac{dS(t)}{dt} = 0 \quad . \tag{35.72}$$

35.5 Conservation Laws and Comoving Congruences

Before discussing some special cases of the solutions of (35.67) in section 35.6 below, it is instructive to rederive the above results in a somewhat more general and covariant manner.[152] Thus we consider a general velocity field $u^\mu(x)$ with $u^\mu u_\mu = -1$, and the perfect fluid energy-momentum tensor (35.15),

$$T_{\mu\nu} = (p + \rho) u_\mu u_\nu + p g_{\mu\nu} \quad , \tag{35.73}$$

[152] For a detailed exposition of cosmology in this covariant framework see e.g. G. Ellis, H. van Elst, *Cosmological Models* (Cargèse Lectures 1998), `arXiv:gr-qc/9812046`, or the even more detailed monograph by G. Ellis, R. Maartens, M. MacCallum, *Relativistic Cosmology*.

with for the time being ρ and p arbitrary functions of the space-time coordinates. Note that $u^\mu\nabla_\mu$ is the covariant derivative along the integral curves of u^μ, the object we denoted D_τ or $D/D\tau$ in section 5.7. Acting on scalars we will simply denote it, as usual, by an overdot, i.e.

$$u^\mu\nabla_\mu\rho \equiv \dot\rho \tag{35.74}$$

etc. Let us now see what the conditon $\nabla_\mu T^{\mu\nu} = 0$ (which has to hold if this energy-momentum tensor is to give us a solution to the Einstein equations) tells us.

1. We first consider the case $p = 0$, so this corresponds to a pressure free perfect fluid (and is used in cosmology to e.g. model cold dark or baryonic matter). Then one has

$$T_{\mu\nu} = \rho u_\mu u_\nu \quad \Rightarrow \quad \nabla_\mu T^{\mu\nu} = (\dot\rho + \rho\nabla_\mu u^\mu)u^\nu + \rho u^\mu\nabla_\mu u^\nu \ . \tag{35.75}$$

   Here $\nabla_\mu u^\mu \equiv \theta$ is (and measures) the *expansion* of the velocity field u^μ (and was introduced previously, in the context of the Raychaudhuri equation, in (12.34)), and the last term $u^\mu\nabla_\mu u^\nu \equiv a^\nu$ is its acceleration (5.98), so that we can also write this equation as

$$(\dot\rho + \theta\rho)u^\nu + \rho a^\nu = 0 \ . \tag{35.76}$$

   Since u^ν and a^ν are orthogonal to each other,

$$u_\mu u^\mu = -1 \quad \Rightarrow \quad u_\mu a^\mu = 0 \ , \tag{35.77}$$

   this equation breaks up into two independent pieces,

$$\nabla_\mu T^{\mu\nu} = 0 \quad \Leftrightarrow \quad \dot\rho + \theta\rho = 0 \quad \text{and} \quad a^\nu = u^\mu\nabla_\mu u^\nu = 0 \ . \tag{35.78}$$

   Its time (energy flow) component is a continuity equation, while its space (momentum flow) part tells us that the particles have to move on geodesics.

2. Now what happens if we include pressure p? This corresponds to adding $p(g^{\mu\nu} + u^\mu u^\nu) \equiv ph^{\mu\nu}$, but this tensor is orthogonal to u^μ (cf. again the discussion in section 12.2 on the Raychaudhuri equation),

$$u^\mu h_{\mu\nu} \equiv u^\mu(g_{\mu\nu} + u_\mu u_\nu) = u_\nu - u_\nu = 0 \ . \tag{35.79}$$

   Therefore the equation $\nabla_\mu T^{\mu\nu} = 0$, with

$$T_{\mu\nu} = (\rho+p)u_\mu u_\nu + pg_{\mu\nu} \quad \Rightarrow \quad \nabla_\mu T^{\mu\nu} = (\dot\rho+\theta(\rho+p))u^\nu + (\rho+p)a^\nu + (\nabla_\mu p)h^{\mu\nu} \tag{35.80}$$

   again breaks up nicely into two orthogonal pieces. The part tangent to u^μ tells us that

$$\dot\rho + \theta(\rho + p) = 0 \ , \tag{35.81}$$

so this is a conservation law, and the part orthogonal to u^μ gives

$$(\rho + p)a^\nu + (\nabla_\mu p)h^{\mu\nu} = 0 \ , \tag{35.82}$$

which is a curved-space relativstic generalisation of the Euler equations of a perfect fluid.

In particular, now the velocity field is not composed of geodesics unless the derivative of p in the directions orthogonal to u^μ (i.e. the spatial derivative) is zero, $h^{\nu\mu}\nabla_\mu p = 0$. This is precisely the situation we are considering in cosmology, where $\rho = \rho(t)$ and $p = p(t)$ depend only on t, which is the proper time of the comoving observers described by the velocity field u^μ.

Returning thus to the cosmological setting, where we have (correctly, and uniquely as we now know) chosen the matter to move along geodesics, we are left with the continuity equation (35.81), which is now the same as (35.67) because for $u^\mu = (1, 0, 0, 0)$ in comoving coordinates one has

$$\theta = \nabla_\mu u^\mu = \frac{1}{\sqrt{g}}\partial_\mu(\sqrt{g}u^\mu) = a(t)^{-3}\partial_t(a(t)^3) = 3\dot{a}(t)/a(t) \ . \tag{35.83}$$

As an aside, note that this equation implies

$$\frac{d}{dt}\theta = 3(\ddot{a}/a - \dot{a}^2/a^2) \ , \tag{35.84}$$

which in terms of the Hubble parameter $H = \dot{a}/a$ and the deceleration parameter $q = -\ddot{a}a/\dot{a}^2$ (34.96) can be written as

$$\frac{d}{dt}\theta = -3H^2(q + 1) \ . \tag{35.85}$$

This equation is a special case of the Raychaudhuri equation (12.36) for timelike geodesic congruences,

$$\frac{d}{d\tau}\theta = -\tfrac{1}{3}\theta^2 - \sigma^{\mu\nu}\sigma_{\mu\nu} + \omega^{\mu\nu}\omega_{\mu\nu} - R_{\mu\nu}u^\mu u^\nu \ . \tag{35.86}$$

Indeed, specialising (35.86) to the family of comoving observers in a Robertson-Walker geometry and noting that

- the proper time τ is the cosmological time t
- the rotation is zero (either by explicit calculation or, more ot the point, because the geodesics are orthogonal to the hypersurfaces $t = const.$, or on symmetry grounds)
- the shear is zero (either by explicit calculation or on symmetry grounds)
- the relevant component of the Ricci tensor is $R_{00} = -3\ddot{a}/a$,

oner sees that (35.86) reduces to

$$\frac{d}{dt}\theta = -\tfrac{1}{3}\theta^2 - R_{00} = -3\dot{a}^2/a^2 + 3\ddot{a}/a \tag{35.87}$$

which is identical to (35.84). Thus we see how the parameters $H(t)$ and $q(t)$, originally introduced to characterise the first terms in a Taylor expansion of the cosmic scale factor also govern the local behaviour of freely falling observers like (clusters of) galaxies. In particular, in an expanding universe the congruences of comoving observers diverge / expand ($\theta > 0$) but the rate of expansion decreases ($\dot{\theta} < 0$) provided that $q > -1$.

35.6 Conservation Laws for Specific Equations of State

We now look at the solutions of (35.67),

$$\dot{\rho} = -3(\rho + p)\frac{\dot{a}}{a} \ , \tag{35.88}$$

for some specific equations of state.

1. For instance, when the pressure of the cosmic matter is negligible, like in the universe today, and we can treat the galaxies (without disrespect) as dust, then one has

$$w = 0 \quad \Rightarrow \quad \frac{\dot{\rho}}{\rho} = -3\frac{\dot{a}}{a} \ , \tag{35.89}$$

and this equation can trivially be integrated to

$$\rho(t)a(t)^3 = \text{const.} \tag{35.90}$$

Thus the (proper) density is proportional to the inverse (proper) spatial volume, an unsurprising (and reassuring) result. This of course also follows on the nose from the conservation law in the form (35.70):

$$p = 0 \quad \Rightarrow \quad \frac{d}{dt}E = 0 \ . \tag{35.91}$$

2. On the other hand, if the universe is dominated by, say, radiation, then one has the equation of state $p = \rho/3$, and the conservation equation reduces to

$$w = 1/3 \quad \Rightarrow \quad \frac{\dot{\rho}}{\rho} = -4\frac{\dot{a}}{a} \ , \tag{35.92}$$

and therefore

$$\rho(t)a(t)^4 = \text{const.} \tag{35.93}$$

The reason why the energy density of photons decreases faster with $a(t)$ than that of dust is of course ... the redshift:

$$\frac{1}{a(t)^4} = \frac{1}{a(t)^3}(\text{number density of photons}) \times \frac{1}{a(t)}(\text{redshift of individual photons}) \ . \tag{35.94}$$

3. More generally, for matter with equation of state parameter w one finds

$$p = w\rho \quad \Rightarrow \quad \rho(t)a(t)^{3(1+w)} = \text{const.} \tag{35.95}$$

The interpretation of this equation becomes somewhat more transparent in terms of the thermodynamic variant (35.70) of the conservation law. For the linear equation of state $p = w\rho$, it reduces to

$$\frac{dE}{dt} = -wE\frac{dV}{dt} \quad \Leftrightarrow \quad \frac{d}{dt}(V^w E) = 0 \ . \tag{35.96}$$

By (35.21), this is simply the statement that the entropy, which is a function of $V^w E$, is constant (cf. (35.72)).

4. In particular, for $w = -1$, $\rho = E/V$ itself is constant,

$$w = -1 \quad \Rightarrow \quad \rho(t) = \text{const.} \tag{35.97}$$

as it should be, in view of its identification (35.31) with a cosmological constant.

5. Quite generally, we see from (35.88) that in an expanding universe (i.e. $\dot{a} > 0$), the energy density of matter satisfying the null energy condition (NEC) $\rho + p \geq 0$ cannot increase (and will necessarily decrease for matter satisfying the strict NEC, with $\rho + p > 0$),

$$\dot{a} > 0 \quad , \quad \rho + p \geq 0 \quad \text{(NEC)} \quad \Rightarrow \quad \dot{\rho} \leq 0 \ . \tag{35.98}$$

6. As the final example, consider the peculiar Chaplygin gas with equation of state $p = -A/\rho$ (35.41) with A constant. In this case (35.88) reads

$$p = -A/\rho \quad \Rightarrow \quad \dot{\rho} = -3\left(\rho - \frac{A}{\rho}\right)\frac{\dot{a}}{a} \ , \tag{35.99}$$

which has the exact solution[153]

$$\rho(t) = \sqrt{A + B/a(t)^6} \tag{35.100}$$

(with integration constant B). In the context of a cosmology of an expanding universe, this has the remarkable property of interpolating between what appears to be a dust-filled universe at early times (small $a(t)$),

$$a(t) \quad \text{small} \quad \Rightarrow \quad \rho(t) \sim \sqrt{B}/a(t)^3 \ , \tag{35.101}$$

and a universe filled with a cosmological constant at late times (large $a(t)$),

$$a(t) \quad \text{large} \quad \Rightarrow \quad \rho(t) \sim \sqrt{A} \quad \Rightarrow \quad p(t) \sim -\sqrt{A} \ . \tag{35.102}$$

Thus this mimics the evolution of an ordinary (matter-filled) universe to one dominated by *dark energy*, resembling the evolution of our universe (cf. section 38.1).

[153] A. Kamenshchik, U. Moschella, V. Pasquier, *An alternative to quintessence*, `arXiv:gr-qc/0103004`.

After these preliminaries, we are now prepared to tackle (hence first to determine) the Einstein equations in this setting.

Allowing for the presence of a cosmological constant, we thus consider the equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ . \tag{35.103}$$

Alternatively we can write these equations as

$$R_{\mu\nu} = 8\pi G_N (T_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} T^{\lambda}_{\ \lambda}) + \Lambda g_{\mu\nu} \ . \tag{35.104}$$

Because of isotropy, there are only two independent equations, e.g. the (00)-component and any one of the non-zero (ij)-components. Using (35.6), we find

$$\begin{aligned} -3\frac{\ddot{a}}{a} &= 4\pi G_N(\rho + 3p) - \Lambda \\ \frac{\ddot{a}}{a} + 2\frac{\dot{a}^2}{a^2} + 2\frac{k}{a^2} &= 4\pi G_N(\rho - p) + \Lambda \ . \end{aligned} \tag{35.105}$$

Alternatively, instead of the spatial components of (35.104) one could have used the (00)-component of (35.103) and the expression in (35.8) for G_{00} to deduce

$$G_{00} - \Lambda = 8\pi G_N \, T_{00} \quad \Rightarrow \quad \frac{\dot{a}^2}{a^2} + \frac{k}{a^2} = \frac{8\pi G_N}{3}\rho + \frac{\Lambda}{3} \tag{35.106}$$

(and this equation could have also been obtained by appropriate subtraction of the previous 2 equations). Either way, we supplement these by the conservation equation

$$\dot{\rho} = -3(\rho + p)\frac{\dot{a}}{a} \ . \tag{35.107}$$

and thus end up with the set of equations

$$\boxed{\begin{array}{lrcl} (F1) & \frac{\dot{a}^2}{a^2} + \frac{k}{a^2} &=& \frac{8\pi G_N}{3}\rho + \frac{\Lambda}{3} \\ (F2) & -3\frac{\ddot{a}}{a} &=& 4\pi G_N(\rho + 3p) - \Lambda \\ (F3) & \dot{\rho} &=& -3(\rho + p)\frac{\dot{a}}{a} \end{array}} \tag{35.108}$$

Together, these are known as the *Friedmann-Lemaître equations* or *Friedmann equations*. These equations form the basis, and govern every aspect, of the standard Friedmann-Lemaître-Robertson-Walker models of cosmology.

Remarks:

1. I will usually refer to the collection of these equations, without disrespect to Lemaître but just out of habit, to the Friedmann equations. When referring to individual equations, rather than quoting equation numbers I will simply refer to them as the equations (F1), (F2), and (F3) respectively.

2. In terms of the Hubble parameter $H(t)$ and the deceleration parameter $q(t)$, these equations can also be written as

$$\begin{array}{lrcl} (F1') & H^2 & = & \frac{8\pi G_N}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3} \\ (F2') & q & = & \frac{1}{3H^2}[4\pi G_N(\rho + 3p) - \Lambda] \\ (F3') & \frac{d}{dt}(\rho a^3) & = & -3Hpa^3 \ . \end{array}$$

3. Commonly, (F1) is referred to as *the Friedmann equation*. As mentioned above, this is precisely the (00)-component of the Einstein equations.

4. (F2), on the other hand, can be interpreted as the Raychaudhuri equation (35.87). Indeed, with $\theta = 3\dot{a}/a$ and

$$R_{00} = 8\pi G_N(T_{00} + \tfrac{1}{2}T^\alpha_\alpha) - \Lambda = 4\pi G_N\ (\rho + 3p) - \Lambda \tag{35.109}$$

one finds that

$$\frac{d}{dt}\theta = -\tfrac{1}{3}\theta^2 - R_{00} \quad \Leftrightarrow \quad -3\ddot{a}/a = 4\pi G_N(\rho + 3p) - \Lambda \tag{35.110}$$

which is precisely (F2).

5. In writing the above equations, I have separated out the cosmological constant Λ from the remaining matter contributions. Of course, using (35.31), it could have just been treated as one other perfect fluid contribution (with $w = -1$). Occasionally either one or the other way of writing these equations is (marginally) more convenient.

6. Note that because of the Bianchi identities, the Einstein equations and the conservation equations should not be independent, and indeed they are not:

   (a) It is easy to see that (F1) and (F3) imply the second order equation (F2) so that, a pleasant simplification, in practice one only has to deal with the two first order equations (F1) and (F3). Sometimes, however, (F2) is easier to solve than (F1), because it is linear in $\ddot{a}(t)$, and then (F1) is just used to fix one constant of integration.

   (b) It is also easy to see that (F1) and (F2) imply (F3), i.e. that the gravity equations of motion imply the matter equations of motion, a general and fundamental feature of general relativity.

   (c) Finally, formally (F2) and (F3) also imply (F1), with k (which only appears in (F1)) arising as an integration constant.

7. Note also that the parameter k appears in the Friedmann equations only in the combination k/a^2. Thus a scaling of k can be compensated by a corresponding scaling of $a(t)$. This reflects the fact, mentioned several times already, that only this particular combination has individual meaning, and that one can exploit the scaling freedom to e.g. set $a_0 = 1$ or $k = 0, \pm 1$.

8. One can use (35.13) to determine the analogue of the Friedmann equations in any space-time dimension $D = n + 1$.

   - The indepedent components of the Einstein equation can then be chosen to be, from $G_{00} = \kappa T_{00}$ (for the reason for now calling the coupling constant κ and not just $8\pi G_N$, as in four dimensions, see the discussion around (19.44)),

   $$\frac{k + \dot{a}^2}{a^2} = \frac{2\kappa}{n(n-1)}\rho \ , \tag{35.111}$$

   and from R_{00}, say, using (19.45) and the fact that the trace of a perfect fluid energy-momentum tensor is now

   $$T^\lambda_\lambda = -\rho + np \ , \tag{35.112}$$

   one finds

   $$\frac{\ddot{a}}{a} = -\frac{\kappa}{n(n-1)}\left[(n-2)\rho + np\right] \ . \tag{35.113}$$

   - Finally the continuity equation takes the form

   $$\dot{\rho} = -n(\rho + p)\frac{\dot{a}}{a} \ . \tag{35.114}$$

9. As an aside note that, as the Robertson-Walker metrics are, in particular, spherically symmetric, and written in the manifestly spherically-symmetric form (24.80), we have the notion of the Misner-Sharp mass (24.82) for spherical symmetry at our disposal. In terms of the area radius (34.42),

   $$\tilde{r}(t, r) = a(t)r \ , \tag{35.115}$$

   the Misner-Sharp mass function is given by (24.82)

   $$M_{MS}(t, r)m(t, r) = \tfrac{1}{2}\tilde{r}\left(1 - g^{ab}\partial_a\tilde{r}\partial_b\tilde{r}\right) \ . \tag{35.116}$$

   With $g^{tt} = -1$ and $g^{rr} = (1 - kr^2)/a(t)^2$, this reduces to

   $$m(t, r) = \tfrac{1}{2}a(t)r^3(\dot{a}(t)^2 + k) \ . \tag{35.117}$$

   Using the Friedmann equation (F1), this can be written in a more informative way as

   $$\begin{aligned} m(t, r) &= \tfrac{1}{2}a(t)^3r^3\left(\frac{8\pi G_N}{3}\rho + \frac{\Lambda}{3}\right) \\ &= \frac{4\pi}{3}\tilde{r}^3 G_N\left(\rho + \rho_\Lambda\right) \ , \end{aligned} \tag{35.118}$$

   with $\rho_\Lambda = \Lambda/8\pi G_N$ the energy density (35.31) associated with the cosmological constant. Note that this result, which again, as in section 24.6, has the interpretation as "mass = coordinate volume (not proper volume) × density", does not depend explicitly on either the pressure p or the curvature k. In particular it is independent of the equation of state relating p and ρ.

For many purposes it is useful to cast the above set of Friedmann-Lemaître equations into a Lagrangian or Hamiltonian form. In particular (and this is the motivation for doing this here), this system of equations is sufficiently simple to provide a concrete illustration of some of the general features of the canonical Hamiltonian formulation of general relativity discussed in a somewhat cursory way in section 21.

Rather than specialising the general results of that section to the case at hand, we will adopt a more pedestrian approach here and derive these results from scratch. This will make contact with and hopefully shed some light on a variety of different issues that have arisen in various parts of these notes, e.g.

- the characteristic constraints of general relativity (sections 19.7 and 21), in particular the *Hamiltonian constraint*
- the *Gibbons-Hawking-York boundary term* discussed in section 20.5
- the *ADM action* discussed in section 23.4
- the role and significance of the *lapse function* (cf. (21.18)) introduced into the Robertson-Walker metric (for what appeared to be no good reason at that point) at the end of section 35.1.

In the previous section, in order to arrive at the Friedmann equations, we made an ansatz for the metric based on symmetry considerations (spatial maximal symmetry), namely

$$ds^2 = -dt^2 + a(t)^2 d\tilde{s}^2 \tag{35.119}$$

and then plugged this (together with a compatible ansatz for the energy-momentum tensor) into the full Einstein equations. Such a reduction procedure is always possible at the level of the equations of motion. In order to develop the Hamiltonian formulation of the Friedmann equations, it is convenient to start with the Lagrangian formulation. However, such a reduction procedure (plugging an ansatz for the solution into the action and then solving the resulting Euler-Lagrange equations) is not necessarily consistent at the level of the Lagrangian or action, i.e. the Euler-Lagrange equations of the reduced Langrangian will not necessarily agree with the reduced equations of motion and will not necessarily give rise to solutions of the original unreduced equations.

Indeed, in the present context it is pretty obvious that, with the single gravitational degree of freedom $a(t)$, associated with the size of the spatial metric, it is impossible to derive both the (one independent) spatial component of the Einstein equations (the Friedmann equation F2, say) and the time-time component of the Einstein equations (the Friedmann equation F1) from a Lagrangian depending just on $a(t)$ (and the matter variables).

It turns out, however, that just by introducing as an additional variable a non-trivial time-time component of the metric (whose variation can then be used to impose the Friedmann equation F1), the reduction is consistent also at the Lagrangian level. We will introduce this additional variable in the form of the lapse function $N(t)$ of (35.10), but we will now simply write t for the time-coordinate and not t', so that the ansatz for the metric is

$$ds^2 = -N^2(t)dt^2 + a^2(t)d\tilde{s}^2 \ , \tag{35.120}$$

and we can always choose the $N(t) = 1$ "t is comoving proper time" gauge at the end of the calculations.

In section 35.1 we had already determined the Ricci scalar of this metric, namely (now again, consistent with $t' \to t$ denoting time-derivatives by overdots rather than primes) (35.12)

$$R = \frac{6}{a^2N^3}(N(a\ddot{a} + \dot{a}^2) - a\dot{a}\dot{N} + kN^3) \ . \tag{35.121}$$

The Einstein-Hilbert Lagangian density is therefore

$$\sqrt{g}R = Na^3\sqrt{\tilde{g}}R = \sqrt{\tilde{g}}\frac{6a}{N^2}(N(a\ddot{a} + \dot{a}^2) - a\dot{a}\dot{N} + kN^3) \ . \tag{35.122}$$

The only dependence on the spatial coordinates is in the spatial volume element $\sqrt{\tilde{g}}$. Therefore, integrating this Lagrangian density over the space-time, one obtains a (potentially infinite) volume factor from the integration over the spatial coordinates which we will simply drop. Thus in the infinite-volume case ($k = -1$ or $k = 0$ without periodic toroidal identifications) this is not really a reduction in the strict technical sense. However, this is not our main concern here. Our aim is simply to obtain a Lagrangian formulation of the Friedmann equations and (as we will see) this can be accomplished by just dropping the integration over the spatial coordinates.

We are thus left with the 1-dimensional (mechanics) gravitational Lagrangian

$$L_{EH} = \frac{6a}{N^2}(N(a\ddot{a} + \dot{a}^2) - a\dot{a}\dot{N} + kN^3) \ . \tag{35.123}$$

Note that this depends not only on $a(t)$ and $\dot{a}(t)$, and $N(t)$ and $\dot{N}(t)$, but also (linearly) on $\ddot{a}(t)$. This of course reflects the general property of the Einstein-Hilbert Lagrangian that it depends on 2nd derivatives of the metric, thus necessitating the introduction of boundary terms etc., as discussed in sections 20.4 and 20.5.

Naively, to eliminate this $\ddot{a}$-term, we can integrate the first term of the Lagrangian by parts,

$$6a^2\ddot{a}N^{-1} = \frac{d}{dt}(6a^2\dot{a}N^{-1}) + 6aN^{-2}(-2\dot{a}^2N + a\dot{a}\dot{N}) \ . \tag{35.124}$$

We see that this has the effect of changing the sign of the 2nd term of (35.123) and cancelling the 3rd term $\sim \dot{N}$, so that we have

$$L_{EH} = \frac{6a}{N^2}(-N\dot{a}^2 + kN^3) + \frac{d}{dt}(6a^2\dot{a}N^{-1}) \ . \tag{35.125}$$

It turns out that the total derivative term is cancelled precisely by the Gibbons-Hawking-York boundary term discussed in section 20.5. This should not come as a surprise: after all, that was its purpose. To see this explicitly, note that with any of the definitions or characterisations of the extrinsic curvature tensor given in section 18, one finds that the extrinsic curvature of the constant time t hypersurfaces (with unit normal vector $(1/N)\partial_t$) in the ambient space-time is

$$K_{ij} = \frac{1}{2N}\partial_t g_{ij} = \frac{\dot{a}}{aN} g_{ij} \tag{35.126}$$

so that the trace is

$$K = g^{ij}K_{ij} = \frac{3\dot{a}}{aN} \ . \tag{35.127}$$

Thus the Gibbons-Hawking-York boundary term (20.61) reduces to (again dropping the spatial volume element $\sqrt{\bar{g}}$ and using $\epsilon = -1$ for spacelike hypersurfaces)

$$2\epsilon\sqrt{h}\, K \to L_{GHY} = -2a^3(3\dot{a}/aN) = -6a^2\dot{a}N^{-1} \ . \tag{35.128}$$

The complete gravitational Lagrangian is now, in analogy with the complete standard gravitational action (20.67),

$$S_g[g_{\alpha\beta}] = S_{EH}[g_{\alpha\beta}] + S_{GHY}[g_{\alpha\beta}] \ , \tag{35.129}$$

the Lagrangian

$$\begin{aligned} L_g &= L_{EH} + \frac{d}{dt}L_{GHY} \\ &= \frac{6a}{N^2}(-N\dot{a}^2 + kN^3) \\ &= 6N(-a(\dot{a}/N)^2 + ka) \ . \end{aligned} \tag{35.130}$$

Inclusion of the cosmological constant Λ,

$$\sqrt{g}R \to \sqrt{g}(R - 2\Lambda) \ , \tag{35.131}$$

leads to

$$L_g = 6N(-a(\dot{a}/N)^2 + ka - \Lambda a^3/3) \ . \tag{35.132}$$

Remarks:

1. Note that, as desired, the Lagrangian now only depends on the "fields" $a(t)$ and $N(t)$ and (at most) their 1st derivatives.

2. Actually, we see that the Lagrangian depends only on $N(t)$, not its time-derivative $\dot{N}(t)$. This will also turn out to be the case for the matter action (which typically does not depend on any derivatives of the metric at all). Thus the role of $N(t)$ is just that of a Lagrange multiplier, and as we will see the constraint it imposes is simply the Friedmann equation F1.

3. One can also obtain (35.130) directly from the Gauss-Codazzi (21.6) or ADM form (21.42) of the action,

$$S_{ADM}[g_{\alpha\beta}] = \int \sqrt{g} d^4x \, (\bar{R} + K^{ij}K_{ij} - K^2) \ . \tag{35.133}$$

Indeed, the 3-dimensional scalar curvature is simply (this follows e.g. from (14.19))

$$\bar{R} = 6k/a^2 \ , \tag{35.134}$$

while from (35.126) one has

$$K^{ij}K_{ij} - K^2 = 3(\dot{a}/aN)^2 - 9(\dot{a}/aN)^2 = -6(\dot{a}/aN)^2 \tag{35.135}$$

so that (again dropping the spatial volume element $\sqrt{\bar{g}}$), one finds

$$L_{ADM} = a^3 N(-6(\dot{a}/aN)^2 + 6k/a^2) = L_g \ . \tag{35.136}$$

We now need to add matter. The phenomenological description of matter in terms of the energy and pressure densities $\rho(t)$ and $p(t)$ is not sufficient for a Lagrangian formulation. The simplest matter model with a "microscopic" Lagrangian description is that of a scalar field in the gravitational field described by the class of metrics (35.120), and this is the model we will consider.

The starting point is thus the action (6.14) for a scalar field with a potential $V(\phi)$,

$$S[\phi, g_{\alpha\beta}] = \int \sqrt{g} d^4x \left[-\tfrac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi - V(\phi) \right] \ . \tag{35.137}$$

Compatibility with the symmetries of the gravitational field, in particular the spatial homogeneity, requires that the scalar field is spatially constant and is thus only a function of time t. With this assumption (and again dropping $\sqrt{\bar{g}}$), the matter Lagrangian reduces to

$$L_m = Na^3(\dot{\phi}^2/2N^2 - V(\phi)) = N(a^3\dot{\phi}^2/2N^2 - a^3V(\phi)) \ . \tag{35.138}$$

Therefore the total gravitational + matter Lagrangian and action are (reinstating the gravitational coupling constant)

$$\begin{aligned} L_{tot} &= \frac{1}{16\pi G_N} L_g + L_m \\ &= N\left(-\frac{3}{4\pi G_N} \frac{a\dot{a}^2}{2N^2} + \frac{a^3\dot{\phi}^2}{2N^2} + \frac{3ka}{8\pi G_N} - \frac{\Lambda a^3}{8\pi G_N} - a^3 V(\phi) \right) \end{aligned} \tag{35.139}$$

and

$$\begin{aligned} S_{tot}[a, \phi, N] &= \int dt \, L_{tot} \\ &= \int dt \, N\left(-\frac{3}{4\pi G_N} \frac{a\dot{a}^2}{2N^2} + \frac{a^3\dot{\phi}^2}{2N^2} + \frac{3ka}{8\pi G_N} - \frac{\Lambda a^3}{8\pi G_N} - a^3 V(\phi) \right) \end{aligned} \tag{35.140}$$

Before analysing the equations of motion and the Hamiltonian arising from this Lagrangian and action, let us note the following points:

1. From this form of the action it is evident that one role of $N(t)$ is to ensure time reparametrisation invariance of the action. Indeed, dt and $N(t)$ only appear in the combinations $N(t)dt$ or $(1/N(t))(d/dt)$. Hence the action remains unchanged under time reparametrisations $t \to \tilde{t}(t)$ if one simultaneously transforms $N(t)$ to a new lapse function $\tilde{N}(\tilde{t})$, say, according to

$$N(t)dt = \tilde{N}(\tilde{t})d\tilde{t} \quad \Leftrightarrow \quad \tilde{N} = N(dt/d\tilde{t}) \ . \tag{35.141}$$

   This is the remnant of the general covariance of the original action, and the mechanism here is the same as that which rendered the "parent" geodesic action (2.111) reparametrisation invariant.

2. We see that this action does not involve the time-derivative of the lapse function $N(t)$. Hence $N(t)$ acts as a Lagrange multiplier enforcing a constraint. This constraint can be regarded as the constraint associated to this reparametrisation invariance, and such constraints are thus a characteristic feature of any reparametrisation invariant or generally covariant system.

3. In this combined gravity plus matter action, we now see very clearly that the gravitational kinetic term has the opposite sign of the matter kinetic term. This is a particular (and particularly obvious) manifestation of the general fact (mentioned in connection with the DeWitt metric at the end of section 21.2) that the gravitational kinetic term is not positive definite and that the negative "direction" in field space is associated with overall spatial volume deformations.

4. Finally, this form of the action makes it particularly obvious that the cosmological constant term can also be regarded as leading to (or arising from) a constant shift of the potential for the matter fields,

$$\frac{\Lambda a^3}{8\pi G_N} + a^3 V(\phi) = a^3(V(\phi) + \rho_\Lambda) \ . \tag{35.142}$$

   We will therefore absorb the cosmological constant term into the scalar potential in the following and not carry it around explicitly.

Now let us look at the Euler-Lagrange equations arising from this action.

- The Euler-Lagrange Equation for $N(t)$

  As mentioned before, $N(t)$ acts as a Lagrange multiplier enforcing the constraint

$$\frac{\delta S_{tot}}{\delta N} = 0 \quad \Rightarrow \quad \frac{\partial L}{\partial N(t)} = 0 \ . \tag{35.143}$$

  Explicitly this is the condition

$$-\frac{3}{4\pi G_N}\frac{a\dot{a}^2}{2N^2} + \frac{a^3\dot{\phi}^2}{2N^2} - \frac{3ka}{8\pi G_N} + a^3 V(\phi) = 0 \ . \tag{35.144}$$

This really is a constraint rather than an equation of motion, because it only depends on the fields and their 1st derivatives, not their 2nd derivatives, and thus constitutes a condition on initial data on some constant time initial hypersurface.

In the gauge $N(t) = 1$ (which we can now choose, after having determined the equation of motion arising from varying N in the action), this constraint becomes

$$\frac{3}{8\pi G_N}(a\dot{a}^2 + ak) = \frac{a^3\dot{\phi}^2}{2} + a^3 V(\phi) \quad , \tag{35.145}$$

or

$$\frac{\dot{a}^2 + k}{a^2} = \frac{8\pi G_N}{3}(\frac{1}{2}\dot{\phi}^2 + V(\phi)) \quad . \tag{35.146}$$

Now recall that for such a scalar field the energy density and pressure are given by (35.38)

$$\rho(t) = \tfrac{1}{2}\dot{\phi}^2 + V(\phi) \quad , \quad p(t) = \tfrac{1}{2}\dot{\phi}^2 - V(\phi) \quad . \tag{35.147}$$

Hence we can write the constraint as

$$\frac{\dot{a}^2 + k}{a^2} = \frac{8\pi G_N}{3}\rho \quad , \tag{35.148}$$

and this we now recognise as precisely the Friedmann equation F1.

Having derived the constraint arising from the variation of the lapse function N, we can now simplify our life by using the reparametrisation invariance to set $N(t) = 1$. Thus we can work with the simpler action

$$S_{tot}[a, \phi] = \int dt \left(-\frac{3}{4\pi G_N}\frac{a\dot{a}^2}{2} + \frac{a^3\dot{\phi}^2}{2} + \frac{3ka}{8\pi G_N} - a^3 V(\phi) \right) \quad . \tag{35.149}$$

- The Euler-Lagrange equation for $a(t)$

  This is the equation

$$\frac{d}{dt}\left(-\frac{3a\dot{a}}{4\pi G_N} \right) = 3\left(-\frac{1}{4\pi G_N}\frac{\dot{a}^2}{2} + \frac{a^2\dot{\phi}^2}{2} + \frac{k}{8\pi G_N} - a^2 V(\phi) \right) \quad , \tag{35.150}$$

  or

$$\frac{\ddot{a}}{a} = -4\pi G_N p - \frac{\dot{a}^2 + k}{2a^2} \quad , \tag{35.151}$$

  where $p = p(t)$ is the pressure (35.147). Using the constraint (the Friedmann equation F1) derived above, this becomes

$$\frac{\ddot{a}}{a} = -\frac{4\pi G_N}{3}(\rho + 3p) \quad , \tag{35.152}$$

  which is precisely the Friedmann equation F2.

- The Euler-Lagrange equation for $\phi(t)$

  Finally, the equation of motion for the scalar field is

$$\frac{d}{dt}(a^3\dot{\phi}) = -a^3 V'(\phi) \quad \Leftrightarrow \quad \ddot{\phi} + 3(\dot{a}/a)\dot{\phi} + V'(\phi) = 0 \ , \tag{35.153}$$

  exhibiting the same friction term coupling to the Hubble parameter as (34.113). Multiplying by $\dot{\phi}$ and noting that

$$\rho + p = \dot{\phi}^2 \tag{35.154}$$

  and

$$\dot{\rho} = \dot{\phi}\ddot{\phi} + V'(\phi)\dot{\phi} \ , \tag{35.155}$$

  it is evident that this equation can equivalently be written as

$$\ddot{\phi} + 3(\dot{a}/a)\dot{\phi} + V'(\phi) = 0 \quad \Leftrightarrow \quad \dot{\rho} + 3(\rho + p)(\dot{a}/a) = 0 \ , \tag{35.156}$$

  which is the Friedmann equation F3.

We have thus verified that we have indeed obtained a Lagrangian description of the complete set of Friedmann equations.

35.9 Hamiltonian Formulation of the Friedmann Equations

Having obtained the Lagrangian formulation, it is now quite straightforward to pass to a Hamiltonian formulation, and this despite the presence of constraints which, in general, can significantly complicate the Hamiltonian formulation (and subsequent canonical quantisation, say) of such systems.[154]

Our starting point is the Lagrangian (35.139) (with Λ absorbed into $V(\phi)$), i.e.

$$L_{tot} = N\left(-\frac{3}{4\pi G_N}\frac{a\dot{a}^2}{2N^2} + \frac{a^3\dot{\phi}^2}{2N^2} + \frac{3ka}{8\pi G_N} - a^3 V(\phi)\right) \ . \tag{35.157}$$

In order to streamline the following discussion, it is convenient to consider the fields $a(t)$ and $\phi(t)$ as the two coordinates

$$Q^A = (a, \phi) \tag{35.158}$$

of a 2-dimensional dynamical system, with metric

$$G_{AB} = \mathrm{diag}(-3a/4\pi G_N, a^3) \ . \tag{35.159}$$

[154] For detailed textbook discussions of this subject, see e.g. M. Henneaux, C. Teitelboim, *Quantisation of Gauge Systems*, or H. Rothe, K. Rothe, *Classical and Quantum Dynamics of Constrained Hamiltonian Systems*.

With this notation, the Lagrangian can be written as

$$L_{tot} = N\left(\frac{1}{2N^2}G_{AB}\dot{Q}^A\dot{Q}^B - V(Q)\right) \tag{35.160}$$

where the potential is

$$V(Q) = a^3V(\phi) - \frac{3ka}{8\pi G_N} \ . \tag{35.161}$$

Were it not for the presence of the lapse function N as an addditional dynamical variable, this would be a completely standard classical mechanics Lagrangian, with an obvious corresponding Hamiltonian. In particular, the momenta conjugate to the variables Q^A take the almost standard form

$$P_A = \frac{\partial L_{tot}}{\partial \dot{Q}^A} = G_{AB}\dot{Q}^B/N \ , \tag{35.162}$$

i.e.

$$P_a = -\frac{3a\dot{a}}{4\pi G_N N} \quad , \quad P_\phi = \frac{a^3\dot{\phi}}{N} \ . \tag{35.163}$$

These relations can, as usual, be used to eliminate the velocities in favour of the momenta. The major novelty is thus the presence of N, whose conjugate momentum vanishes,

$$P_N = \frac{\partial L_{tot}}{\partial \dot{N}} = 0 \ . \tag{35.164}$$

This is our first constraint (and a *primary constraint* in the terminology of constrained systems). While this relation does not allow us to eliminate $\dot{N}$ in terms of the momentum P_N, this is not an issue here since the Lagrangian does not depend on $\dot{N}$ in the first place.

We can now follow the standard procedure to construct the Hamiltonian, via (in the case at hand, whether or not we include the $P_N\dot{N}$-term evidently makes no difference)

$$H_{tot} = P_N\dot{N} + P_A\dot{Q}^A - L_{tot} \ . \tag{35.165}$$

Then one finds the Hamiltonian.

$$H_{tot}(N, Q, P) = N\left(\frac{1}{2}G^{AB}P_AP_B + V(Q)\right) \equiv N\mathcal{H}(Q, P) \ . \tag{35.166}$$

As far as its dependence on (Q, P) is concerned, this presents no surprises: time evolution is given by the Hamilton equations

$$\begin{aligned} \dot{Q}^A &= +\frac{\partial H_{tot}}{\partial P_A} = \{Q^A, H_{tot}\} \\ \dot{P}_A &= -\frac{\partial H_{tot}}{\partial Q_A} = \{P_A, H_{tot}\} \ , \end{aligned} \tag{35.167}$$

and the 1st of these just reproduces the definition of the momenta (35.162), while the 2nd then reproduces the Euler-Lagrange equations for the fields $Q^A = (a, \phi)$ discussed in the previous section.

Turning to N, consistency of the primary constraint $P_N = 0$ (i.e. the condition that it be preserved in time) requires

$$\dot{P}_N = \{P_N, H_{tot}\} = -\mathcal{H} = 0 \ . \tag{35.168}$$

Thus the primary constraint $p_N = 0$ gives rise to the *secondary constraint*

$$\mathcal{H}(Q, P) = 0 \tag{35.169}$$

(and there are no further constraints in this class of examples). This *Hamiltonian constraint* is precisely the Friedmann equation F1, i.e. the condition that was imposed in the Lagrangian formulation by the Lagrange multipler N,

$$\frac{\partial L_{tot}}{\partial N} = 0 \quad \Leftrightarrow \quad \mathcal{H} = 0 \ . \tag{35.170}$$

A painless way to see this is to note that the N-dependence in the Lagrangian (35.160),

$$L_{tot} = \frac{1}{2N} G_{AB} \dot{Q}^A \dot{Q}^B - N V(Q) \ , \tag{35.171}$$

is precisely such that differentiation with respect to N changes the relative sign between the 2 terms and thus essentially implements the Legendre transformation from the Lagrangian to the Hamiltonian (expressed as a function of the velocities),

$$-\frac{\partial L_{tot}}{\partial N} = \frac{1}{2N^2} G_{AB} \dot{Q}^A \dot{Q}^B + V(Q) = \mathcal{H} \ . \tag{35.172}$$

This structure

$$H = N\mathcal{H} \tag{35.173}$$

of the Hamiltonian, with the constraint $\mathcal{H} = 0$, is again characteristic of parametrisation invariant or generally covariant systems. In particular, this provides a concrete illustration of the general features of the gravitational Hamiltonian mentioned in section 21.7.

36 Cosmology IV: Qualitative Analysis

A lot can be deduced about the solutions of the Friedmann-Lemaître equations, i.e. the evolution of the universe in the Friedmann-Lemaître-Robertson-Walker cosmologies, without solving the equations directly and even without specifying a precise equation of state, i.e. a relation between p and ρ. In the following we will, in turn, discuss the Big Bang, the age of the universe, and its long term behaviour, from this qualitative point of view. I will then introduce the notions of critical density and density parameters, and discuss some global and causal aspects of these cosmological models (Penrose diagrams, horizons, ...).

36.1 The Past I: Big Bang (Existence of an Initial Singularity)

One amazing thing about the Friedmann-Lemaître-Robertson-Walker models is that all of them (provided that the matter content is reasonably physical - I will be more precise about this below) predict an initial singularity, commonly known as a Big Bang. This is very easy to see.

The Friedmann equation (F2),

$$-3\frac{\ddot{a}}{a} = 4\pi G_N(\rho + 3p) \quad \Leftrightarrow \quad q = \frac{4\pi G_N}{3H^2}(\rho + 3p) \ , \tag{36.1}$$

shows that, as long as the right-hand side is positive, one has $q > 0$, i.e. $\ddot{a} < 0$ so that the universe is decelerating due to gravitational attraction. This is the case for standard matter ($\rho > 0$) when it satisfies the *strong energy condition* (SEC) strictly (cf. the discussion in section 35.2, in particular (35.34)),

$$\text{strict SEC} \quad \Rightarrow \quad \rho + 3p > 0 \quad \Rightarrow \quad \ddot{a} < 0 \ . \tag{36.2}$$

It is also true for a negative cosmological constant (its negative energy density being outweighed by 3 times its positive pressure). It need not be true, however, in the presence of a positive cosmological constant which provides an accelerating contribution to the expansion of the universe. We will, for the time being, continue with the assumption that Λ is zero or, at least, non-positive, even though, as we will discuss later, recent evidence (strongly) suggests the presence of a non-negligible positive cosmological constant in our universe today (which is, however, totally irrelevant for the energy budget of the early universe).

Since $a > 0$ by definition, $\dot{a}(t_0) > 0$ because we observe a redshift, and $\ddot{a} < 0$ because $\rho + 3p > 0$, it follows that there cannot have been a turning point in the past and $a(t)$ must be concave downwards. Therefore $a(t)$ must have reached $a = 0$ at some *finite* time in the past. We will call this time $t = 0$, $a(0) = 0$.

As ρa^4 is constant for radiation (an apppropriate description of earlier periods of the universe), this shows that the energy density grows like $1/a^4$ as $a \to 0$ so this leads to quite a singular situation.

Once again, as in our discussion of black holes, it is natural to wonder at this point if the singularities predicted by General Relativity in the case of cosmological models are generic or only artefacts of the highly symmetric situations we were considering. And again there are singularity theorems applicable to these situations which state that, under reasonable assumptions about the matter content, singularities will occur independently of assumptions about symmetries.

36.2 The Past II: Age of the Universe

With the normalisation $a(0) = 0$, it is fair to call t_0 the age of the universe. If $\ddot{a}$ had been zero in the past for all $t \leq t_0$, then we would have

$$\ddot{a} = 0 \quad \Rightarrow \quad a(t) = a_0 t/t_0 \ , \tag{36.3}$$

and

$$\dot{a}(t) = a_0/t_0 = \dot{a}_0 \ . \tag{36.4}$$

This would determine the age of the universe to be

$$\ddot{a} = 0 \quad \Rightarrow \quad t_0 = \frac{a_0}{\dot{a}_0} = H_0^{-1} \ , \tag{36.5}$$

where H_0^{-1} is the *Hubble time*. However, provided that $\ddot{a} < 0$ for $t \leq t_0$ (as discussed above, this holds under suitable conditions on the matter content - which may or may not be realised in our universe), the actual age of the universe must be smaller than this,

$$\ddot{a} < 0 \quad \Rightarrow \quad t_0 < H_0^{-1} \ . \tag{36.6}$$

Thus the Hubble time H_0 sets an upper bound on the age of the universe. See Figure 46 for an illustration of this. In particular, this means that one can obtain an upper limit on the age of the universe by determining the leading (linear) term in the Hubble relation (34.104) from observations of redshifts of galaxies!

36.3 The Future: Long Term Behaviour

Let us now try to take a look into the future of the universe. Again we will see that it is remarkably simple to extract relevant information from the Friedmann equations without ever having to solve an equation.

We will assume that $\Lambda = 0$ and that we are dealing with matter with $\rho > 0$ and $w > -1/3$ (the SEC). The Friedmann equation (F1) can be written as

$$\dot{a}^2 = \frac{8\pi G_N}{3}\rho a^2 - k \ . \tag{36.7}$$

The left-hand side is manifestly non-negative. Let us see what this tells us about the right-hand side. Focus on the first term $\sim \rho a^2$. This term is strictly positive and, according to (35.95), behaves as

$$\rho a^2 \sim a^{-3(1+w)+2} = a^{-(1+3w)} \quad . \tag{36.8}$$

Thus for $w > -1/3$ the exponent is negative, so that if and when the cosmic scale factor $a(t)$ goes to infinity, one has

$$\lim_{a\to\infty} \rho a^2 = 0 \quad . \tag{36.9}$$

Now let us look at the second term on the right-hand side of (36.7), and analyse the 3 choices for k. For $k = -1$ or $k = 0$, the right hand side of (36.7) is strictly positive. Therefore $\dot{a}$ is never zero and since $\dot{a}_0 > 0$, we must have

$$\dot{a}(t) > 0 \ \ \forall\, t \quad . \tag{36.10}$$

Thus we can immediately conclude that open and flat universes must expand forever, i.e. they are open in space *and* time.

By taking into account (36.9), we can even be somewhat more precise about the long term behaviour. For $k = 0$, we learn that

$$k = 0: \qquad \lim_{a\to\infty} \dot{a}^2 = 0 \quad . \tag{36.11}$$

Thus the universe keeps expanding but more and more slowly as time goes on. By the same reasoning we see that for $k = -1$ we have

$$k = -1: \qquad \lim_{a\to\infty} \dot{a}^2 = 1 \quad . \tag{36.12}$$

Thus the universe keeps expanding, reaching a constant limiting velocity.

For $k = +1$, validity of (36.9) would lead us to conclude that $\dot{a}^2 \to -1$, but this is obviously a contradiction. Therefore we learn that the $k = +1$ universes never reach $a \to \infty$ and that there is therefore a maximal radius a_{max}. This maximal radius occurs for $\dot{a} = 0$ and therefore

$$k = +1: \qquad a_{max}^2 = \frac{3}{8\pi G_N \rho} \quad . \tag{36.13}$$

Note that intuitively this makes sense. For larger ρ or larger G_N the gravitational attraction is stronger, and therefore the maximal radius of the universe will be smaller. Since we have $\ddot{a} < 0$ also at a_{max}, again there is no turning point and the universe recontracts back to zero size leading to a *Big Crunch*. Therefore, spatially closed universes ($k = +1$) with physical matter are also closed in time. All of these findings are summarised in Figure 46.

If the cosmological constant Λ is not zero, this poetic correspondence "(open/closed) in space $\Leftrightarrow$ (open/closed) in time" is no longer necessarily true since the cosmological constant (with $\rho_\Lambda a^2 \sim a^2$) will dominate over other forms of matter for sufficiently large $a(t)$ - the asymptotic solutions in this case are given in section 37.5.

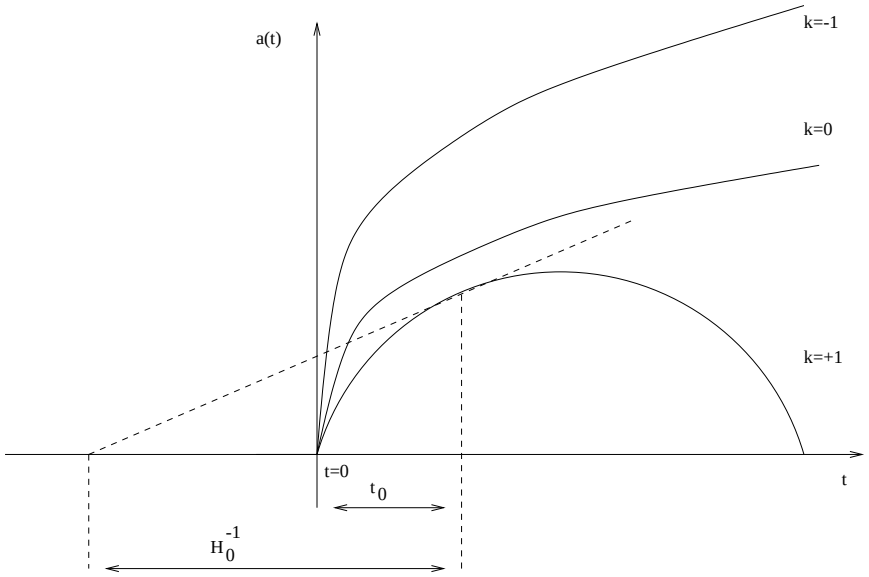


Figure 46: Qualitative behaviour of the Friedmann-Lemaître-Robertson-Walker models for $\Lambda = 0$. All models start with a Big Bang. For $k = +1$ the universe reaches a maximum radius and recollapses after a finite time. For $k = 0$, the universe keeps expanding but the expansion velocity tends to zero for $t \to \infty$ or $a \to \infty$. For $k = -1$, the expansion velocity approaches a non-zero constant value. Also shown is the significance of the Hubble time for the $k = +1$ universe showing clearly that H_0^{-1} gives an upper bound on the age of the universe.

In order to make the dependence of the Friedmann equation (F1) on the equation of state parameters w_b and on $a(t)$ more manifest, it is useful to use the conservation law (35.95,36.8) to write

$$\frac{8\pi G_N}{3}\rho_b(t)a(t)^2 = C_b a(t)^{-(1+3w_b)} \quad \Leftrightarrow \quad \Omega_b \dot{a}(t)^2 = C_b a(t)^{-(1+3w_b)} \tag{36.14}$$

for some constant C_b. Then the Friedmann equation takes the more explicit (in the sense that all the dependence on the cosmic scale factor $a(t)$ is explicit) form

$$\dot{a}^2 = \sum_b C_b a^{-(1+3w_b)} - k + \frac{\Lambda}{3}a^2 \ . \tag{36.15}$$

In addition to the vacuum energy (and pressure) provided by Λ, there are typically two other kinds of matter which are relevant in our approximation, namely matter in the form of dust ($w = 0$) and radiation ($w = 1/3$). Denoting the corresponding constants by C_m and C_r respectively, the Friedmann equation that we will be dealing with takes the form

$$(F1'') \quad \dot{a}^2 = \frac{C_m}{a} + \frac{C_r}{a^2} - k + \frac{\Lambda}{3}a^2 \ , \tag{36.16}$$

illustrating the qualitatively different contributions to the time-evolution.

One can then characterise the different eras in the evolution of the universe by which of the above terms dominates, i.e. gives the leading contribution to the equation of motion for a. This already gives some insight into the physics of the situation. We will call a universe

1. *matter dominated* if C_m/a dominates
2. *radiation dominated* if C_r/a^2 dominates
3. *curvature dominated* if k dominates
4. *cosmological constant dominated* if Λa^2 dominates

Here are some immediate consequences of the Friedmann equation (F1"):

1. No matter how small C_r is, provided that it is non-zero, for sufficiently small values of a that term will dominate and one is in the radiation dominated era. In that case, one finds the characteristic behaviour

$$\dot{a}^2 = \frac{C_r}{a^2} \quad \Rightarrow \quad a(t) = (4C_r)^{1/4} t^{1/2} \ . \tag{36.17}$$

It is more informative to trade the constant C_r for the condition $a(t_0) = a_0$, which leads to

$$a(t) = a_0 (t/t_0)^{1/2} \ . \tag{36.18}$$

2. On the other hand, if C_m dominates, one has the characteristic behaviour

$$\dot{a}^2 = \frac{C_m}{a} \quad \Rightarrow \quad a(t) = (9C_m/4)^{1/3} t^{2/3} \tag{36.19}$$

or

$$a(t) = a_0 (t/t_0)^{2/3} \tag{36.20}$$

Both this and the previous example illustrate the characteristic late-time behaviour $a(t) \to \infty$ with $\dot{a}(t) \to 0$ of the $k = 0$ models.

3. For general equation of state parameter $w \neq -1$, one similarly has

$$a(t) = a_0 (t/t_0)^h \quad , \quad h = \frac{2}{3(1+w)} \quad . \tag{36.21}$$

This describes a decelerating universe ($h(h-1) < 0 \Rightarrow 0 < h < 1$) for $w > -1/3$ and an accelerating universe ($h > 1$) for $-1 < w < -1/3$. This evidently reflects, and agrees with, the fact that from (F2) one has $\ddot{a} < 0$ for $\rho + 3p > 0$ and $\ddot{a} > 0$ for $\rho + 3p < 0$.

4. For the special case $w = -1/3$, one has $h = 1$ and thus the linear evolution $a(t) \sim t$. Since, as noted in (36.38) below, one can formally attribute an equation of state parameter $w_k = -1/3$ to the curvature contribution to the Friedmann equation, this solution arises not only for an exotic matter component with $w = -1/3$ and $k = 0$, but also for an empty universe with $k = -1$. We will look at the latter (the *Milne universe*) in more detail in section 37.1 (evidently an empty universe with $k = +1$, governed by $\dot{a}^2 = -1$, is not possible).

In spite of sharing the same Friedmann equation (F1') and the same solution $a(t)$, these two universes with $w = -1/3$ are decidedly not identical for the obvious reason that one is empty and the other one is not, and thus they solve the Einstein equations with very different energy-momentum tensors (alternatively, e.g. their Misner-Sharp masses (35.118) are different). More dramatically, the scalar curvature $R(t)$ for $a(t) = t$ is

$$R(t) = \frac{6}{a^2}(a\ddot{a} + \dot{a}^2 + k) = \frac{6(1+k)}{t^2} \quad . \tag{36.22}$$

Thus there is a singularity at $t = 0$ for $k = 0$ while $R(t) = 0$ for the Milne metric with $k = -1$ (which turns out to be just Minkowski space in disguise).

Note, as an side, that one cannot conclude from the fact alone that one has $k = -1$ and the other one has $k = 0$ that they are different, since it is possible that a given universe can be foliated in different ways by spatial hypersurfaces with different curvatures. An example of this is provided by the de Sitter universe, the solution to the Friedmann equations with a positive cosmological constant (and no other matter) - see section 37.5.

5. For sufficiently large a, the cosmological constant Λ, if not identically zero, will always dominate, no matter how small the cosmological constant may be, as all the other energy-content of the universe gets more and more diluted. In particular, for $k = 0$, the Friedmann equation for a positive cosmological constant reduces to

$$\dot{a}^2 = (\Lambda/3)a^2 \quad \Rightarrow \quad a(t) = a(t_0)\mathrm{e}^{\pm\sqrt{\Lambda/3}(t-t_0)} \ , \tag{36.23}$$

with Hubble parameter

$$H(t) = \frac{\dot{a}}{a} = \pm\sqrt{\Lambda/3} = H_0 \ . \tag{36.24}$$

This gives the $k = 0$ metric of de Sitter space,

$$ds^2 = -dt^2 + \mathrm{e}^{2H_0 t}d\vec{x}^2 \ . \tag{36.25}$$

6. Only for $\Lambda = 0$ does k dominate for large a and one obtains, as we saw before, a constant expansion velocity (for $k = 0, -1$).

7. We will find and discuss various other exact solutions in section 37.

36.5 Density Parameters and the Critical Density

The primary purpose of this section is to introduce some convenient and commonly used notation and terminology in cosmology associated with the Friedmann equation (F1'). We will now include the cosmological constant in our analysis. For starters, however, let us again consider the case $\Lambda = 0$ (or include ρ_Λ as one contribution to ρ). (F1') can be written as

$$\frac{8\pi G_N \rho}{3H^2} - 1 = \frac{k}{a^2 H^2} \ . \tag{36.26}$$

If one defines the *critical density* ρ_{cr} by

$$\rho_{cr} = \frac{3H^2}{8\pi G_N} \ , \tag{36.27}$$

and the *density parameter* Ω by

$$\Omega = \frac{\rho}{\rho_{cr}} = \frac{8\pi G_N \ \rho}{3H^2} \ , \tag{36.28}$$

then (F1') becomes

$$\Omega - 1 = \frac{k}{a^2 H^2} \tag{36.29}$$

Thus the sign of k is determined by whether the actual energy density ρ in the universe is greater than, equal to, or less than the critical density,

$$\begin{array}{ccccccc} \Omega < 1 & \Leftrightarrow & \rho < \rho_{cr} & \Leftrightarrow & k = -1 & \Leftrightarrow & \text{open} \\ \Omega = 1 & \Leftrightarrow & \rho = \rho_{cr} & \Leftrightarrow & k = 0 & \Leftrightarrow & \text{flat} \\ \Omega > 1 & \Leftrightarrow & \rho > \rho_{cr} & \Leftrightarrow & k = +1 & \Leftrightarrow & \text{closed} \end{array}$$

Note that, in particular, $\rho_{cr} = 0$ at a turning point (maximal radius, $H = 0$), and correspondingly $\Omega \to \infty$ there.

This can be generalised to several species of (not mutually interacting) matter, characterised by equation of state parameters w_b, subject to the condition $w_b > 0$ or $w_b > -1/3$, with density parameters

$$\Omega_b = \frac{\rho_b}{\rho_{cr}} \ . \tag{36.30}$$

The total matter contribution Ω_M is then

$$\Omega_M = \sum_b \Omega_b \ . \tag{36.31}$$

Along the same lines we can also include the cosmological constant Λ. Inspection of the Friedmann equations reveals that the presence of a cosmological constant is equivalent to adding matter $(\rho_\Lambda, p_\Lambda)$ with

$$\Lambda \quad \Leftrightarrow \quad \rho_\Lambda = -p_\Lambda = \frac{\Lambda}{8\pi G_N} \quad w_\Lambda = -1 \ . \tag{36.32}$$

in agreement with what we had already deduced in (35.31). Note that this identification is consistent with the conservation law (F3), since Λ is constant.

Then the Friedmann equation (F1') with a cosmological constant can be written as

$$(F1') \quad \Leftrightarrow \quad \Omega_M + \Omega_\Lambda = 1 + \frac{k}{a^2 H^2} \ , \tag{36.33}$$

where

$$\Omega_\Lambda = \frac{\rho_\Lambda}{\rho_{cr}} = \frac{\Lambda}{3H^2} \ . \tag{36.34}$$

One can also formally attribute a density parameter

$$\Omega_k = -\frac{k}{a^2 H^2} \tag{36.35}$$

to the curvature, so that the Friedmann equation (F1) can now succinctly (if somewhat obscurely) be written as the condition that the sum of all density parameters be equal to 1,

$$(F1') \quad \Leftrightarrow \quad \Omega_M + \Omega_\Lambda + \Omega_k = 1 \ . \tag{36.36}$$

Note that the Friedmann equation can thus be regarded as a kind of balance equation among the contributions of matter, dark energy and (negative) curvature: if you have more of one, you need less of the other and vice versa.

The corresponding curvature energy density ρ_k can now be determined from

$$\Omega_k = \frac{8\pi G_N}{3H^2}\rho_k \stackrel{!}{=} -\frac{k}{a^2 H^2} \ , \tag{36.37}$$

and from (F2), which does not depend on k, one deduces that the corrsponding pressure is $p_k = w_k \rho_k$ with $w_k = -1/3$, so that $\rho_k + 3p_k = 0$. Thus the curvature contribution can be described as

$$k \quad \Leftrightarrow \quad \rho_k = -3p_k = \frac{-3k}{8\pi G_N a^2} \quad w_k = -1/3 \ . \tag{36.38}$$

The continuity equation (F3) is identically satisfied in this case (or, if you prefer, requires that k is constant).

Finally, the 2nd order equation (F2') can also be written in terms of the density parameters,

$$q = \tfrac{1}{2} \sum_b (1 + 3w_b)\Omega_b - \Omega_\Lambda \ . \tag{36.39}$$

Denoting the values of the parameters today, at time $t = t_0$, by a subscript 0, the two key equations relating the cosmological parameters are

$$\begin{aligned} (\Omega_M)_0 + (\Omega_\Lambda)_0 + (\Omega_k)_0 &= 1 \\ \tfrac{1}{2}(1 + 3w_0)(\Omega_M)_0 - (\Omega_\Lambda)_0 &= q_0 \end{aligned} \tag{36.40}$$

Clearly, it is of utmost importance to determine the various contributions ρ_a to the matter density ρ of the universe (and to determine ρ_{cr} e.g. by measurements of the Hubble parameter $H(t)$, i.e. of H_0, q_0 etc.). From H_0 and q_0 and $(\Omega_M)_0$ one can then in principle determine $(\Omega_\Lambda)_0$ and $(\Omega_k)_0$.

36.6 Causal Structure and Penrose Diagrams for $k = 0$

In order to gain a better understanding of a space-time, it is always useful to study its null geodesics, the behaviour of lightcones and thus its causal structure. In this section, as a first step towards this, we will shed light on the global structure of (spatially flat, $k = 0$) cosmological space-times by constructing Penrose diagrams for them. The reason for the restriction to $k = 0$ is that we already know the exact solution to the Friedmann equations in this case, and with a single matter component characterised by the equation of state parameter w, namely

$$ds^2 = -dt^2 + a(t)^2 d\vec{x}^2 \tag{36.41}$$

with (36.21)

$$k = 0 \quad , \quad p = w\rho \quad \Rightarrow \quad a(t) = a_0 (t/t_0)^h \quad , \quad h = \frac{2}{3(1+w)} \tag{36.42}$$

(for $w \neq -1$). The case of a cosmological constant ($w = -1$), i.e. (anti-)de Sitter space, needs to be treated separately. Penrose diagrams for some other ($k = +1$) solutions will be given in section 37.

One crucial feature that will emerge from this analysis is the possible presence of *cosmological horizons* which delimit the regions of space-time that can be in causal contact at a given time or that are visible at a given time or in principle in the infinite future. This will be discussed further in section 36.7 below. This will also allow us to then, in sections 36.8 and 36.9, better understand the significance (or lack thereof) of the Hubble sphere or Hubble radius $R_H(t)$ (34.40) introduced in section 34.3.

For later reference, we note that the acceleration is

$$\ddot{a}(t) = h(h-1)\frac{a(t)}{t^2} \quad , \quad h(h-1) = -\frac{2(w+1/3)}{3(1+w)^2} \quad , \tag{36.43}$$

so that

$$h(h-1) \qquad \begin{cases} < 0 \;\; \text{(decelerating)} & \text{for} \;\; 0 < h < 1 \quad w > -1/3 \\ > 0 \;\; \text{(accelerating)} & \text{for} \;\; h > 1 \qquad\;\; w < -1/3 \end{cases} \tag{36.44}$$

Now the 1st step will be the introduction of the conformal time coordinate η (cf. section 34.5) defined by

$$d\eta(t) = dt/a(t) \tag{36.45}$$

in terms of which the metric can be written as

$$ds^2 = a(\eta)^2(-d\eta^2 + d\vec{x}^2) \;\; . \tag{36.46}$$

This makes it manifest that the metric is conformally flat, and thus its causal structure is particularly easy to analyse. All we need to pay attention to is the range of η.

In the case at hand, we have

$$a(t) = a_0 t_0^{-h} t^h \quad \Rightarrow \quad d\eta = a_0^{-1} t_0^h t^{-h} dt \tag{36.47}$$

and therefore

$$\eta(t) = a_0^{-1} t_0^h t^{1-h}/(1-h) \tag{36.48}$$

for $h \neq 1$ and

$$\eta(t) = a_0^{-1} t_0^h \log|t| \tag{36.49}$$

for $h = 1, w = -1/3$. Since the range of t is $(0, \infty)$, the range of η is

$$t \in (0, +\infty) \quad \Rightarrow \quad \begin{cases} \eta \in (0, +\infty) & \text{for} \;\; 0 < h < 1 \\ \eta \in (-\infty, +\infty) & \text{for} \;\; h = 1 \\ \eta \in (-\infty, 0) & \text{for} \;\; h > 1 \;\; . \end{cases} \tag{36.50}$$

This shows that the causal structure and Penrose diagram of the space-time for $h = 1, w = -1/3$ are identical to that of Minkowski space in Figure 23, with the only crucial difference that there is now a singularity at i^-, i.e. at $t \to 0, \eta \to -\infty$ (recall that the scalar curvature is $R(t) \sim t^{-2}$ (36.22)).

For $h \neq 1$, on the other hand, as indicated in Figure 48, the Penrose diagram is given by the upper (lower) half of the Minkowski Penrose diagram respectively, with the addition of the initial spacelike singularity. This has drastic implications for the global causal structure (and the existence of cosmological horizons) to be explored below.

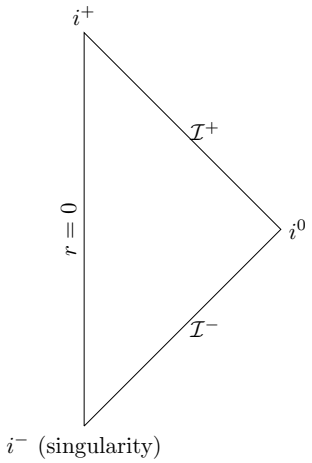


Figure 47: Penrose Diagram for the $k = 0, w = -1/3$ solution.

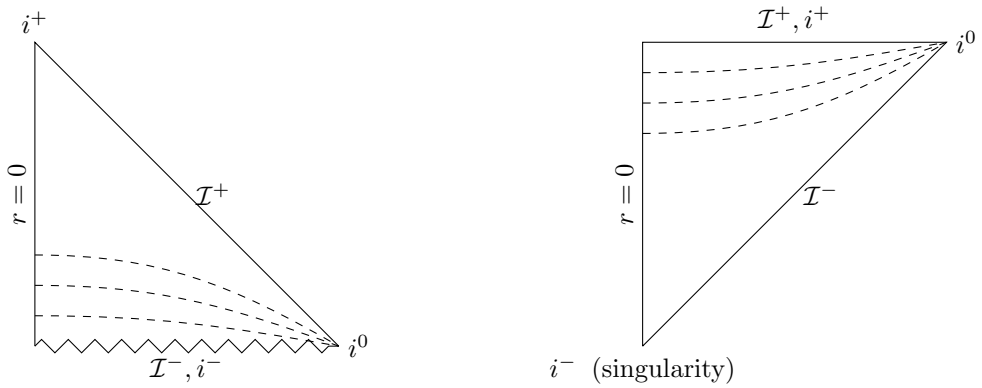


Figure 48: Penrose Diagram for the $k = 0, w \neq -1/3, -1$ solutions. On the left, the diagram for decelerating solutions with $0 < h < 1$, $w > -1/3$, on the right that for acelerating solutions with $h > 1$. Also schematically indicated (as dashed lines) are some surfaces of constant time t or η.

36.7 Lightrays and Cosmological Particle and Event Horizons

We now turn to one particularly relevant implication of the preceding analysis, namely the existence of (observer-dependent) *horizons* in cosmological space-times. Thus we will mainly consider the spatially flat case $k = 0$ in the following, but in principle everything can easily and immediately be extended to $k \neq 0$, e.g. by working in the polar coordinates (34.10), with $r \to \psi$. In the following, the term "comoving distance" will refer to the coordinate distance as measured by the comoving radial coordinate r.

Let us first (re-)analyse the evolution of lightrays in the Robertson-Walker geometry. We have already studied one aspect of light propagation in Robertson-Walker geometries in section 34.8, namely the cosmological redshift. Here we will look at the propagation

of light from a more global perspective (or at least with more global intentions).

By spatial isotropy, we can without loss of generality choose the lightrays to be purely radial. Then they are governed by the equation

$$dt^2 = a(t)^2 dr^2 \quad \Leftrightarrow \quad \frac{dr_\gamma(t)}{dt} = \pm\frac{1}{a(t)} \ . \tag{36.51}$$

Here, in order to avoid a proliferation of objects with the same anonymous name r, we call the comoving photon path $r_\gamma(t)$. Since $a(t)$ is non-negative, lightrays propagate either in the direction of increasing comoving radial coordinate distance r, or in the direction of decreasing r. In particular, lightrays coming towards us (at $r = 0$) and reaching $r = 0$ at $t = t_0$ are governed by the equation

$$\frac{dr_\gamma(t)}{dt} = -\frac{1}{a(t)} \quad \Rightarrow \quad r_\gamma(t) = \int_t^{t_0} \frac{dt}{a(t)} \ . \tag{36.52}$$

We will occasionally also denote this solution by

$$r_\gamma(t) \equiv r_\gamma(t; t_0) \quad \text{with} \quad r_\gamma(t_0, t_0) = 0 \ , \tag{36.53}$$

when we want to make the dependence of the solution on t_0 more explicit. We see that this solution can be directly expressed in terms of conformal time η as

$$r_\gamma(t; t_0) = \eta(t_0) - \eta(t) \ . \tag{36.54}$$

One can also, if one wishes, express light propagation in terms of the (instantaneous) proper spacelike distance

$$R_\gamma(t) = a(t) r_\gamma(t) \ , \tag{36.55}$$

i.e. the spatial proper distance to the photon as measured by the maximally symmetric spatial geometry on the slice of constant time t but, as we will see in sections 36.8 and 36.9, this is not without its pitfalls.

As an example, for matter with the equation of state parameter $w \neq -1$ and cosmic scale factor $a(t) = a_0(t/t_0)^h$ (36.21), the equation for lightrays is

$$dr_\gamma(t)/dt = -t_0^h t^{-h}/a_0 \quad , \quad h = \frac{2}{3(1+w)} \ . \tag{36.56}$$

Either from this, or from the result for $\eta(t)$ from (36.48), one finds that for $h \neq 1$

$$r_\gamma(t) = \frac{t_0}{a_0(1-h)}(1 - (t/t_0)^{1-h}) \ . \tag{36.57}$$

and

$$R_\gamma(t) = \frac{1}{1-h}(t_0^{1-h} t^h - t) \tag{36.58}$$

(while for $h = 1$ one finds a logarithmic behaviour). We now look at 2 limiting cases of these lightrays.

1. Cosmological Particle Horizon

Backtracking lightrays that reach us today, the largest comoving distance of an object about which we can have received any (causal) information by today or that can have had any causal influence on us is given by

$$r_{ph}(t_0) = \lim_{t\to 0_+} r_\gamma(t;t_0) = \eta(t_0) - \eta(0) \quad . \tag{36.59}$$

(with $t = 0$ the time of the initial singularity). If this integral converges, this means that there is a region of the universe, namely that at comoving distance $r > r_{ph}(t_0)$, with which we could not have had any causal contact until today. In that case $r_{ph}(t_0)$ defines the *cosmological particle horizon* (or just *particle horizon*).

Expressed in terms of conformal time, whether or not there is a particle horizon is equivalent to the question if $\eta(t)$ is finite or diverges as $t \to 0$. From the analysis of the previous section for the single-component models with equation of state $p = w\rho$, $w \neq -1$, we can thus conclude that

- for the non-decelerating models, $h \geq 1$, one has $\eta(t) \to -\infty$, and therefore the space-time is causally equivalent to Minkowski space in the past, and there is no particle horizon;
- for the decelerating models with $0 < h < 1$, $w > -1/3$, or those that are dominated by ordinary matter or radiation at early times, say, $\eta(0)$ is finite, the causal structure is as in the left-hand diagram of Figure 48 and there is a particle horizon.

Remarks:

(a) It is also common to express the particle horizon in terms of

$$R_{ph}(t_0) = a(t_0) r_{ph}(t_0) = a(t_0) \int_0^{t_0} \frac{dt}{a(t)} \quad , \tag{36.60}$$

giving the present instantaneous proper distance to the particle horizon. At any time t_0, the particle horizon is a spatial 2-sphere with radius $R_{ph}(t_0)$ around an observer, no information being available about objects outside that 2-sphere. This particle horizon is observer-dependent, i.e. other comoving observers at other values of r will also have a particle horizon, but it will be a 2-sphere of radius $R_{ph}(t_0)$ around them.

(b) As (36.59) shows, $r_{ph}(t_0)$ grows monotonically with t_0 (so that more and more of the universe becomes visible as time goes on). A good way of visualising the particle horizon is to draw $r_{ph}(t)$ as a function of t in a (t,r)- or (η,r)-diagram.[155] For a comoving observer at $r = r_0$, this is simply the future

[155] Note that often the particle horizon is depicted in a different and somewhat less informative, or at

lightcone of the point $(0, r_0)$, and the particle horizon size at any given time t_0 is given by the intersection of this lightcone null surface with the spacelike constant $t = t_0$ surface.

(c) This is indicated in a Penrose diagram in Figure 49. It is evident from this diagram that the past lightcone of an event at some finite time $\eta > \eta(0)$ cannot cover the entire space, and that at early times suffficiently spatially separated events have non-intersecting past lightcones, i.e. have never been in causal contact in the past. We will come back to one problematic aspect and consequence of this in section 38.3 in our brief discussion of the (aptly) so-called horizon problem in cosmology.

In interpreting this diagram, it is also good to keep in mind the highly distorted nature of constant time hypersurfaces at late times, which all end up at i^0. Thus the particle horizon really reaches infinity $\mathcal{I}^+$ at $t \to \infty$ or $\eta \to +\infty$.

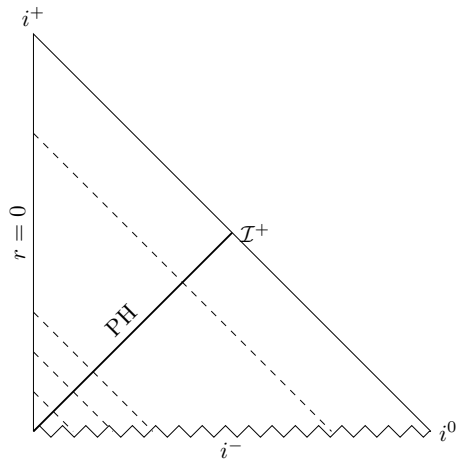


Figure 49: Cosmological Particle Horizon: indicated are the past lightcones of a comoving observer at $r = 0$ (dashed lines) as well as that observer's particle horizon (thick solid line).

This diagram also shows that the particle horizon can be regarded as the boundary of the region that can be influenced by the observer.

(d) One can of course also define these horizons for the models with $k \neq 0$. The case $k = +1$ is different in the sense that space is compact. E.g. the range of ψ in

$$ds^2 = a(\eta)^2(-d\eta^2 + d\psi^2 + \sin^2\psi d\Omega^2) \tag{36.61}$$

least potentially more confusing, way - see Figures 1 and 3 and section 3.4 of T. Davis, C. Lineweaver, *Expanding Confusion: common misconceptions of cosmological horizons and the superluminal expansion of the Universe*, `arXiv:astro-ph/0310808` for a nice discussion and illustration of this.

is finite, so that even if there is a particle horizon initially, it may be possible to see all of space (and later on all around the space at the back of one's head) at a later time, namely when the η-life-time of the universe is $\geq \pi$ (respectively $\geq 2\pi$). An example of this is provided by the matter dominated $k = +1$ universe, whose conformal diagram is given in Figure 51 of section 37.3.

(e) The particle horizon is a horizon in the sense that it is null and acts as a one-way membrane: any object that is inside the particle horizon or has crossed the particle horizon towards us will remain inside the particle horizon at all future times. This is also evident from the diagram.

2. Cosmological Event Horizon

From the above discussion it is also clear that the cosmological space-time is causally inequivalent to Minkowski space in the far future if η is bounded from above as $t \to \infty$. If it is bounded, then the past lightcone of an observer will never be able to cover all of space-time even as $t \to \infty$, i.e. there will be regions of space-time from which that observer can never ever receive any information, and in this case the past lightcone is known as a *cosmological event horizon*.

The radial coordinate size of the past lightcone from $t_0 = \infty$ at some time t is given by the future partner of (36.59), namely

$$r_{eh}(t) = \lim_{t_0 \to \infty} r_\gamma(t; t_0) = \eta(\infty) - \eta(t) \ , \tag{36.62}$$

and there is an event horizon iff this integral is finite, i.e. iff $\eta(\infty)$ is finite.

Going back to our $k = 0$ single-component matter example, it is now precisely the accelerating cosmologies with $w < -1/3$ that exhibit an event horizon. This is iluustrated in Figure 50.

REMARKS:

(a) Since current observations suggest that our universe will be dominated by a positive cosmological constant in the future, i.e. that it will be asymptotically de Sitter in the future, while it was dominated by standard types of matter in the past, the standard hot big bang / Λ-CDM model of our universe has both a particle horizon and an event horizon.

(b) This cosmological event horizon (or the issue whether or not one exists) is not particularly relevant for observational cosmology today (or in the foreseeable future), but it is of great theoretical interest since, at least in the (asymptotically) de Sitter case, in some respects this (again observer-dependent) horizon appears to have more in common with a true black hole horizon

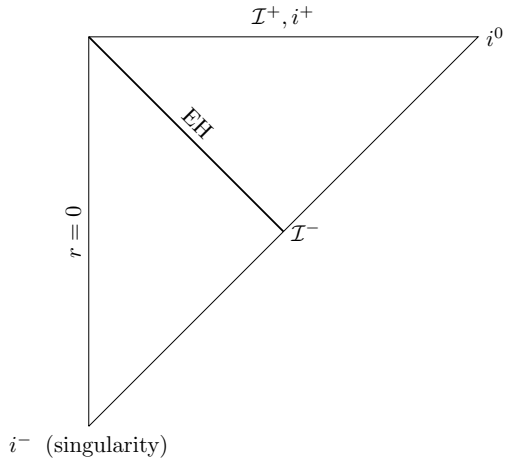


Figure 50: Cosmological Event Horizon: the region outside the event horizon (above the line EH) is invisible to a comoving observer at $r = 0$.

(like a temperature and an entropy) than one would have had the right to expect.[156]

(c) The event horizon is a horizon in the sense that it is null and acts as a one-way membrane: any object that is outside the event horizon or has crossed the event horizon away from us will remain outside the event horizon at all future times. Given that the event horizon is the past lightcone at a point at future infinity, these statements are almost a tautology, and are also evident from the diagram.

36.8 Lightrays and the Hubble Sphere

Recall that the Hubble radius $R_H(t)$ was introduced in section 34.3 as the (instantaneous) proper distance at which the recessional velocity (34.29)

$$V_{rec}(t) \equiv \frac{d}{dt} R_p(t) = H(t) R_p(t) \tag{36.63}$$

of a comoving object with $R_p(t) = a(t) r_1$ equals the speed of light,

$$V_{rec}(t) = c \quad \Leftrightarrow \quad R_H(t) = c/H(t) = ca(t)/\dot{a}(t) \tag{36.64}$$

For a contracting universe this should of course be replaced by $R_H = -c/H = c/|H|$, but until further notice we will consider the expanding case in the following (and we will from now on set $c = 1$ again).

[156] See e.g. M. Spradlin, A. Strominger, A. Volovich, *Les Houches Lectures on de Sitter Space*, `arXiv:hep-th/0110007` and references therein.

I also mentioned in section 34.3 that the Hubble sphere is occasionally also referred to as the Hubble horizon. Let us see if, or to which extent, it has such a role and how it is related to the cosmological particle and event horizons discussed in the previous section.

The most direct connection between the Hubble radius and the propgation of light arises from describing a photon path via its proper distance $R_\gamma(t)$ (36.55). This proper distance of an (out- or ingoing) lightray changes with time according to

$$\frac{d}{dt}R_\gamma(t) = \dot{a}(t)r_\gamma(t) \pm 1 = \frac{R_\gamma(t)}{R_H(t)} \pm 1 \ . \tag{36.65}$$

Thus expressed (or plotted) in terms of proper distance, an ingoing lightray exhibits (or appears to exhibit) quite a different behaviour from the strictly monotonous $\dot{r}_\gamma < 0$ evolution of $r_\gamma(t)$. In particular, for such ingoing lightrays the proper distance reaches a maximum at a time $t = t_m$ when

$$r_\gamma(t_m) = 1/\dot{a}(t_m) = r_H(t_m) \quad \Leftrightarrow \quad R_\gamma(t_m) = R_H(t_m) \ , \tag{36.66}$$

i.e. when the lightray crosses the Hubble sphere, and then decreases towards $r = 0$. Here

$$r_H(t) = R_H(t)/a(t) = 1/\dot{a}(t) \ . \tag{36.67}$$

is the time-dependent comoving radial coordinate of the Hubble radius. This behaviour can intuitively (but with caution) be attributed to the fact that at the time when the lightray crosses the Hubble radius $R_H(t)$ it has recessional velocity (i.e. away from us) equal to c and is thus momentarily at rest with respect to us.

Before discussing how (not) to interpret this result, let us determine how the Hubble radius evolves with time. The evolution equation for $R_H(t)$ is

$$\frac{d}{dt}R_H(t) = \frac{d}{dt}(a(t)/\dot{a}(t)) = 1 + q(t) \ , \tag{36.68}$$

while that for $r_H(t)$ is

$$\frac{d}{dt}r_H(t) = \frac{d}{dt}(1/\dot{a}(t)) = q(t)/a(t) \ . \tag{36.69}$$

Using (36.39) in the schematic form

$$q(t) = \tfrac{1}{2}(1 + 3w)\Omega_M - \Omega_\Lambda \tag{36.70}$$

and

$$k = 0 \quad \Rightarrow \quad \Omega_M + \Omega_\Lambda = 1 \ , \tag{36.71}$$

these equations can be written as

$$\begin{aligned} \frac{d}{dt}R_H(t) &= \tfrac{3}{2}(1 + w)\Omega_M \\ a(t)\frac{d}{dt}r_H(t) &= \tfrac{3}{2}(1 + w)\Omega_M - 1 \ . \end{aligned} \tag{36.72}$$

In particular, if one has a single species of matter (or just a cosmological constant), i.e. $\Omega_M = 1$ (or $\Omega_M = 0$) one has

$$q = \tfrac{1}{2}(1+3w) \ , \tag{36.73}$$

which is constant, and

$$\begin{aligned} \frac{d}{dt}R_H(t) &= \tfrac{3}{2}(1+w) \\ a(t)\frac{d}{dt}r_H(t) &= \tfrac{1}{2}(1+3w) \ . \end{aligned} \tag{36.74}$$

This implies that the Hubble radius increases monotonically in time for

$$\begin{aligned} w > -1 \quad &\Rightarrow \quad \frac{d}{dt}R_H(t) > 0 \\ w > -\frac{1}{3} \quad &\Rightarrow \quad \frac{d}{dt}r_H(t) > 0 \ . \end{aligned} \tag{36.75}$$

Evidently for $w > -1$ and $a(t) \sim t^h$ the solution for $R_H(t)$ reproduces the definition $R_H = a/\dot{a}$,

$$w > -1 \quad \Rightarrow \quad R_H(t) = \tfrac{3}{2}(1+w)t = t/h \ , \tag{36.76}$$

while for $w = -1$,

$$w = -1 \quad \Rightarrow \quad R_H(t) = R_H(t_0) = 1/H_0 \ . \tag{36.77}$$

With these results at our disposal, we can now make the following elementary observations:

1. Since $R_H(t)$ is non-decreasing for $w \geq -1$, it is a triviality that the maximal proper distance $R_\gamma(t_m) = R_H(t_m)$ a lightray reached in the past that arrives today is bounded from above by the value of the Hubble radius today,

   $$R_\gamma(t_m) = R_H(t_m) < R_H(t_0) \ . \tag{36.78}$$

   (It is also easy to sharpen this upper bound somewhat for $w > -1/3$, but we will not need this).

2. Thus we see that the objects whose light we receive today at $r = 0$ cannot be further away from us today than at a proper distance $R_H(t_0)$ so that the Hubble radius $R_H(t_0)$ provides some kind of (rough) upper bound on the distance of such objects. This may suggest (to some) that therefore the Hubble radius provides a limit to what we can see (or can have seen) of the universe at time t_0.[157] However, this appears to me to be at best an extremely misleading way of phrasing things.

[157] See e.g. F. Melia, *The Cosmic Horizon*, `arXiv:0711.4181 [astro-ph]`, or the plots and numerical calculations performed in O. Bikwa, F. Melia, A. Shevchuk, *Photon Geodesics in FRW Cosmologies*, `arXiv:1112.4774 [astro-ph.CO]` in order to confirm (36.78) (which requires no numerical confirmation) in some examples.

In fact, even though $R_\gamma(t)$ reaches a maximum at $t = t_m$, this is not the case for the comoving radial coordinate $r_\gamma(t)$ along the photon path. The monotonous behaviour $\dot{r}_\gamma < 0$ for ingoing lightrays implies that

$$t < t_m \quad \Rightarrow \quad r_\gamma(t) > r_\gamma(t_m) \tag{36.79}$$

so that information about objects at larger distances $r > r_H(t_m)$ than the Hubble radius can easily reach us. This distance is bounded only by the *particle horizon* $r_{ph}(t)$ which is what one gets when one tracks $r_\gamma(t)$ back to $t = 0$. Therefore it is precisely the particle horizon which tells us about which points / comoving objects in the constant time spatial surface we can already have obtained information, and not the Hubble radius.

3. The metric induced on the Hubble surface Σ_H defined by $r = r_H(t)$ is

$$\begin{aligned} ds^2|_{\Sigma_H} &= -dt^2 + a(t)^2(dr_H(t)^2 + r_H(t)^2 d\Omega^2) \\ &= -(1 - q(t)^2)dt^2 + R_H(t)^2 d\Omega^2 \ . \end{aligned} \tag{36.80}$$

Therefore this surface Σ_H is null iff $q = \pm 1$. In these two cases,

$$q = -1 \quad \Rightarrow \quad w = -1 \quad , \quad q = +1 \quad \Rightarrow \quad w = +\frac{1}{3} \ , \tag{36.81}$$

(i.e. a pure positive cosmological constant or radiation) the Hubble radius agrees with the event horizon or the particle horizon respectively,

$$\begin{aligned} w = -1 \quad &\Rightarrow \quad R_H(t) = R_{eh}(t) = R_H(t_0) \\ w = +\frac{1}{3} \quad &\Rightarrow \quad R_H(t) = R_{ph}(t) \ . \end{aligned} \tag{36.82}$$

In a universe which is dominated by radiation in the distant past and a cosmological constant in the far future, the Hubble radius will interpolate between the particle hoirzon in the past and the event horizon in the future, but it has no particular significance inbetween.

4. Indeed, more generally one has

$$\Sigma_H \quad \text{is} \quad \begin{cases} \text{spacelike} & \text{for} & w > 1/3 \\ \text{null} & \text{for} & w = 1/3 \\ \text{timelike} & \text{for} & -1 < w < 1/3 \\ \text{null} & \text{for} & w = -1 \\ \text{spacelike} & \text{for} & w < -1 \end{cases} \tag{36.83}$$

Thus for the physically relevant intermediate range $-1 < w < 1/3$ the Hubble surface is timelike, causally there is nothing strange or interesting going on there, and nothing prevents one from crossing it mulitple times in both directions. In particular it cannot and will not act like a one-way membrane.

36.9 Hubble "Horizon": Is our Universe (like) a Black Hole?

Hinchcliff's rule:
Whenever the title of a paper (section) is a question with a yes/no answer, the answer is "no".

In the previous section we collected some elementary properties of the Hubble radius (Hubble sphere). In particular, we saw that the Hubble sphere is only null in 2 special cases, in which it coincides with the particle or event horizon respectively.

Nevertheless, in spite of these elementary and well-known facts, strangely there is some debate in the current literature about the significance of the Hubble "horizon" (or lack thereof), and even the bizzarre idea that the "visible universe" (defined by $R_H(t)$) is somehow like the inside of a black hole.

As far as I can tell one of the contributing factors to this is the fact that in cosmology, distances to distant objects are commonly expressed in terms of their (somewhat fictitious) instantaneous proper spacelike distance $R(t_0) = a(t_0)r$ from us today, and not (for instance) in terms of their (approximately constant) comoving coordinate distance, or some measure of distance at the time the objects emitted the light that we receive today. In principle, this is perfectly fine, of course, and with due care myths about the Hubble radius can also be exorcised from this point of view.[158]

In practice, however, this use of $R(t)$ leads to a somewhat allegorical (and therefore potentially misleading) way of talking about perfectly mundane things:

- For example, as we have seen above in our discussion of lightrays, one can say (referring to $R_\gamma(t)$) that lightrays reach a maximum distance and then turn around to come towards us, but viewed in terms of comoving coordinates the lightrays just continue in the direction of decreasing r. Thus care needs to be taken to separate the physical motion of objects through space from artefacts arising from describing them in terms of their instantaneous proper spacelike distance.

- Let us consider the spatially flat decelerating case $k = 0$ and $w > -1/3$, so that one has $a(t) \sim t^h$ for some $0 < h < 1$. Then

$$a(t) \sim t^h \quad \Rightarrow \quad R_H(t) = t/h \ . \tag{36.84}$$

  In this setting one frequently encounters the following kind of argument to "explain" from this point of view why light can reach us from outside the Hubble sphere: for $0 < h < 1$ the Hubble sphere expands faster than the universe; thus

[158] See e.g. P. van Oirschot, J. Kwan, G. Lewis, *Through the Looking Glass: Why the "Cosmic Horizon" is not a horizon*, `arXiv:1001.4795 [astro-ph.CO]`, G. Lewis, P. van Oirschot, *How does the Hubble Sphere limit our view of the Universe?*, `arXiv:1203.0032` for illuminating discussions and enjoyable dissections of these issues from this perspective.

light emitted from a receding galaxy initially outside the Hubble sphere can eventually be overtaken by the Hubble radius and can then become visible to us. True, and very figurative, in terms of some radius overtaking some other radius, but this totally obscures the fact that the Hubble sphere has nothing to do with this and that the only thing that matters for what is visible to us today is if the object is inside the particle horizon or not, i.e. has comoving coordinate $r < r_{ph}(t_0)$ or not.

The above argument also seems to suggest that somehow things change when one has an accelerating universe with $h > 1$, and this is again true, but the thing that changes is that (as we have seen) for $h > 1$ there is simply no particle horizon and therefore no obstruction to seeing objects at any time. This simple fact is again obscured by the above argument.

Nevertheless, as I mentioned already in section 34.3, some people (and not just laymen who understand neither cosmology nor black holes) appear to be obsessed with the idea that our visible universe, defined (counterfactually, as we saw above) as the interior of the Hubble sphere, is somehow like the inside of a Schwarzschild black hole, the Hubble sphere playing the role of its horizon.

This is usually based on some variant of one of the following arguments.[159]

1. Friedmann Equation and the Schwarzschild Radius

   The Friedmann equation
   $$H^2 = \frac{8\pi G_N}{3}\rho - \frac{k}{a^2} \tag{36.85}$$
   can (by dividing by H^3) be written as
   $$R_H = 2G_N \frac{4\pi R_H^3}{3}\rho - \frac{k}{a^2 H^3} \quad . \tag{36.86}$$
   Therefore the statement that $k = 0$ is equivalent to the statement that the Hubble radius R_H is equal to the (would-be) Schwarzschild radius $R_s = 2G_N M$ associated with the total mass
   $$M(R_H) = \frac{4\pi R_H^3}{3}\rho \tag{36.87}$$
   contained in the Hubble sphere,
   $$k = 0 \quad \Leftrightarrow \quad R_H = R_s = 2G_N\, M(R_H) \quad . \tag{36.88}$$
   There may be something profound in this, I don't know, but just saying "hey, it's the Schwarzschild radius, hence I have a black hole" is not!

[159] See e.g. F. Melia (footnote 157) or F. Melia, M. Abdelquader, *The Cosmological Spacetime*, `arXiv:0907.5394 [astro-ph.CO]`.

- In particular, as we have seen above, for reasonable equations of state the "horizon" $r = r_H$ is timelike and we (and lightrays) can simply leave this would-be interior of the black hole. The region we cannot in principle escape from is that bounded by the particle horizon, but the above numerology does not work in that case (unless $w = 1/3$, so that $R_{ph} = R_H$).
- Moreover, as discussed at length in section 32, a black hole is not defined by "mass m enclosed in a sphere of radius $r < 2m$" but by reference to the outside of the black hole and the existence of an *event horizon* and the fact that there is a region of space-time causally sealed off from the asymptotic region.
- Other quasi-local characterisations of black hole-like objects also do not refer directly to mass and radius but to trapped surfaces and the behaviour of lightrays (and more on this below).
- In particular, we will see below that for $k = 0$ (but only for $k = 0$), the Hubble "horizon" is an apparent horizon, but one with properties that are quite different to those of the apparent horizon of a black hole.[160]
- Last but not least, the interior of a black hole exhibits a future spacelike singularity while our universe appears to have emerged from an initial (past) spacelike singularity (and, no, saying "I meant *white* hole, not black hole" will not help, see the discussion below).

2. Writing the FRW Metric in Schwarzschild-like Form

   In section 34.4 it was shown that the metric can be written in a form which resembles that of the Schwarzschild metric in Painlevé-Gullstrand-coordinates. Using the current notation ($R = a(t)r$ was called $\tilde{r}$ in section 34.4 and we use $R_H = 1/H$) we can write the general Robertson-Walker metric as (34.43)

$$\begin{aligned} ds^2 &= -(1 - R^2H(t)^2)dt^2 - 2RH(t)dtdR + (dR^2 + R^2d\Omega^2) \\ &= -(1 - (R/R_H)^2)dt^2 - 2(R/R_H)dtdR + (dR^2 + R^2d\Omega^2) \\ &= -\Phi(dt + (R/\Phi R_H)dR)^2 + \Phi^{-1}dR^2 + R^2d\Omega^2 \quad , \end{aligned} \tag{36.89}$$

   where

$$\Phi(t, R) = 1 - (R/R_H(t))^2 \quad . \tag{36.90}$$

   This form of the metric shows that there is an infinite time-dilation between the proper time of an observer at constant $R = R_0$ as $R_0 \to R_H$ and the cosmological time t, vaguely suggestive of the same phenomenon in the Schwarzschild metric. However, this is clearly a problem of the observers moving at constant $R(t) =$

[160] Moreover, there is nothing new about the observation (36.88) regarding the relation between the Hubble radius and the Schwarzschild-like radius R_s. It is e.g. already pointed out in section 10.1 of S. Hawking and G. Ellis, *The large scale structure of space-time*, that the numerology leading to (36.88) provides an "intuitive way of viewing this result".

$a(t)r$, not of the space-time itself. These observers at constant R are not the standard comoving observers (at constant r). Their 4-velocity has the form

$$u^t = \Phi(t,R_0)^{-1/2} \quad , \quad u^R = 0 \tag{36.91}$$

in the above (t,R)-coordinates, and

$$u^t = \Phi(t,R_0)^{-1/2} \quad , \quad u^r = \Phi(t,R_0)^{-1/2}(-R_0H(t)/a(t)) \tag{36.92}$$

in the standard comoving (t,r) coordinates, with

$$\Phi(t,R_0) = 1 - R_0^2H(t)^2 \ . \tag{36.93}$$

so that $u^t \to \infty$ for $R_0 \to R_H = a(t)/\dot{a}(t)$, while comoving observers have $u^t = 1$. Calculating the acceleration of these observers (in comoving coordinates, say), one finds

$$a^\alpha = u^\beta\nabla_\beta u^\alpha = \Phi^{-2}(q + H^2R_0^2)(R_0H^2)(-R_0H, 1/a) \tag{36.94}$$

with norm

$$g_{\alpha\beta}a^\alpha a^\beta = \Phi^{-3}(q + H^2R_0^2)^2(R_0H^2)^2 \ . \tag{36.95}$$

Clearly this diverges as $R_0 \to R_H = 1/H$, and therefore this is exactly like the space-time seen by ∞ acceleration observers in a Rindler space-time who detect a fictitious Rindler horizon.

Thus R_H is not like an event horizon and there is nothing like a black hole in sight. In fact, the best and most informative way of saying what $R_H(t)$ is and what its significance is (or is not) is that (for $k = 0$) $R_H(t)$ is a spherically symmetric marginally trapped tube (MTT) foliated by spherically symmetric marginally trapped surfaces (MTSs). In other words, with respect to a spherically symmetric foliation of space-time, for $k = 0$ (but only for $k = 0$) the surface $r = r_H(t)$ is an *apparent (3-)horizon* (cf. the discussion in section 32.9). Moreover, as the hypersurface $r = r_H(t)$ is timelike in the range $-1 < w < 1/3$, the horizon-terminology is really not very appropriate and, as mentioned in section 32.9, in current terminology such a hypersurface is referred to as a *timelike membrane*.

Concretely, and in elementary terms, this means the following. Consider (for any k) the equation

$$a(t)\frac{dr}{\sqrt{1-kr^2}} = \pm dt \tag{36.96}$$

for out- respectively ingoing lightrays, and write this in terms of the area radius $R(t) = a(t)r$,

$$dR_\pm = \sqrt{1-kR^2/a^2}\left(\pm 1 + RH/\sqrt{1-kR^2/a^2}\right)dt \ . \tag{36.97}$$

This can of course also be deduced directly from the PG-like metric (36.89) and its $k \neq 0$ counterpart (34.46).

For sufficiently small values of R,

$$R|H|/\sqrt{1-kR^2/a^2} < 1 \quad \Leftrightarrow \quad R(t) < \frac{1}{\sqrt{H(t)^2 + k/a(t)^2}} \equiv R_{ah}(t) \ , \tag{36.98}$$

this exhibits the standard behaviour

$$R(t) < R_{ah}(t) \quad \Rightarrow \quad \begin{cases} dR_+/dt > 0 \\ dR_-/dt < 0 \end{cases} \tag{36.99}$$

that one solution, R_+, propagates to larger values of R, while the other, R_-, propagates to smaller values of R. Evidently something special happens at $R = R_{ah}$ and for larger values one has

$$R(t) > R_{ah}(t) \quad \Rightarrow \quad \begin{cases} H(t) > 0: & dR_+/dt > 0 \quad \text{and} \quad dR_-(t)/dt > 0 \\ H(t) < 0: & dR_+/dt < 0 \quad \text{and} \quad dR_-(t)/dt < 0 \end{cases} \tag{36.100}$$

Thus, for a contracting universe this means that the spheres of constant R and t are *trapped* for $R > R_{ah}(t)$, like the 2-spheres inside the event horizon of the Schwarzschild black hole, with both ingoing and (would-be) outgoing radial lightrays moving towards smaller radii. However, far from indicating the presence of a black hole in this case, these trapped spheres and the apparent horizon $R_{ah}(t)$ indicate (together with an energy condition) a future cosmological singularity.

In an expanding universe, on the other hand, for $R > R_{ah}$ both in- and outgoing lightrays move to larger values of R. The spheres with $R > R_{ah}$ are thus the opposite of trapped surfaces, i.e. anti-trapped or trapped towards the past (and reflect the existence of a big bang singularity in the past).

The fact that both in- and outgoing expansions are positive is also a characteristic feature of the (unphysical) "white hole" region of a black hole in the region before the past event horizon. However, in that case it is the region around $r = 0$, and with $r < 2m$, that contains the anti-trapped surfaces while sufficiently large spheres show normal behaviour. Here, on the contrary, it is sufficiently large spheres around the comoving observer that are anti-trapped while in a sufficiently small region around any cmoving observer there are no (anti-)trapped surfaces.

Noting that for $k = 0$ the apparent horizon is equal to the Hubble radius,

$$k = 0 \quad \Rightarrow \quad R_{ah}(t) = R_H(t) \ , \tag{36.101}$$

all of this is simply a restatement of the discussion about the behaviour of lightrays in the previous section, and nothing seems to be gained by coining a new name for this well-established concept of an apparent horizon. Moreover, for $k \neq 0$ the Hubble sphere is not even an apparent horizon or some other marginally trapped tube, so it seems best not to associate the word "Hubble" with the word "horizon" at all.

37 Cosmology V: Some Exact Solutions

We have seen that a lot can be learnt about the Friedmann-Lemaître-Robertson-Walker models without ever having to solve a differential equation. On the other hand, more precise information can be obtained by specifying an equation of state for the matter content and solving explicitly the Friedmann equations. We have already seen some special exact solutions ($k = \Lambda = 0$ and only one species of matter) in section 36.4. Solutions with $k \neq 0$ but $\Lambda = 0$ are also easy to obtain. Together with the exact solutions for just $k \neq 0$ and $\Lambda \neq 0$, but with no other matter, these are the most useful solutions in practice, but it is also possible to solve the equations explicitly in some other cases.[161]

37.1 Milne Universe

Let us start our excursion into exact solutions by looking at the totally unphysical case of a completely empty universe, i.e. $\rho = p = \Lambda = 0$, and only k possibly not zero. As trivial as this may be, it has its pedagogical value, which is why I am including this case here.

In this case (F1) reduces to

$$\dot{a}^2 + k = 0 \quad \Rightarrow \quad k \leq 0 \ . \tag{37.1}$$

For $k = 0$, one has $\dot{a} = 0$ and thus $a(t) = a_0$, and the metric is

$$ds^2 = -dt^2 + a_0^2 d\vec{x}^2 \ . \tag{37.2}$$

It should not come as a suprise that the metric we have found is just that of Minkowski space (the constant a_0 can be absorbed into a rescaling of the spatial coordinates).

The case $k = -1$ is a bit more interesting. In that case the equation to solve is (I will call the time-coordinate τ instead of t for reasons that will perhaps become apparent below)

$$\dot{a}^2 = +1 \quad \Rightarrow \quad a(\tau) = \pm(\tau - \tau_0) \ . \tag{37.3}$$

and the resulting space-time metric is (choosing without loss of generality $\tau_0 = 0$)

$$ds^2 = -d\tau^2 + \tau^2 \, d\tilde{\Omega}_3^2 \tag{37.4}$$

where $d\tilde{\Omega}_3^2$ is the line-element on the unit 3-hyperboloid (14.33).

This appears to describe a non-trivial universe (known as the *Milne universe*) with either (for $a(\tau) = +\tau$) a big bang at $\tau = 0$ and with the universe subsequently expanding

[161] For explicit solutions of the Friedmann equations for $\Lambda \neq 0$ and non-trivial matter content (generalising the solution (38.26)) see e.g. B. Aldrovandi, R. Cuzinatto, L. Medeiros, *Analytic solutions for the Λ-FRW Model*, `arXiv:gr-qc/0508073`.

linearly with $\tau > 0$, or (for $a(\tau) = -\tau$) with a linearly contracting universe for $\tau < 0$, ending in a "big crunch" at $\tau = 0$, and all this in spite of the fact that the universe is empty.

This is deceptive, however (not that it is empty, but that it is non-trivial). As always, we need to be careful to disentangle coordinate artefacts from genuine geometrical statements. Indeed, this space-time is again nothing other than (a part of) Minkowski space-time. We had already anticipated this in section 35.1, in connection with (35.9) which expresses the Riemann tensor of a Robertson-Walker metric in terms of its Ricci tensor (and which therefore implies that a vacuum solution will necessarily be flat).

Nevertheless, it will be instructive to see this explicitly. To that end we start with the Minkowski metric

$$ds^2 = -dt^2 + d\vec{x}^2 \tag{37.5}$$

and introduce coordinates that are adapted to the family of space-like hyperboloids

$$t^2 - \vec{x}^2 = \tau^2 > 0 \ . \tag{37.6}$$

For $t > 0$ these hyperboloids fill (foliate) the interior of the future lightcone at the origin (and the interior of the past lightcone for $t < 0$), and if you draw these hyperboloids you can perceive what appears to be a non-trivial dynamical evolution of these surfaces. To show that this "fake" dynamics is precisely the dynamics exhibited by the Milne metric, introduce new coordinates $(\tau, \rho, \text{angles})$ via

$$t = \tau \cosh \rho \quad , \quad \vec{x} = \vec{n}\tau \sinh \rho \tag{37.7}$$

where $\vec{n}$ is a unit vector on S^2, so that

$$\vec{n}.\vec{n} = 1 \quad , \quad \vec{n}.d\vec{n} = 0 \quad , \quad d\vec{n}.d\vec{n} = d\Omega_2^2 \ . \tag{37.8}$$

Then the Minkowski metric becomes precisely the Milne metric (37.4),

$$ds^2 = -d\tau^2 + \tau^2 d\rho^2 + \tau^2 \sinh^2 \rho \, d\Omega_2^2 \ . \tag{37.9}$$

Remarks:

1. These coordinates only cover a part of Minkowski space-time, namely the interior of the future (and past) lightcone of the origin. They are adapted to the comoving (and thus geodesic) observers of the Milne metric, i.e. to families of observers with constant values $\rho = \rho_0$, $\vec{n} = \vec{n}_0$. Thus in terms of Minkowski coordinates, the worldlines of these observers are described by

$$t(\tau) = \tau \cosh \rho_0 \quad , \quad \vec{x}(\tau) = \vec{n}_0 \tau \sinh \rho_0 \tag{37.10}$$

or

$$\vec{x}(t) = (\vec{n}_0 \tanh \rho_0) t \equiv \vec{v}_0 t \tag{37.11}$$

with

$$(v_0)^2 = \tanh^2 \rho_0 \leq 1 \quad . \tag{37.12}$$

Thus these are geodesic observers emanating radially with different initial directions and velocities from the origin, the constant-time hyperboloids of the Milne metric being the surfaces which these observers reach at their proper time τ.

2. The coordinate singularity at $\tau = 0$ is simpy due to the fact that the worldlines of these observers all intersect at the origin (and thus do not provide good coordinates there).

3. These coordinates are, as may have occurred to you by now, the future and past relatives of the Rindler coordinates discussed (way back) in sections 1.3 and 3.4 (and then again in the context of the near-horizon geometry of the Schwarzschild metric in section 26.6), in particular of the spherical Rindler space discussed at the end of section 3.4:

   Introducing Rindler-like coordinates $(\tau, \rho, \text{angles})$ adapted to the timelike hyperboloids

   $$\vec{x}^2 - t^2 = \rho^2 \tag{37.13}$$

   via

   $$t = \rho \sinh \tau \quad , \quad \vec{x} = \vec{n} \rho \cosh \tau \quad , \tag{37.14}$$

   with $\vec{n}$ a spatial unit vector, $\vec{n}.\vec{n} = 1$, the Minkowski metric takes the form (3.100)

   $$\begin{aligned} ds^2 &= -\rho^2 d\tau^2 + d\rho^2 + \rho^2 \cosh^2 \tau \, d\Omega_2^2 \\ &= d\rho^2 + \rho^2 (-d\tau^2 + \cosh^2 \tau \, d\Omega_2^2) \quad . \end{aligned} \tag{37.15}$$

   This is the spherical Rindler metric. The metric in brackets in the 2nd line will reappear below, in the guise of the (2+1)-dimensional de Sitter metric.

4. Thus, we see that we can foliate Minkowski space in different ways:

   - in the standard inertial coordinates, slices of constant Minkowski time t are Euclidean spaces $\mathbb{R}^3$, and slices of constant x^3, say, are Minkowski spaces $\mathbb{R}^{1,2}$;
   - in terms of Minkowski time and spatial spherical coordinates,

     $$ds^2 = -dt^2 + dr^2 + r^2 \, d\Omega_2^2 \tag{37.16}$$

     slices of constant r are of the form $S^2 \times \mathbb{R}$ (and we will recognise these below, in more fancy terminology, as (2+1)-dimensional versions of the Einstein static universe);
   - in Milne coordinates, slices of constant τ are hyperboloids H^3;
   - in the Rindler-like coordinates (37.15), slices of constant ρ are are de Sitter spaces dS_3.

5. The "singularity" of the Milne metric is of course just a coordinate singularity, but with some discrete identifications on the hyperboloid this becomes an honest conical singularity at $\tau = 0$. This is a nice (and much studied) toy-model of a big bang (everywhere flat except at $\tau = 0$).

37.2 Einstein Static Universe

This particular solution is mainly of historical interest. Before the discovery and understanding of the Hubble expansion of the universe, it was natural to assume that the universe can be described by a static solution compatible with the cosmological principle (homogeneity and isotropy). This led Einstein to introduce the (in-)famous cosmological constant. Let us see how this comes about.

Static means that $a(t) = a_0, \dot{a}(t) = \ddot{a}(t) = 0$. (F2) with $\ddot{a} = 0$ and a zero cosmological constant leads to

$$\text{(F2)}(\Lambda = 0) \quad \Rightarrow \quad 4\pi G_N(\rho + 3p) = 0 \ . \tag{37.17}$$

This equation is simply not satisfied for ordinary matter, and therefore no static solution of the Friedmann equations with $\Lambda = 0$ exists. (Re-)introducing Λ, however, (F2) then tells us that

$$\text{(F2)} \quad \Rightarrow \quad 4\pi G_N(\rho + 3p) = \Lambda \ . \tag{37.18}$$

Moreover, the conservation law (F3) with $\dot{a} = 0$ implies

$$\text{(F3)} \quad \Rightarrow \quad \dot{\rho} = 0 \ , \tag{37.19}$$

which now implies that also $\dot{p} = 0$. (37.18) then simply fixes the constant Λ in terms of the constants ρ and p. For a standard matter content of the universe, say $\rho = \rho_m + \rho_r$, this requires that $\Lambda > 0$. Finally, the first Friedmann equation (F1) now becomes an algebraic equation for $a(t) = a_0$, namely

$$\begin{aligned} \text{(F1)} \quad &\Rightarrow \quad \frac{k}{a^2} = \frac{8\pi G_N}{3}\rho + \frac{\Lambda}{3} = 4\pi G_N(\rho + p) \\ &\Rightarrow \quad k = +1 \quad \text{and} \quad a_0^2 = (4\pi G_N(\rho + p))^{-1} \ . \end{aligned} \tag{37.20}$$

This is thus a static solution of the Friedmann equations,

$$ds^2 = -dt^2 + a_0^2 \, d\Omega_3^2 \ , \tag{37.21}$$

the *Einstein Static Universe*.

Remarks:

1. The topology of the solution is $\mathbb{R} \times S^3$, the radius a_0 of the S^3 being smaller for larger energy and pressure density (bigger gravitational attraction) and vice-versa.

2. In this solution, the gravitational attraction is precisely balanced by the positive cosmological constant. Note that even though a positive cosmological constant has a positive energy density $\rho_\Lambda > 0$, it has a negative pressure, and the net effect of a positive cosmological constant in (F2) is that of gravitational repulsion rather than attraction,
$$\rho_\Lambda + 3p_\Lambda = -2\rho_\Lambda < 0 \ . \tag{37.22}$$

3. The precise balance between matter and cosmological constant required by this solution also implies that it is unstable to perturbations of either Λ or ρ, p: a slight increase in Λ relative to $\rho + p$, no matter how small, will make the universe expand, and a slight decrease will make it collapse. This alone is enough to make this particular solution unphysical and shows that, even with inclusion of a cosmological constant, an expanding or collapsing universe is practically inevitable, thus undermining the original motivation for introducing the cosmological constant in the first place.

37.3 Matter Dominated Era

We now return to somewhat more realistic solutions of the Friedmann equations. In a matter dominated era (but retaining the curvature term) we have to solve the equation
$$\dot{a}^2 = \frac{C_m}{a} - k \ . \tag{37.23}$$
We will now look in turn at the cases $k = 0, +1, -1$.

1. For $k = 0$, this is the equation we already discussed above, leading to the solution (36.20),
$$a(t) = a_0 (t/t_0)^{2/3} \tag{37.24}$$
This solution is also known as the Einstein - de Sitter universe. It describes a universe expanding from an initial singularity at $t = 0$. Another solution, arising from taking the negative square root of the Friedmann equation $\dot{a}^2 = C_m/a$,
$$a^{1/2}\dot{a} = -(C_m)^{1/2} \quad \Rightarrow \quad a(t) = a_0 (t_f - t)^{2/3}/(t_f - t_0)^{2/3} \ , \tag{37.25}$$
describes a universe collapsing to a big crunch at $t = t_f$.

2. For $k = +1$, the equation is
$$\dot{a}^2 = \frac{C_m}{a} - 1 \ . \tag{37.26}$$
We recall that in this case we will have a recollapsing universe, with $a_{max} = C_m$ attained for $\dot{a} = 0$,
$$\dot{a} = 0 \quad \Rightarrow \quad a = a_{max} = C_m \ . \tag{37.27}$$

The equation (37.26) can be solved in closed form for t as a function of a, and the solution to

$$\frac{dt}{da} = \left(\frac{a}{a_{max} - a}\right)^{1/2} \tag{37.28}$$

is

$$t(a) = \frac{a_{max}}{2} \arccos(1 - 2a/a_{max}) - \sqrt{a a_{max} - a^2} \ , \tag{37.29}$$

as can easily be verified. The universe starts at $t = 0$ with $a(0) = 0$, reaches its maximum $a = a_{max}$ at

$$t_{max} = a_{max} \arccos(-1)/2 = a_{max}\pi/2 \ , \tag{37.30}$$

and ends in a Big Crunch at $t = 2t_{max}$.

The curve $a(t)$ is a cycloid, as is most readily seen by writing the solution in parametrised form. For this it is convenient to use the conformal time coordinate η introduced in (34.48) through

$$d\eta = dt/a(t) \quad \Rightarrow \quad \partial_t = a(\eta)^{-1}\partial_\eta \quad , \quad \partial_\eta = a(t)\partial_t \ . \tag{37.31}$$

Denoting a derivative with respect to η by a prime, and noting that $\dot{a} = a'/a$, one then finds that for $k \neq 0$ the Friedman equation (37.23) can be written as

$$\begin{aligned} \dot{a}^2 + k = \frac{C_m}{a} \quad &\Leftrightarrow \quad (a')^2 + ka^2 = C_m a \\ &\Leftrightarrow \quad ((a - C_m/2k)')^2 + k(a - C_m/2k)^2 = kC_m^2/4 \ . \end{aligned} \tag{37.32}$$

Thus for $k = +1$, the solution to the Friedmann equation can be written as

$$a(\eta) - C_m/2 = (C_m/2)\cos(\eta - \eta_0) \ . \tag{37.33}$$

Choosing $\eta_0 = \pi$ (so that $a(\eta = 0) = 0$), and integrating the relation $dt/d\eta = a(\eta)$ to find $t(\eta)$, one then finds the solution

$$\begin{aligned} a(\eta) &= \frac{a_{max}}{2}(1 - \cos\eta) \\ t(\eta) &= \frac{a_{max}}{2}(\eta - \sin\eta) \ , \end{aligned} \tag{37.34}$$

which makes it transparent that the curve is indeed a cycloid, roughly as indicated in Figure 46.

The maximal radius is reached at

$$t_{max} = t(a = a_{max}) = t(\eta = \pi) = a_{max}\pi/2 \tag{37.35}$$

(with $a_{max} = C_m$), as before, and the total lifetime of the universe is $2t_{max}$.

3. Analogously, for $k = -1$ the Friedmann equation in parametrised form (37.32) can be solved in terms of hyperbolic (rather than trigonometric) functions,

$$\begin{aligned} a(\eta) &= \frac{C_m}{2}(\cosh\eta - 1) \\ t(\eta) &= \frac{C_m}{2}(\sinh\eta - \eta) \ . \end{aligned} \tag{37.36}$$

REMARKS:

1. We see that for small times (for which matter dominates over curvature) the solutions for $k \neq 0$ reduce to $t \sim \eta^3, a \sim \eta^2$ and therefore $a \sim t^{2/3}$ which is indeed the exact solution for $k = 0$.

2. Analogously, for late times in the $k = -1$ model one finds that $a(\eta) \sim t(\eta)$, reproducing the expected late-time behaviour $\dot{a} \to 1$ of section 36.3.

3. In section 29 we will use the exact solutions of this matter dominated phase to describe the interior geometry of collapsing stars.

It is instructive to display the causal structure of the $k = +1$ metric in a conformal diagram. Recall that in terms of conformal time the metric takes the form

$$ds^2 = a(\eta)^2(-d\eta^2 + d\Omega_3^2) = a(\eta)^2(-d\eta^2 + d\psi^2 + \sin^2\psi d\Omega_2^2) \ , \tag{37.37}$$

Suppressing the transverse 2-sphere and eliminating the conformal factor $a(\eta)^2$, we are thus led to consider the metric

$$d\tilde{s}^2 = -d\eta^2 + d\psi^2 \ . \tag{37.38}$$

Noting that in the case at hand the range of both η and ψ is finite,

$$\eta \in [0, 2\pi] \quad , \quad \psi \in [0, \pi] \ , \tag{37.39}$$

we see that no further redefinition of the coordinates is required to obtain a conformal diagram of this solution (Figure 51).

Here are some of the characteristic features of this diagram:

1. Since this solution is spatially compact, there is no analogue here of spatial infinity i^0. As a consequence, this diagram a priori looks very different from Penrose diagrams for asymptotically flat space-times, in particular also from those for spatially flat cosmologies in section 36.6.

2. All timelike and null geodesics begin at the initial (spacelike) singularity at $\eta = 0$ and end at the final singualrity at $\eta = 2\pi$.

3. A lightray sent out at the Big Bang will reach the antipodal point of the sphere exactly at the time $\eta_{max} = \pi$ the universe reaches its maximal radius, and will have circled around the universe exactly once precisely at the time $2\eta_{max}$ of the final big crunch.

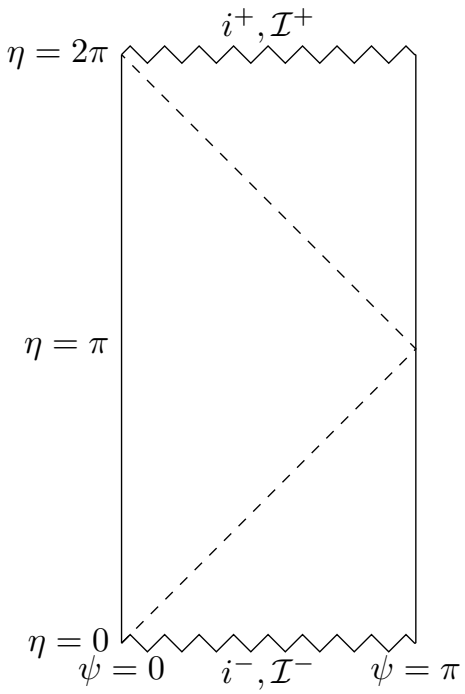


Figure 51: Conformal Diagram of the $k = +1$ matter dominated universe.

37.4 Radiation Dominated Era

We now consider the situation when radiation is dominant (as is expected during some time in the very early universe). In this case we need to solve

$$a^2\dot{a}^2 = C_r - ka^2 \quad . \tag{37.40}$$

For $k = 0$ we had already seen the solution in (36.18),

$$a(t) = a_0(t/t_0)^{1/2} \tag{37.41}$$

(and there is also evidently a corresponding collapsing solution). For $k = +1$, on the other hand, there will be a maximal radius a_{max} at

$$k = +1: \quad \dot{a} = 0 \quad \Rightarrow \quad a_{max} = C_r^{1/2} \quad . \tag{37.42}$$

Here are 2 ways to solve the Friedmann equation for $k \neq 0$:

1. Because $a(t)$ appears only quadratically, it is convenient to make the change of variables $b = a^2$. Then one obtains

$$\frac{\dot{b}^2}{4} + kb = C_r \quad . \tag{37.43}$$

For $k = \pm 1$, one necessarily has $b(t) = b_0 + b_1 t + b_2 t^2$. Fixing $b(0) = 0$, one easily finds the solution

$$a(t) = [2C_r^{1/2}t - kt^2]^{1/2} \quad . \tag{37.44}$$

As expected this reduces to $a(t) \sim t^{1/2}$ for small times where the curvature term is irrelevant.

- For $k = +1$ one has
$$a(0) = a(2C_r^{1/2}) = 0 \ , \tag{37.45}$$
and $a = a_{max}$ at the time
$$t_{max} = C_r^{1/2} \quad \Rightarrow \quad a(t_{max}) = a_{max} = C_r^{1/2} \ . \tag{37.46}$$
- For $k = -1$, on the other hand, the universe expands forever, the late-time behaviour being given by
$$a(t) \to (-kt^2)^{1/2} = t \ , \tag{37.47}$$
again as expected.

All this is of course in agreement with the results of the qualitative discussion given earlier.

2. It is also instructive to solve these equations in terms of conformal time η. Using $\dot{a} = a'/a$, the Friedmann equation becomes
$$(a')^2 + ka^2 = C_r \ . \tag{37.48}$$
Thus the solution for $k = +1$ is obviously (with the choice $a(\eta = 0) = 0$)
$$k = +1 \quad \Rightarrow \quad a(\eta) = C_r^{1/2} \sin \eta \tag{37.49}$$
and for $k = -1$ one has
$$k = -1 \quad \Rightarrow \quad a(\eta) = C_r^{1/2} \sinh \eta \ . \tag{37.50}$$
Using $dt/d\eta = a(\eta)$ one can also find $t(\eta)$. E.g. for $k = +1$ one has
$$t(\eta) = C_r^{1/2}(1 - \cos \eta) \ , \tag{37.51}$$
where the integration constant has been chosen such that
$$\eta = 0 \quad \Rightarrow \quad a(\eta = 0) = 0 \quad , \quad t(\eta = 0) = 0 \ . \tag{37.52}$$
One also has $\eta_{max} = \pi/2$ and the recollapse to $a(\eta) = 0$ at $\eta = \pi$ corresponding to $t = 2C_r^{1/2}$, as in the 1st derivation.

Again it is instructive to display the $k = +1$ solution in a conformal diagram (Figure 52) and to compare it with that of the matter dominated solution (Figure 51). The main difference is that here the range of η is equal to the range of ψ,
$$\eta \in [0, \pi] \quad , \quad \psi \in [0, \pi] \ . \tag{37.53}$$
As a consequence, the conformal diagram is a square, and any lightray sent out at the Big Bang can only travel half-way around the universe during the lifetime of the universe.

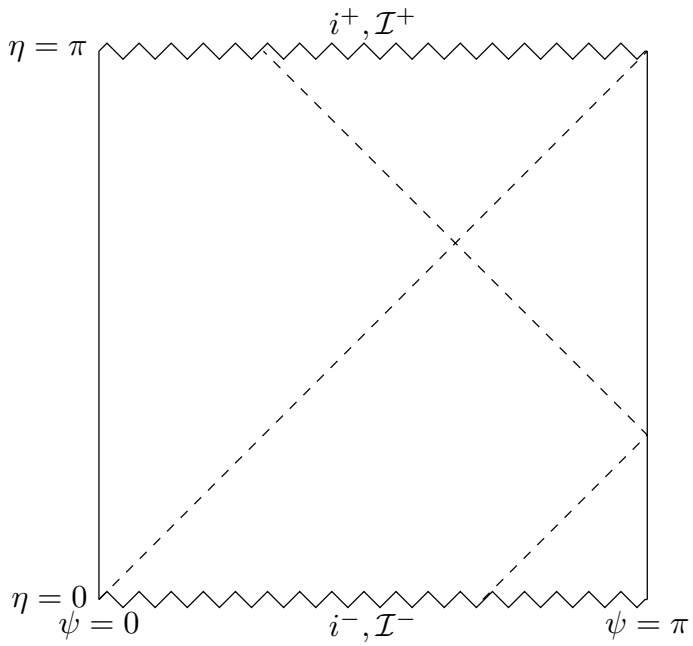


Figure 52: Conformal Diagram of the $k = +1$ radiation dominated universe.

37.5 Cosmological Constant Dominated Era: (Anti-) de Sitter Space

This case is of considerable interest for at least two reasons. On the one hand, as we know, Λ is the dominant driving force for $a(t)$ very large, and may therefore, if current observations are to be believed (see section 38.1), dominate the late-time behaviour of our universe.

On the other hand, the currently most popular cosmological models trying to also address and solve the so-called *horizon problem* and *flatness problem* (cf. the discussion in section 38.3) of the standard FLRW model of cosmology (as well as a number of other issues) use a mechanism called *inflation* based on an era of exponential expansion during some time in the very early universe. This is typically generated by something that acts effectively like a cosmological constant.

Thus the equations to solve are, setting $\rho = 0$ and $p = 0$ but retaining Λ and k,

$$\dot{a}^2 = -k + \frac{\Lambda}{3} a^2 \quad . \tag{37.54}$$

We see immediately that Λ has to be positive for $k = +1$ or $k = 0$, whereas for $k = -1$ both positive and negative Λ are possible,

$$\dot{a}^2 = -k + \frac{\Lambda}{3} a^2 \quad \Rightarrow \quad \begin{cases} \text{if } \Lambda > 0 & k = 0, \pm 1 \text{ possible} \\ \text{if } \Lambda < 0 & k = -1 \end{cases} \tag{37.55}$$

This is one instance where the solution to the second order equation (F2),

$$\ddot{a} = \frac{\Lambda}{3} a \quad , \tag{37.56}$$

is more immediate, namely trigonometric functions for $\Lambda < 0$ (only possible for $k = -1$) and hyperbolic functions for $\Lambda > 0$. The first order equation then fixes the constants of integration according to the value of k.

It will be convenient to express Λ in terms of a length-scale ℓ (which will turn out to be the curvature radius of the space-time) through

$$|\Lambda|/3 = \ell^{-2} \tag{37.57}$$

so that the two Friedmann equations read

$$\dot{a}^2 = -k \pm a/\ell^2 \quad , \quad \ddot{a} = \pm a/\ell^2 \quad , \tag{37.58}$$

the plus/minus referring to $\Lambda > 0$ and $\Lambda < 0$ respectively.

Solutions for $\Lambda > 0$:

1. $k = 0$: The solutions are evidently (cf. (36.23))

$$a_\pm(t) \sim \mathrm{e}^{\pm t/\ell} \tag{37.59}$$

leading to the two metrics (related by $t \to -t$)

$$ds^2 = -dt^2 + \mathrm{e}^{\pm 2t/\ell} d\vec{x}^2 \tag{37.60}$$

2. $k = +1$: The solution is

$$ds^2 = -dt^2 + \ell^2 \cosh^2 t/\ell \; d\Omega_3^2 \quad . \tag{37.61}$$

3. $k = -1$: The solution is

$$ds^2 = -dt^2 + \ell^2 \sinh^2 t/\ell \; d\tilde{\Omega}_3^2 \quad . \tag{37.62}$$

Remarks:

1. It turns out that all 3 metrics actually represent the same space-time metric, just written in different coordinates. This space-time is known as the *de Sitter space-time*. Thus these 3 metrics exhibit different slicings of the de Sitter space-time (or, henceforth, de Sitter space for short, or just dS space), with the t = const. slices being $\mathbb{R}^3$, S^3 and H^3 respectively.

2. One way to establish this would be to directly exhibit the coordinate transformations that map one metric to the other, but this is messy and does not provide any additional insight. We will proceed in a different way in section 39 below. Indeed, it turns out that the de Sitter solution is the unique maximally symmetric spacetime with positive curvature (cf. the discussion in sections 14 and 39), and this perspective will provide us with a more efficient and insightful way of constructing different coordinate systems and exploring the relations among them.

3. All 3 metrics exhibit an exponential expansion or collapse at early and/or late times. The $k = -1$ metric also appears to exhibit a big bang singularity as $t \to 0$, but this is an illusion. Indeed, as $t \to 0$ the metric approaches the metric (37.4) of the Milne universe, and we already know that the singularity at $\tau = 0$ of the Milne metric is just a coordinate singularity.

4. The coordinates appearing in the $k = +1$ metric turn out to cover de Sitter space globally. Thus the global picture of de Sitter space is that of a 3-sphere that

   - started out with infinite radius in the infinite past $t \to -\infty$,
   - undergoes an exponential contraction to a sphere of radius ℓ at time $t = 0$
   - and then again undergoes an exponential expansion to infinite size as $t \to +\infty$.

   It is clear from this description that e.g. the 2 metrics for $k = 0$ only cover the contracting (-) or expanding (+) period of the de Sitter universe.

5. For later purposes it is useful to write the expanding $k = 0$ metric in terms of conformal time η (cf. sections 34.5 and 36.6) as

$$ds^2 = \mathrm{e}^{2t/\ell}(-dt^2\mathrm{e}^{-2t/\ell} + d\vec{x}^2) = \frac{\ell^2}{\eta^2}(-d\eta^2 + d\vec{x}^2) \tag{37.63}$$

   where

$$\eta = -\ell\mathrm{e}^{-t/\ell} \ , \tag{37.64}$$

   with $\eta \to -\infty$ for $a(t(\eta)) \to 0$. This clearly exhibits the simple Minkowskian causal structure of de Sitter space in the expanding phase / patch.

6. The (2+1)-dimensional counterpart of the $k = +1$ metric is the metric that appeared in the Rindler-like coordinate system (37.15). Thus we can say that these Rindler-like coordinates provide a (radial) foliation of the left- and right Rindler wedges by de Sitter spaces.

7. A detailed discussion of many different coordinate systems for de Sitter space is given in section 39.2 below.

Solution for $\Lambda < 0$:

In this case, the only possibility (within the Robertson-Walker ansatz for the metric) is $k = -1$, and the solution is

$$ds^2 = -dt^2 + \ell^2 \sin^2 t/\ell \, d\tilde{\Omega}_3^2 \ . \tag{37.65}$$

Remarks:

1. This solution is nowadays known as the *anti-de Sitter space-time* (or *anti-de Sitter space* or AdS space for short).

2. As its $\Lambda > 0, k = -1$ counterpart, it appears to exhibit a big bang singularity as $t \to 0$, but this is again an illusion since this metric also approaches the non-singular Milne metric (37.4) as $t \to 0$.

3. AdS space turns out to be the unique maximally symmetric space-time with negative curvature.

4. The above cosmological coordinates do not provide a complete covering of anti-de Sitter space. In (one choice of) global coordinates the metric turns out to take the form (39.75)

$$ds^2 = -\cosh^2 \rho d\tau^2 + d\rho^2 + \sinh^2 \rho d\Omega_2^2 \tag{37.66}$$

   with $0 \leq \rho < \infty$ and $-\infty < \tau < \infty$. In these coordinates the metric is time-independent and therefore (unlike dS) AdS has a global timelike Killing vector, namely ∂_τ.

5. A detailed discussion of many different coordinate systems for anti-de Sitter space is given in section 39.3 below.

38 Cosmology VI: The Universe Today - Insights and Puzzles

38.1 Λ-CDM Cold Dark Matter + Dark Energy Model

Let us recall the key equations governing the evolution of the universe which, in the notation introduced in section 36.5, take the compact (albeit somewhat obscure) form

$$\begin{aligned} \Omega_M + \Omega_\Lambda + \Omega_k &= 1 \\ \tfrac{1}{2}(1+3w)\Omega_M - \Omega_\Lambda &= q \end{aligned} \tag{38.1}$$

Their version today (at $t = t_0$) is (36.40)

$$\begin{aligned} (\Omega_M)_0 + (\Omega_\Lambda)_0 + (\Omega_k)_0 &= 1 \\ \tfrac{1}{2}(1+3w_0)(\Omega_M)_0 - (\Omega_\Lambda)_0 &= q_0 \end{aligned} \tag{38.2}$$

In the universe today, the radiation contribution to the matter content is negligible and the only non-negligible matter content appears to be that of $w = 0$ pressureless matter, and thus

$$q_0 = \tfrac{1}{2}(\Omega_M)_0 - (\Omega_\Lambda)_0 \quad . \tag{38.3}$$

In that case Ω_M is related to $(\Omega_M)_0$ by

$$\Omega_M = \frac{\Omega_M}{(\Omega_M)_0}(\Omega_M)_0 = \frac{\rho_M}{(\rho_M)_0}\frac{(\rho_{cr})_0}{\rho_{cr}}(\Omega_M)_0 = \frac{a_0^3}{a^3}\frac{H_0^2}{H^2}(\Omega_M)_0 \quad . \tag{38.4}$$

Likewise, for the cosmological constant and curvature contributions one has

$$\Omega_\Lambda = \frac{H_0^2}{H^2}(\Omega_\Lambda)_0 \quad , \quad \Omega_k = \frac{a_0^2 H_0^2}{a^2 H^2}(\Omega_k)_0 \quad . \tag{38.5}$$

Thus the Friedmann equation can be written as

$$H^2 = H_0^2\left(\frac{a_0^3}{a^3}(\Omega_M)_0 + \frac{a_0^2}{a^2}(\Omega_k)_0 + (\Omega_\Lambda)_0\right) \quad . \tag{38.6}$$

Using the constraint $(\Omega_M)_0 + (\Omega_\Lambda)_0 + (\Omega_k)_0 = 1$ to eliminate $(\Omega_k)_0$ from this equation, one finds

$$H^2 = \frac{H_0^2 a_0^2}{a^2}\left(1 + (\frac{a_0}{a} - 1)(\Omega_M)_0 + (\frac{a^2}{a_0^2} - 1)(\Omega_\Lambda)_0\right) \quad . \tag{38.7}$$

This equation is also frequently written as an equation for H as a function of the redshift z, $H = H(z)$, which can be obtained by the substitution $a_0/a = 1 + z$,

$$H(z) = H_0(1+z)\left[1 + (\Omega_M)_0 z + (\Omega_\Lambda)_0\left(\frac{1}{(1+z)^2} - 1\right)\right]^{1/2} \quad . \tag{38.8}$$

This expression is useful because it is expressed directly in terms of observable quantities, with $H(z)$ the value of the Hubble parameter at the time an object emitted the light that we now observe with redshift z. Using

$$1 + z = \frac{a_0}{a(t)} \quad \Rightarrow \quad \frac{dz}{H(z)} = -\frac{a_0 dt}{a(t)} \tag{38.9}$$

it is also possible to convert other evolution equations in t (or integrals over t) into evolution equations in z (or integrals over z).

For a long time it was believed that $\Omega_\Lambda = 0$ (or at least negligibly small) today. In that case one would have

$$(\Omega_M)_0 + (\Omega_k)_0 = 1 \tag{38.10}$$

and

$$q_0 = \tfrac{1}{2}(\Omega_M)_0 > 0 \ . \tag{38.11}$$

Therefore the curvature parameter k would be directly related to the value q_0 of the deceleration parameter today:

$$\begin{array}{lllll} q_0 > 1/2 & \Rightarrow & \rho_0 > (\rho_{cr})_0 & \Rightarrow & k = +1 \\ q_0 < 1/2 & \Rightarrow & \rho_0 < (\rho_{cr})_0 & \Rightarrow & k = -1 \end{array} \tag{38.12}$$

While there were large uncertainties about q_0, observations indicated a value of ρ_0 much smaller than the critical density $(\rho_{cr})_0$ (even taking into account not just visible matter but also some invisible ("dark") matter whose existence had been deduced indirectly from astrophysical observations like the rotations of galaxies). Thus this strongly suggested a decelerating open $k = -1$ universe. While perhaps not the most hospitable place in the long run, at least this scenario had the virtue of simplicity.

However, exciting recent developments and observations in cosmology and astrophysics have provided strong evidence for a very different and extremely intriguing and puzzling picture of the universe today. Since I am not an expert on these matters, I will just summarise the results here.[162]

1. Estimates for the current matter contribution $\Omega_M = \sum_b \Omega_b$ are

$$(\Omega_M)_0 \approx 0.3 \ . \tag{38.13}$$

2. Ordinary (visible, baryonic) matter only accounts for a small fraction of this, namely

$$(\Omega_{M,\text{visible}})_0 \approx 0.04 \ . \tag{38.14}$$

   Most of the matter density of the universe must therefore be due to some form of (as yet ill-understood, non-relativistic, weakly interacting) *Dark Matter* or *Cold Dark Matter* (CDM).

3. A detailed analysis of the fine structure of the anisotropies of the *Cosmic Microwave Background* suggests that the universe is spatially flat, $k = 0$, i.e. that the total density of whatever fills the universe is very close to the critical density.

[162] For excellent recent introductions to cosmology which explain the significance of these results, how they were obtained and much more, see e.g. M. Trodden and S. Carroll, *TASI Lectures: Introduction to Cosmology*, `arXiv:astro-ph/0401547`, and J. Frieman, M. Turner, D. Huterer, *Dark Energy and the Accelerating Universe*, `arXiv:0803.0982v1 [astro-ph]`.

4. Observations of redshifts of particular types of *Supernovae* in distant galaxies lead to the conclusion that the universe must currently be in a phase of *accelerated expansion*, with the data well explained by assuming that the *Dark Energy* responsible for this acceleration (thus with equation of state parameter $w < -1/3$) is due to the existence of a positive cosmological constant ($w = -1$) with

$$q_0 = \tfrac{1}{2}(\Omega_M)_0 - (\Omega_\Lambda)_0 < 0 \tag{38.15}$$

and, more specifically,

$$(\Omega_\Lambda)_0 \approx 0.7 \ . \tag{38.16}$$

Since then

$$(\Omega_M)_0 + (\Omega_\Lambda)_0 \approx 1 \quad \Leftrightarrow \quad (\Omega_k)_0 \approx 0 \ , \tag{38.17}$$

these last two (completely independent) observations are evidently compatible with each other.

Remarks:

1. Approximate numerical values for the parameters Λ and H_0 characterising the universe today are

$$H_0 \approx 2.3 \times 10^{-18}\ \mathsf{s}^{-1} \quad , \quad \Lambda \approx 10^{-35}\ \mathsf{s}^{-2} \ . \tag{38.18}$$

2. Somewhat more informative are perhaps the current estimates for the age of the universe,

$$t_0 \approx 13.8 \times 10^9 \text{years} \tag{38.19}$$

and the energy density of dark energy, approximately

$$\rho_\Lambda \approx (10^{-3}\ \mathsf{eV})^4 \ . \tag{38.20}$$

3. The equation of state parameter for dark energy is known to very close to $w = -1$,

$$w \approx -0.98 \pm 0.05 \ , \tag{38.21}$$

so a cosmological constant indeed appears to be a very plausible candidate for dark energy.

4. The fact that there is something like a cosmological constant is perhaps not particularly puzzling as such (in the absence of a good reason why it should not have been there in the first place), but nevertheless there are a number of puzzling issues related to the value of the cosmological constant - see section 38.4 for a brief discussion.

38.2 Exact Λ-CDM Solution

The exact solution for a spatially flat universe dominated by non-interacting (cold dark) matter was already given in (36.20),

$$a(t) = a_0 (t/t_0)^{2/3} \quad . \tag{38.22}$$

and that for a spatially flat universe dominated by a cosmological constant was given in (36.23),

$$a(t) = a_0 \mathrm{e}^{\sqrt{\Lambda/3}(t-t_0)} \quad . \tag{38.23}$$

An exact solution of the Friedmann equations can also be written down when both types of energy/matter are present simultaneously (as in the universe today). In this case the Friedmann equation is just

$$\Omega_M + \Omega_\Lambda = 1 \quad , \tag{38.24}$$

so that the Friedmann equation of the Λ-*CDM Model* (cold dark matter with a cosmological constant) can be written as

$$H^2 = H_0^2 \left(\frac{a_0^3}{a^3} (\Omega_M)_0 + (\Omega_\Lambda)_0 \right) \quad . \tag{38.25}$$

An exact solution of this equation is

$$\begin{aligned} a(t) &= a_0 \left(\frac{(\Omega_M)_0}{(\Omega_\Lambda)_0} \right)^{1/3} \left(\sinh((3/2)\sqrt{(\Omega_\Lambda)_0} H_0 t) \right)^{2/3} \\ &\equiv a_0 \left(\frac{(\Omega_M)_0}{(\Omega_\Lambda)_0} \right)^{1/3} (\sinh(t/t_\Omega))^{2/3} \quad . \end{aligned} \tag{38.26}$$

This evidently reproduces the above power-law (exponential) behaviour at early (late) times. The transition between the decelerating matter dominated and accelerating Λ-dominated phases occurs at the time $t = t_\Lambda$ at which

$$\ddot{a}(t_\Lambda) = 0 \quad . \tag{38.27}$$

Calculating

$$\frac{d^2}{dt^2} \sinh^{2/3}(t/t_\Omega) = (2/3) t_\Omega^{-2} \left(-(1/3) \sinh^{-4/3}(t/t_\Omega) \cosh^2(t/t_\Omega) + \sinh^{2/3}(t/t_\Omega) \right) \tag{38.28}$$

one finds that at $t = t_\Lambda$

$$\sinh^2(t_\Lambda/t_\Omega) = 1/2 \quad , \tag{38.29}$$

and plugging this back into (38.26), one sees that

$$\frac{a(t_\Lambda)}{a_0} = \left(\frac{(\Omega_M)_0}{2(\Omega_\Lambda)_0} \right)^{1/3} \quad . \tag{38.30}$$

With the values for the density parameters given above, $(\Omega_M)_0 = 0.3$, $(\Omega_\Lambda)_0 = 0.7$, this is roughly

$$a(t_\Lambda) \approx 0.6\, a_0 \quad . \tag{38.31}$$

Thus the transition occurred at a time when the universe was roughly speaking "half as big" as today, and at a corresponding value

$$z = \frac{a_0}{a(t_\Lambda)} - 1 \approx 0.66 \tag{38.32}$$

of the redshift parameter.

38.3 Flatness and Horizon Problems

The currently favoured Λ-CDM scenario with $(\Omega_M)_0 = 0.3$, $(\Omega_\Lambda)_0 = 0.7$ and $k = 0$ has been tested and confirmed in various independent ways. Nevertheless, it is a mystery and raises all kinds of questions and puzzles, in particular because the emergence of this particular universe appears to be somewhat unnatural (although it may perhaps be difficult to quantify this sentiment) and to require an incredible amount of fine-tuning.

I will not say anything about the dark matter component, since an explanation presumably needs to be found in the realm of particle physics (every model of physics beyond the standard model worth its salt has its own dark matter candidates), and since the story is in any case sufficiently strange and interesting even without worrying about, or having to put up with, things like WIMPS (weakly interacting massive particles), or MACHOS (massive compact halo objects), or *binos*, *winos* and other *neutralinos*.

Here, in a nutshell, are some of these problems or puzzles (and for more in-depth and far-reaching discussions of these issues please consult the extensive literature on cosmology). The first two (discussed in this section) are independent of the recent discovery of dark energy, and appear to require some fine-tuning or other mechanism to intervene in the very early universe. The third, on the other hand, has to do with dark energy / the cosmological constant today, and with the relatively recent period of accelerated expansion of the universe and is discussed in section 38.4.

I should stress that, while I generally make an attempt in these notes to present just the (well-established) facts, this is not entirely possible in this and the subsequent section, since there is no general consensus on how to resolve these issues (in particular those arising in connection with dark energy). As a consequence, these sections contain not just facts but certainly and unavoidably also some opinions (whereas there was no point in sharing with you my opinion about the Riemann curvature tensor, say - I don't even have one).

1. The Flatness Problem

The flatness problem is the problem that a universe with a value of the density parameter $\Omega = \Omega_M + \Omega_\Lambda \approx 1$ today, equivalently $(\Omega_k)_0 \approx 0$, appears to be "unnatural".

To see in simple terms what the issue is, note that from the Friedmann equation we can read off that

$$H^2 = \frac{8\pi G_N}{3}\rho - \frac{k}{a^2} \quad \Rightarrow \quad \Omega_k = -\frac{k}{a^2 H^2} \ . \tag{38.33}$$

Now for a matter or radiation dominated universe we have $a(t) \sim t^{2/3}$ or $a(t) \sim t^{1/2}$ (with subleading corrections when $k \neq 0$), and therefore

$$a^2 H^2 = \dot{a}^2 \sim \begin{cases} t^{-2/3} & \text{for } \ w = 0 \\ t^{-1} & \text{for } \ w = 1/3 \end{cases} \tag{38.34}$$

so that

$$\Omega_k(t) \sim \begin{cases} t^{+2/3} & \text{for } \ w = 0 \\ t & \text{for } \ w = 1/3 \end{cases} \tag{38.35}$$

Thus a small value of $(\Omega_k)_0 \approx 0$ today (whether to within a few percent, as observations suggest, or even give or take a couple of orders of magnitude) seems to require $\Omega_k = 0$ with an enormous accuracy at some given time in the early universe.

It is more common to phrase this issue in terms of the critical density ρ_{cr} and the density parameter $\Omega = \Omega_M + \Omega_\Lambda$ for matter / radiation plus dark energy (but the contribution of the latter is negligible in the early universe). To that end we rewrite the Friedmann equation as

$$H^2 = \frac{8\pi G_N}{3}\rho - \frac{k}{a^2} \quad \Leftrightarrow \quad (\rho_{cr} - \rho)a^2 = -\frac{3k}{8\pi G_N} \ , \tag{38.36}$$

the point of rewriting it in this way being that what now appears on the right-hand side is a constant. In terms of the density parameter Ω we have

$$(\Omega^{-1} - 1)\rho a^2 = -\frac{3k}{8\pi G_N} \ . \tag{38.37}$$

Since the right-hand side is a constant, the product of the two terms on the left-hand side is a constant. However, as we have already seen (and used) in section 36.3, for ordinary matter ($w > -1/3$) the factor ρa^2 decreases (enormously) as the universe expands. This has to be (and will be, by the above equation) compensated and accompanied by a huge increase in the first factor. Thus, to have $(\Omega)_0 \approx 1$ today appears to require the density $\rho(t)$ of the universe to be equal to the critical density $\rho_{cr}(t)$ with an enormous accuracy at some given time in the early universe.

For example, in the matter dominated phase one has $\rho a^2 \sim a^{-1}$. In this case one can write (38.37) as

$$(\Omega^{-1} - 1)/a = (\Omega_0^{-1} - 1)/a_0 \quad \Leftrightarrow \quad \Omega = \frac{\Omega_0}{\Omega_0 + (a/a_0)(1 - \Omega_0)} \ , \tag{38.38}$$

or

$$\Delta\Omega \equiv \frac{\Omega - 1}{\Omega} = \frac{a}{a_0}\Delta\Omega_0 \ . \tag{38.39}$$

Going back to the beginning of the matter-dominated phase when radiation decoupled from matter, $z = a_0/a \approx 10^3$, one sees that

$$\Delta\Omega \approx 10^{-3}\Delta\Omega_0 \ , \tag{38.40}$$

so that at that time the deviation from flatness (as measured by $\Delta\Omega$) was smaller by a factor of 10^{-3} than it is today.

However, this may begin to look like a more serious fine-tuning issue when one compares with the very early universe and specifies a particular ("earliest") time in the past. Concretely, let us (for the sake of argument) assume that

- the relevant early time is at the Planck scale, $t_P \approx 10^{-43}\mathtt{s}$
- and that matter consists of dust and radiation, with radiation dominating at early times, so that

$$\rho_r(t)a(t)^2 \sim a(t)^{-2} \sim t^{-1} \ . \tag{38.41}$$

With the age of the universe t_0 taken to be (only orders of magnitude will be relevant for this argument) $t_0 \approx 10^{17}\mathtt{s}$, one finds that

$$(\Omega^{-1} - 1)_{t=t_0} \approx 10^{-2} \quad \Rightarrow \quad (\Omega^{-1} - 1)_{t=t_p} \approx 10^{-62} \ . \tag{38.42}$$

In particular, any miniscule deviation from this tiny value at $t = t_p$, i.e. a miniscule difference between the actual and the critical density of the universe at that time, would have led to a universe completely incompatible with obervations. The universe would have most likely either recollapsed after a very short time ($\Omega \to \infty$), or expanded so quickly as to prevent the formation of any structure in the universe ($\Omega \to 0$). In fact, $\Omega = 0$ and $\Omega = \infty$ are the only attractors (attractive fixed points) of the theory in the terminology of dynamical systems, while $\Omega = 1$ is a repeller (unstable fixed point).

What is one supposed to make of this?

Now it should be borne in mind that the fact that Ω was very close to 1 in the past is not in itself a fine-tuning issue: after all, $\Omega \to 1$ in the past is *implied* by the Friedmann equations anyway, no matter what the size of Ω today. So one option is to declare that the above argument regarding finetuning is bogus, in particular since extrapolating all the way back to the Planck time (or the GUT era or ...), and maintaining the assumptions of isotropy and homogeneity that underlie the Friedmann equations requires some enormous leap of faith.[163]

[163] For various critical examinations of this flatness problem, see e.g. P. Helbig, *Is there a flatness problem in classical cosmology?*, `arXiv:1112.1666 [astro-ph.CO]` and M. Holman, *How Problematic is the Near-Euclidean Spatial Geometry of the Large-Scale Universe?*, `arXiv:1803.05148 [gr-qc]`, and references therein and thereto.

But if the flatness problem is perceived as a real problem, then no explanation for this can be found in the standard model of cosmology we have been discussing here so far (FLRW with standard matter content at early times). Occasionally, therefore, weakly anthropic arguments of the kind "if Ω had not been fine-tuned to this value by some (unknown) mechanism/deity, we couldn't be here in the first place to ask the question why Ω takes the value it does today" have been advocated.

While this cannot be ruled out (this is the whole problem with this entire anthropic reasoning business), there is actually a mechanism which naturally leads to the required tiny value of $\Omega - 1$ at early times, namely *inflation*.[164] As this mechanism is also invoked to solve other problems of the standard model of cosmology (such as the horizon problem, see below) which are (a) possibly more serious and (b) not obviously of anthropic significance, one could perhaps consider the anthropophile value of Ω_0 as an unintentional (but serendipitous) side-effect of inflation.

What inflation does is to postulate, for a number of reasons, a brief but highly significant period of exponential expansion in the very early universe, as could be triggered by the presence of a cosmological constant (36.23),

$$H^2 = (\Lambda/3) \quad \Rightarrow \quad a(t) \sim \mathrm{e}^{Ht} \ . \tag{38.43}$$

or more precisely by something that acts like a cosmological constant during a certain period, e.g. an almost constant scalar field (35.39). During such a period of exponential expansion, the factor ρa^2 grown enormously, and correspondingly $\Omega^{-1} - 1$ can be driven arbitrarily close to zero. When, after the end of inflation, the universe resumes its normal (radiation-dominated, say) decelerating expansion (this requires a process called *reheating*, since the universe has cooled down significantly during the inflationary period), the initial value of $\Omega^{-1} - 1$ for this phase can easily be small enough to account for $\Omega \approx 1$ to within a few percent today.

2. The Horizon Problem

   The key issue here is the high degree of isotropy of the universe that we observe, in particular as manifested in the cosmic microwave background radiation (CMBR, see the references in section 34.8), which is isotropic to better than one part in 10^5. This is a puzzle because, as we will see below, according to the standard model of cosmology, the different regions of the spatial surface at $t = t_{ls}$ (the time of

[164] For introductions to inflation, see e.g. A. Linde, *Particle Physics and Inflationary Cosmology*, `arXiv:hep-th/0503203`; D. Baumann, *TASI Lectures on Inflation*, `arXiv:0907.5424 [hep-th]`; S. Tsujikawa, *Introductory review of cosmic inflation*, `arXiv:hep-ph/0304257`; A. Liddle, *An introduction to cosmological inflation*, `arXiv:astro-ph/9901124`; A. Linde, *Inflationary Cosmology*, `arXiv:0705.0164v2 [hep-th]`; R. Brandenberger, *Inflationary Cosmology: Progress and Problems*, `arXiv:hep-ph/9910410`.

"last scattering") from which the CMBR photons that we observe today originated were not ever in causal contact since the beginning of the universe. Thus standard causal microphysics provides no explanation for why these different regions were so precisely at the same temperature. This is schematically illustrated in Figure 53.

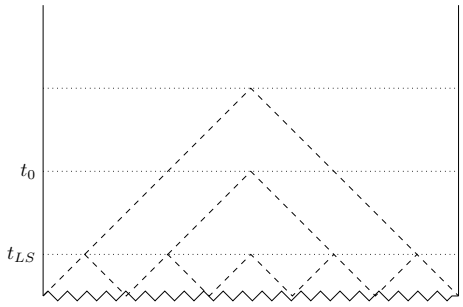


Figure 53: Horizon Problem ($k = +1$ cartoon): causally disconnected patches at the time t_{LS} of last scattering become simultaneously visible at later times.

To quantify this somewhat, note that the maximal spatial coordinate distance photons could have travelled between the big bang at $t = 0$ and $t = t_{ls}$ is (setting $k = 0$ for simplicity, the k-dependence is insignificant for this argument)

$$dt^2 = a(t)^2 dr^2 \quad \Rightarrow \quad r(t_{ls}) = r_{ph}(t_{ls}) = \int_0^{t_{ls}} \frac{dt}{a(t)} \ , \tag{38.44}$$

the particle horizon at time $t = t_{ls}$. Converting this to proper distance at time $t = t_{ls}$, one obtains the quantity

$$R_{ph}(t_{ls}) = a(t_{ls}) \int_0^{t_{ls}} \frac{dt}{a(t)} \ , \tag{38.45}$$

and for a radiation-dominated universe with $a(t) \sim t^{1/2}$ one finds

$$R_{ph}(t_{ls}) = t_{ls}^{1/2} \int_0^{t_{ls}} dt \ t^{-1/2} = 2t_{ls} \ . \tag{38.46}$$

Its significance in the present context is that the past light cone of events that are further than $2R_{ph}(t_{ls})$ apart on the surface $t = t_{ls}$ do not intersect, so that they are so far apart that they were never before in causal contact. In particular this means that no causal interaction can be responsible for the temperature being the same at the two events.

In the (matter dominated) meantime (i.e. between t_{ls} and t_0, today) the size of such a causal patch of size $\sim R_{ph}(t_{ls})$ on the last scattering surface has expanded to proper size

$$\frac{a(t_0)}{a(t_{ls})} R_{ph}(t_{ls}) \sim (t_0/t_{ls})^{2/3} t_{ls} = t_0^{2/3} t_{ls}^{1/3} \ . \tag{38.47}$$

On the other hand, the distance over which the CMBR photons could have travelled since $t = t_{ls}$ is

$$a(t_0) \int_{t_{ls}}^{t_0} \frac{dt}{a(t)} = 3t_0^{2/3}(t_0^{1/3} - t_{ls}^{1/3}) \sim t_0 \ , \tag{38.48}$$

where we have dropped the second term since $t_0 \sim 10^{10}$ years while $t_{ls} \sim 10^5$ years (once again all numbers here and below are just meant to be order of magnitude estimates). Thus the region of the last scattering surface from which we receive the CMBR photons today is much larger than a causal patch, their ratio being (one can of course calculate this either at $t = t_{ls}$ or at $t = t_0$, here we have chosen the latter option)

$$t_0/(t_0^{2/3} t_{ls}^{1/3}) = (t_0/t_{ls})^{1/3} \approx 10^{5/3} \ . \tag{38.49}$$

What this means is that the sky splits into roughly $4\pi 10^{10/3} \gtrsim 10^4$ disconnected patches, that were never in communication before sending light to us. In view of this the observed isotropy of the CMBR is not only astounding but utterly implausible.

Again, inflation solves this in an extremely natural way. Inflation operates at a time $t_i \ll t_{ls}$ (perhaps some time between 10^{-36}s and 10^{-32}s after the big bang) and it can easily inflate a tiny causally connected patch at that time $t = t_i$ to such a size that at time $t = t_{ls}$ it is (more than) large enough to explain the isotropy of the CMBR.

This is easy to understand by looking at the expanding de Sitter metric in conformal time (37.63),

$$ds^2 = \mathrm{e}^{2t/\ell}(-dt^2 \mathrm{e}^{-2t/\ell} + d\vec{x}^2) = \frac{\ell^2}{\eta^2}(-d\eta^2 + d\vec{x}^2) \tag{38.50}$$

with

$$\eta = -\ell \mathrm{e}^{-t/\ell} \ . \tag{38.51}$$

This shows that in a conformal diagram whatever came before inflation can now be pushed to arbitrarily large negative values of η, and therefore the horizon problem as depicted in Figure 53 does not arise.

The majority view among cosmologists appears to be that for these and other reasons inflation should be considered to be part of the standard model of cosmology. However, science is not decided by opinion polls and it is good scientific practice to keep an open mind. In particular, one should keep in mind the possibility that for instance the two (flatness and horizon) problems (or perhaps better: puzzles) may indicate problems with the approximation of the early universe by a FLRW cosmology and are thus not

something that should (or needs to, or perhaps even can) be solved by (inflationary) modifications within the class of FLRW models.[165]

The most significant success of inflation, however, is that it not only "explains" the high degree of isotropy of the CMBR but that it also provides a mechanism for, and a precise quantitative account of, density perturbations and the small inhomogeneities exhibited by the CMBR (namely as arising from quantum vacuum fluctuations). Simple models of inflation appear to be able to account for the latest (2013) precision measurements and data from the Planck satellite[166], but the debate over this and other aspects of inflation continues.[167] I will not try to constantly update these notes as this discussion evolves.[168]

Ultimately, in order for inflation to be considered a natural solution to the above problems (and others I have not mentioned - see the literature cited above), one needs to be able to show that inflation arises fairly naturally (in some precise sense) and does not itself require a comparably huge amount of fine-tuning in order to resolve these issues. This is a complicated and intensely debated issue, and one I don't feel sufficiently competent and knowledgeable about to have an informed opinion, let alone utter an opinion in public. It will perhaps only be settled once one has a better understanding of the physics in the very early, pre-inflationary, era that is supposed to be responsible for setting the initial conditions for inflation.

38.4 Aspects of the Cosmological Constant Problem(s)

This is a complex, confusing and multi-faceted problem that has been around for a long time, and I will not remotely be able to do justice to it here.[169] It has been sharpened and brought to the forefront again by the discovery of dark energy.[170]

We have remarked before that the cosmological constant looks like a vacuum energy contribution to the energy-momentum tensor. It is perhaps better to turn this around and

[165] For example, there are claims that in certain inhomogeneous Lemaître-Tolman cosmological models the horizon problem can be avoided without invoking inflation - see e.g. chapter 18.17 of J. Plebanski, A. Krasinski, *An Introduction to General Relativity and Cosmology*.

[166] A. Linde, *Inflationary Cosmology after Planck 2013*, arXiv:1402.0526 [hep-th].

[167] A. Ijjas, P. Steinhardt, A. Loeb, *Inflationary schism after Planck2013*, arXiv:1402.6980 [astro-ph.CO].

[168] For a recent (at the time of writing this) assessment of inflation see J. Martin, *Cosmic Inflation: Trick or Treat?*, arXiv:1902.05286 [astro-ph.CO].

[169] The classic reference for this is the authoritative and influential article by S. Weinberg, *The cosmological constant problem*, Rev. Mod. Phys. 61 (1989) 1-23.

[170] For recent reviews see e.g. S. Carroll, *The Cosmological Constant*, Living Rev. Relativity 4 (2001) 1, http://www.livingreviews.org/lrr-2001-1; J. Polchinski, *The Cosmological Constant and the String Landscape*, arXiv:hep-th/0603249; T. Padmanabhan, *Cosmological Constant - the Weight of the Vacuum*, arXiv:hep-th/0212290; R. Bousso, *TASI Lectures on the Cosmological Constant*, arXiv:0708.4231 [hep-th], J. Martin, *Everything You Always Wanted To Know About The Cosmological Constant Problem (But Were Afraid To Ask)*, arXiv:1205.3365 [astro-ph.CO].

to say that vacuum energy is one potential contribution to the cosmological constant, in the sense that the associated energy density ρ_Λ can be written as

$$\rho_\Lambda = (\rho_\Lambda)_{\text{bare}} + (\rho_\Lambda)_{\text{vac}} = \rho_{\Lambda_0} + \rho_{\text{vac}} \quad . \tag{38.52}$$

Here

$$\rho_{\Lambda_0} = \frac{\Lambda_0}{8\pi G_N} \tag{38.53}$$

is associated with a "bare" cosmological constant Λ_0, some parameter in the action, while ρ_{vac} is a quantum contribution arising from the energy in the ground state or vacuum, i.e. it is the vacuum expectation value of the Hamiltonian density operator $\hat{T}_{00}$,

$$\rho_{\text{vac}} = < \text{vac}|\hat{T}_{00}|\text{vac} > \quad . \tag{38.54}$$

In a Poincaré-invariant field theory in Minkowski space, with a Poincaré-invariant ground state, this will give rise to a contribution

$$< \text{vac}|\hat{T}_{ab}|\text{vac} > = -\rho_{\text{vac}}\eta_{ab} \quad , \tag{38.55}$$

so the equation of state

$$p_{\text{vac}} = -\rho_{\text{vac}} \tag{38.56}$$

is implied by Lorentz invariance.

This suggests that this extends to quantum field theory in a gravitational background (as described e.g. in the references in footnote 87 in section 27.7) in the form

$$< \text{vac}|\hat{T}_{\alpha\beta}|\text{vac} > = -\rho_{\text{vac}} g_{\alpha\beta} \quad . \tag{38.57}$$

In discussions of the cosmological constant problem, it is usually assumed that (38.57) holds, but it is worth bearing in mind that this is a non-trivial (and possibly incorrect) assumption:

1. First of all, the semi-classical approach of treating quantised matter fields in a classical gravitational background is not something that is completely internally consistent, but it is generally believed to be a valid approximation for sufficiently weak gravitational fields (for which one expects whatever quantum gravitational effects may be present in principle to be completely negligible).

2. Even in Minkowski space an expression like (38.55) requires some sort of regularisation procedure, and the procedures that are common or privileged in the case of Poincaré-invariant field theories may either not be available or may not be in any way privileged when one considers a general curved background. Thus there are ambiguities in the calculation of $< \hat{T}_{\alpha\beta} >$ which may effect the validity of (38.57).

One (relatively harmless) ambiguity of this kind would be the addition of a term proportional to the Einstein tensor $G_{\alpha\beta}$ to the right-hand side of (38.57),

$$< \text{vac}|\hat{T}_{\alpha\beta}|\text{vac} >= -\rho_{\text{vac}} g_{\alpha\beta} + \frac{\gamma}{8\pi G_N} G_{\alpha\beta} \tag{38.58}$$

for some constant γ. This would be compatible with $\nabla^\alpha T_{\alpha\beta} = 0$ and with the Minkowskian limit, and would amount to a renormalisation of Newton's constant G_N.

3. Finally, and perhaps most crucially, while reasonable Poincaré-invariant quantum field theories in Minkowski space typically have a preferred and unique Poincaré-invariant ground state, the vacuum, the key feature of quantum field theory in a curved background is that quantum field theory can be extended to this case but that this uniqueness is lost (and it is precisely this non-uniqueness that is at the heart of the characteristic phenomena of quantum field theory in a gravitational field like particle production and Hawking radiation). As a consequence, the state unceremoniously called $|\text{vac} >$ in (38.57) is not unique, and there is no *a priori* reason to believe that (38.57) will hold for any putative vacuum state.

In spite of all this it is usually assumed that something like (38.57) is at least approximately true for sufficiently weak fields, and we will proceed with this assumption.

With the vacuum-energy contributing to the cosmological constant, it is natural to imagine that contributions to the cosmological constant can arise from many quantum field theory processes such as phase-transitions etc, and are of the order of the momentum cut-off, say. Now in field theory in a fixed background, only energy-differences matter, not absolute energies, but the inclusion of gravity and its presumed universal coupling to all forms of energy change this. It is then not clear if one can simply drop finite (let alone infinite) energy-differences from one's equations.

Depending on the physics or physical process one is looking at, natural estimates for the energy scale of a cosmological constant produced in this way via some phase transition, say, might be in the MeV or GeV range, or perhaps (via some wild extrapolation) even at the (ultimate Planck cut-off) scale of

$$E_P = \sqrt{\frac{\hbar c^5}{8\pi G_N}} \approx 10^{18}\ \text{GeV} \tag{38.59}$$

(with the inclusion of the factor of 8π in the definition this is known as the reduced Planck energy). But one does not really have to go that far to see that there is an issue: after all, any of these values are many many orders of magnitude above what was and is compatible with observation, or even with the very existence of our universe.

One cannot simply "solve" this problem by using the bare value Λ_0 of the cosmological constant to cancel the vacuum contribution by hand, because

- this would require an enormous (and unexplained) fine-tuning of this bare value
- contributions to the vacuum energy are expected to arise at various instances during the evolution of the universe while this cancellation could at best be achieved at one point in time during the evolution of the universe (if this cancellation is chosen to take place too early, it will be incompatible with observations today while if it happens too late it will be incompatible with the well-established thermal history of the universe).

It therefore seemed natural to seek some mechanism that would simply make the net cosmological constant identically zero, and a lot of effort went into finding or inventing some mechanism responsible for this. Nowadays, with the strong evidence for dark energy, this cannot be the solution and the original question/problem has morphed into at least 3 distinct questions, namely

1. Why is the cosmological constant Λ not huge?
2. What is the origin of Dark Energy, with $\rho_{DE} \sim (10^{-3}\ \mathsf{eV})^4$?
3. Why is ρ_Λ (constant in time) of the same order of magnitude as ρ_M (today)?

and we will briefly address these in turn.

1. Why is Λ not huge?

   This is the same problem (described above) as before the discovery of dark energy, and is essentially the question "why the vacuum does not gravitate" (Polchinski; see his article in footnote 170 for a very illuminating discussion of this issue).

   This whole business points to some loop-hole in the reasoning leading to (38.57) and the estimates for the size of this quantity.[171] It may even indicate some fundamental lack of (or perhaps mis-) understanding about how gravity couples to quantum fields.[172]

   One on the face of it rather radical, but actually quite conservative, possibility that has been explored (and that I mention here not because I think it is the most plausible explanation but because it is easy to explain and also quite interesting in its own right) is to modify the Einstein equations in such a way that the

[171] See e.g. Q. Wang, Z. Zhu, W. Unruh, *How the huge energy of quantum vacuum gravitates to drive the slow accelerating expansion of the Universe*, `arXiv:1703.00543 [gr-qc]` where it is argued that, because of quantum fluctuations, the quantum vacuum actually gravitates substantially differently from a cosmological constant.

[172] See S. Hollands, R. Wald, *Quantum Field Theory Is Not Merely Quantum Mechanics Applied to Low Energy Effective Degrees of Freedom*, `arXiv:gr-qc/0405082` for some reflections on this issue.

gravitational field does not couple directly to any contribution to the energy-momentum tensor $\sim g_{\alpha\beta}$, i.e. such that they are invariant under shifts

$$T_{\alpha\beta} \to T_{\alpha\beta} + c g_{\alpha\beta} \tag{38.60}$$

for constant c. To that end one postulates the so-called *trace-free Einstein equations*

$$R_{\alpha\beta} - \tfrac{1}{4} g_{\alpha\beta} R = 8\pi G_N \left(T_{\alpha\beta} - \tfrac{1}{4} g_{\alpha\beta} T\right) \ , \tag{38.61}$$

supplemented by the usual covariant energy-momentum tensor conservation law

$$\nabla_\alpha T^{\alpha\beta} = 0 \tag{38.62}$$

(which is not implied by the modified Einstein equations (38.61)).[173] Note that both sides of (38.61) are manifestly traceless. In particular, therefore, any contribution to the energy-momentum tensor $\sim g_{\alpha\beta}$ (like a cosmological constant) does not contribute to (38.61),

$$T_{\alpha\beta} \sim g_{\alpha\beta} \quad \Rightarrow \quad T_{\alpha\beta} - \tfrac{1}{4} g_{\alpha\beta} T = 0 \ , \tag{38.63}$$

and therefore does not couple directly to gravity (said differently, the source term is invariant under the shift (38.60), as required).

While this looks like a major departure from the usual (and well-tested) Einstein equations, and this might make one believe that the equations (38.61) can easily be ruled out experimentally, what actually happens is more subtle and somewhat surprising. Namely, writing (38.61) as

$$R_{\alpha\beta} - \tfrac{1}{2} g_{\alpha\beta} R = 8\pi G_N \left(T_{\alpha\beta} - \tfrac{1}{4} g_{\alpha\beta} T\right) - \tfrac{1}{4} g_{\alpha\beta} R \tag{38.64}$$

and using the contracted Bianchi idenities $\nabla_\alpha G^{\alpha\beta} = 0$ as well as the postulated $\nabla_\alpha T^{\alpha\beta} = 0$, one deduces that

$$\nabla_\alpha (8\pi G_N \, T + R) = 0 \quad \Rightarrow \quad 8\pi G_N \, T = -R + 4\Lambda \tag{38.65}$$

for some integration constant Λ. Plugging this result back into (38.64), one finds

$$\begin{aligned} R_{\alpha\beta} - \tfrac{1}{2} g_{\alpha\beta} R &= 8\pi G_N \, T_{\alpha\beta} - \tfrac{1}{4} g_{\alpha\beta} [8\pi G_N \, T + R] \\ &= 8\pi G_N \, T_{\alpha\beta} - g_{\alpha\beta} \Lambda \ . \end{aligned} \tag{38.66}$$

This is nothing other than the usual Einstein equations with a cosmological constant,

$$R_{\alpha\beta} - \tfrac{1}{2} g_{\alpha\beta} R + \Lambda g_{\alpha\beta} = 8\pi G_N \, T_{\alpha\beta} \ , \tag{38.67}$$

[173] See e.g. G. Ellis, H. van Elst, J. Murugan, J.-P. Uzan, *On the Trace-Free Einstein Equations as a Viable Alternative to General Relativity*, `arXiv:1008.1196v2 [gr-qc]` and references therein for a more detailed discussion and the history of this and other variants of *unimodular gravity*, originally even considered by Einstein himself in 1919.

the crucial difference being that here Λ is *not* determined by the matter content and its vacuum energy but arises solely as an integration constant. While this does not explain the observed tiny value of the cosmological constant, it separates this issue from that of the vacuum fluctuations. Note also that this scenario does not rule out a gravitational coupling to general quantum corrections to states of some physical system (Lamb shift, Casimir energies, ...) which are not Lorentz invariant, as required e.g. by precision tests of the equivalence principle (see the discussion regarding "do vacuum fluctuations gravitate?" in the reviews by Polchinski and Martin in footnote 170).

2. What is the origin of Dark Energy, with $\rho_{DE} \sim (10^{-3}\ \mathsf{eV})^4$?

   Perhaps this is the wrong question altogether. After all, it could just be that Λ happens to be a new constant of nature, like G_N, that has now been found to be non-zero and that has been experimentally determined, and whose precise value cannot currently (or perhaps even in principle) be derived from some underlying theory.

   But a priori it is also possible that the value can be derived, and/or that dark energy is due to something other than a cosmological constant. Again anthropic reasoning (shudder!) can be invoked to argue that the above is a plausible value for the cosmological constant. It would however of course be very desirable to find alternative explanations.

   Numerous models have been proposed that do give rise to a late-time acceleration of the universe for one reason or another (one of the buzzwords here is *quintessence*). Most of them, however, assume (explicitly or implicitly) that somehow the first problem has been solved and that there is no (bare or combined) cosmological constant source term in the Einstein equations.

   It is not clear at all, however, what kind of mechanism could produce on the nose an effect of the desired size. One curious observation, which may provide us with a clue, is that the energy scale E_Λ associated with the cosmological constant,

$$E_\Lambda \approx 10^{-3}\ \mathsf{eV} = 10^{-12}\ \mathsf{GeV}\ , \tag{38.68}$$

   which is much much smaller than the "natural" Planck energy scale,

$$E_\Lambda \approx 10^{-30} E_P\ , \tag{38.69}$$

   is however of the order of the geometric mean of the (current-day) Hubble scale and the Planck scale (I do not know who (if anybody) should be credited with this observation which has probably been made independently multiple times). In energy units this is the curious relation

$$H_0 \approx 10^{-42}\ \mathsf{GeV}\ ,\ E_P \approx 10^{18}\mathsf{GeV} \quad \Rightarrow \quad E_\Lambda \approx (E_P H_0)^{1/2}\ , \tag{38.70}$$

and in time-units, expressed in terms of the (reduced) *Planck time*

$$t_P = \sqrt{\frac{\hbar 8\pi G_N}{c^5}} \approx 10^{-43}\ \mathtt{s}\ , \tag{38.71}$$

this is the equally baffling statement that

$$(H_0)^{-1} \approx 10^{60} t_P \quad , \quad \Lambda \approx 10^{-120} (t_P)^{-2} \approx (H_0)^2\ . \tag{38.72}$$

This is certainly intriguing, as it appears to relate the cosmological constant to both a fundamental UV cut-off (the Planck scale) and a current IR cut-off (the Hubble scale), which has led to some "holographic" ruminations.[174] This curious relation is also one aspect of the so-called "coincidence problem", and we will briefly come back to this below.

A more conservative realisation of the scenario described by (38.70) is also possible.[175] For concreteness, consider the scalar field in a cosmological background discussed in section 34.10, with a UV momentum cutoff at a momentum $k = k_{UV}$. Namely if, for one reason or another, one can argue that the leading quartic divergence in the vacuum energy

$$\rho_{\rm vac} \sim \tfrac{1}{2} \int \frac{d^3k}{(2\pi)^3} \sqrt{k^2 + m_{eff}^2} = \frac{(k_{UV})^4}{16\pi^2} + \dots \tag{38.73}$$

(which would also be present in, and hence destroy, Minkowski space) should be subtracted, then the subleading term is quite generically of the form

$$\rho_{\rm vac} \sim (k_{UV})^2 (H_0)^2 \tag{38.74}$$

(in the model in section 34.10 this H_0^2-term would arise from the scalar curvature term in the effective mass). With a cut-off at the Planck scale, this gives a vacuum energy contribution of the order of the observed dark energy density,

$$k_{UV} \sim k_P \quad \Rightarrow \quad \rho_{\rm vac} \sim (E_P)^2 (H_0)^2 \sim (\rho_\Lambda)_{\rm observed}\ . \tag{38.75}$$

In designing such scenarios, care should be taken that the vacuum contribution thus determined really has an equation of state parameter equal (or very close to) $w = -1$, so that $\rho_{\rm vac}$ calculated today is really (approximately) constant (and does not behave like $\rho_{\rm vac}(t) \sim H(t)^2$, say, which would be in conflict with many cosmological observations). Exploring scenarios of this kind is clearly a constructive alternative to anthropic incantations.

[174] See e.g. A. Cohen, D. Kaplan, A. Nelson, *Effective Field Theory, Black Holes, and the Cosmological Constant*, `arXiv:hep-th/9803132`, M. Li, *A Model of Holographic Dark Energy*, `arXiv:hep-th/0403127`, T. Padmanabhan, *Vacuum Fluctuations of Energy Density can lead to the observed Cosmological Constant*, `arXiv:hep-th/0406060`, S. Hsu, A. Zee, *A speculative relation between the cosmological constant and the Planck mass*, `arXiv:hep-th/0406142`, and references thereto for theoretical speculations along these lines.

[175] See e.g. M. Maggiore, *Zero-point quantum fluctuations and dark energy*, `arXiv:1004.1782 [astro-ph.CO]`, and D. Bernard, A. LeClair, *Scrutinizing the Cosmological Constant Problem and a possible resolution*, `arXiv:1211.4848 [hep-th]`.

3. Why is ρ_Λ (constant in time) of the same order of magnitude as ρ_M (today)?

   This is weird, and one version of what is known as the *Coincidence Problem* in cosmology. To see why this is peculiar, note that ρ_Λ is constant while $\rho_M \sim a^{-3}$ so that the ratio of the two behaves as

$$\frac{\rho_\Lambda(t)}{\rho_M(t)} \sim a(t)^3 \ . \tag{38.76}$$

   It is therefore, one might reason, extremely unlikely to find oneself at a time when the two happen to be comparable in size, but nevertheless this is precisely the time we find ourselves at (how precisely depends among other things on what one means by "comparable in size"). However, any such statement has to also rely on some probability distribution for something (the probability that observers exist at some time t_0?) that allows one to quantify this statement somewhat. But it is not clear if this is an issue that really requires an explanation.

   If one thinks that this coincidence is indeed a problem that requires a solution, then one can try to come up with models in which dark energy does not behave like a cosmological constant at all times. However:

   > "Why obfuscate? If a poet sees something that walks like a duck and swims like a duck and quacks like a duck, we will forgive him for entertaining more fanciful possibilities, It could be a unicorn in a duck suit - who's to say! But we know that more likely, it's a duck."[176]

   Fair enough.[177]

   Note by the way that this coincidence issue is equivalent to the empirical observation (38.70) mentioned above. Namely, with $\rho_\Lambda \sim \rho_M$, and both of order of the critival density because $\Omega_M + \Omega_\Lambda = 1$, one has

$$\rho_\Lambda = \frac{\Lambda}{8\pi G_N} \sim (\rho_c)_0 = \frac{3H_0^2}{8\pi G_N} \sim H_0^2 E_P^2 \tag{38.77}$$

   which is just (38.70). One can also eliminate any reference to Planck units and the Planck scale and write this as (38.72)

$$\Lambda \approx (H_0)^2 \ . \tag{38.78}$$

   Thus another (and perhaps more objective and constructive) way of stating the coincidence issue is as the curious fact that the two a priori completely unrelated time-scales, one set by $(H_0)^{-1}$ (which turns out to be remarkably close to the current best estimates for the age of the universe) and the other set by the energy density or curvature radius of dark energy, are approximately equal. Why should

[176] R. Bousso, *TASI Lectures on the Cosmological Constant*, `arXiv:0708.4231 [hep-th]`.

[177] See also E. Bianchi, C. Rovelli, *Why all these prejudices against a constant?*, `arXiv:1002.3966 [astro-ph.CO]` for further reflections on this and related issues.

the cosmological constant today be related to the age of the universe today??? Perhaps this is indeed not a coincidence and is trying to tell us something.

It is not clear if a fundamental physics (non-multiverse, non-anthropic) explanation for dark energy can be found that is capable of resolving all 3 of the cosmological constant problems at once, but perhaps one should at least be humble and continue to entertain the possibility, alluded to above, that this issue is telling us that we are missing something fundamental about the coupling of gravity and quantum fields.

This is not to say that there is not a strong case to be made for what appears to be an extreme fine-tuning of other (standard model) parameters to support everything from primordial nucleosynthesis to chemistry and life as we know it.[178] However, among other things precisely because of the proviso "as we know it", and also because of the (as far as I know) limited understanding of chemistry based on other than standard model gauge theories, this begs the question and the significance of these findings is far from clear and difficult to assess.

[178] See e.g. L. Barnes, *The Fine-Tuning of the Universe for Intelligent Life*, `arXiv:1112.4647 [physics.hist-ph]` for a recent assessment of the situation.